EE 3072
System Dynamics and Control

Lecture 6
Frequency-Domain Analysis

Contents

- Frequency Response (or Frequency Characteristics)
- Bode Diagrams
- Frequency-Domain Specifications
- Nyquist Plots
- Nyquist Stability Criterion

Frequency Response

A Perspective on the Frequency-Response Design Method

The design of feedback control systems in industry is often accomplished using frequency-response methods.

Advantages of frequency-response design:

-It provides good designs in the face of uncertainty in the plant model

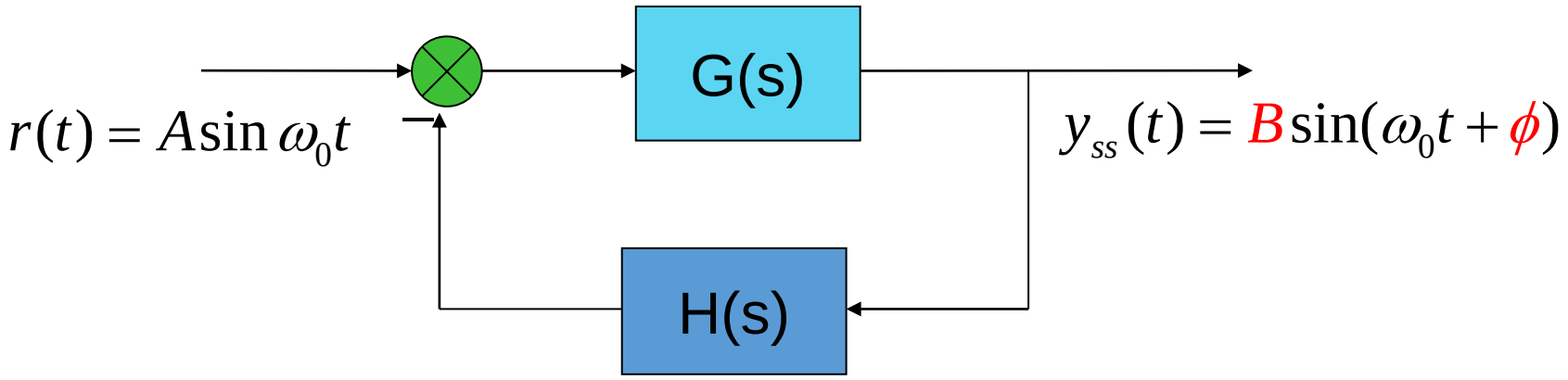
-Experimental information can be used for design purposes.

Raw measurements of the output amplitude and phase of a plant undergoing a sinusoidal input excitation are sufficient to design a suitable feedback control.

-No intermediate processing of the data (such as finding poles and zeros) is required to arrive at the system model.

Frequency response

The frequency response of a system is defined as the steady-state response of the system to a sinusoidal input signal.



For a stable LTI system, when the input to it is a sinusoid signal, the resulting output, as well as all signals throughout the system, are also sinusoidal and have the same frequency as the input signal at steady-state.

The output differs from the input waveform only in amplitude and phase.

Frequency response

To see this, consider a system with the TF

$$T(s) = \frac{K\prod_{i=1}^{m}(s+z_i)}{\prod_{i=1}^{n}(s+p_i)}$$

with an input

$$r(t) = A\sin(\omega_0 t)$$

$$R(s) = \frac{A\omega_0}{s^2+\omega_0^2}$$

The output can be written as

$$Y(s) = T(s)R(s) = \frac{A_1}{s+p_1} + \cdots + \frac{A_n}{s+p_n} + \frac{\alpha s+\beta}{s^2+\omega_0^2}$$

Taking the inverse Laplace transform yields

$$y(t) = A_1 e^{-p_1 t} + \cdots + A_n e^{-p_n t} + L^{-1}\left\{\frac{\alpha s+\beta}{s^2+\omega_0^2}\right\}, t \geq 0$$

Frequency response

For a stable system, at steady state we have

$$y_{ss} = \lim_{t\to\infty} y(t) = \lim_{t\to\infty} L^{-1}\left\{\frac{\alpha s+\beta}{s^2+\omega_0^2}\right\} = A\left|T(j\omega_0)\right|\sin(\omega_0 t+\phi)$$

where

$\left|T(j\omega_0)\right| \Rightarrow$ is the magnitude $\quad \phi = \measuredangle T(j\omega_0) \Rightarrow$ is the phase

of $T(s)$ at $s = j\omega_0$.

Therefore, for a stable LTI system with TF T(s), the response to an input $r(t) = A\sin\omega_0 t$ at steady state is $y(t) = B\sin(\omega_0 t+\phi)$ where $B = A\left|T(j\omega_0)\right|$ and $\phi = \measuredangle T(j\omega_0)$

The steady-state output signal depends only on the magnitude and phase of $T(j\omega)$ at $\omega=\omega_0$.

Fourier Transform and Laplace Transform Relationship

Recall that the Fourier transform of a signal $y(t)$ defined for $t \geq 0$, i.e., satisfying $y(t) = 0, t < 0$, is

$$Y(j\omega) = F\{y(t)\} = \int_{-\infty}^{+\infty} y(t)e^{-j\omega t}dt = \int_{0}^{+\infty} y(t)e^{-j\omega t}dt$$

and the inverse Fourier transform of $Y(j\omega)$ is

$$y(t) = F^{-1}\{Y(j\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{+\infty} Y(j\omega)e^{j\omega t}d\omega$$

For a signal $y(t)$ satisfying $y(t) = 0, t < 0$, we have $Y(j\omega) = Y(s)_{|s=j\omega}$ since the limits of the of integration of the Laplace and Fourier integral are the same.

This shows the close relationship between the Laplace and Fourier transforms.

Fourier Transform and Laplace Transform Relationship

The Laplace transform allows us to investigate the s-plane pole and zero locations of a given TF and therefore the time response characteristics of the system.

The Fourier transform (i.e., the frequency response methods) allows us to investigate the steady state amplitude and phase characteristics of the system.

In other words, frequency response methods allow us to investigate the steady state characteristic of the response to sinusoidal input signals with different frequencies.

Magnitude and Phase of a TF

Consider a TF G(s) of some system. The frequency-domain (sinusoidal) TF $G(j\omega)$ is a complex variable (of ω) and can be described by

$$G(j\omega) = G(s)_{|s=j\omega} = G_R(\omega) + jG_I(\omega)$$

where $G_R(\omega) = \mathrm{Re}\{G(j\omega)\}$ $G_I(\omega) = \mathrm{Im}\{G(j\omega)\}$

In polar notation it is given by

$$G(j\omega) = |G(j\omega)|e^{-j\measuredangle G(j\omega)} = |G(j\omega)|\measuredangle G(j\omega)$$

magnitude phase

where $|G(j\omega)| = \sqrt{(G_R(\omega))^2 + (G_I(\omega))^2}$ is its magnitude

$$\phi = \measuredangle G(j\omega) = \tan^{-1}\left(\frac{G_I(\omega)}{G_R(\omega)}\right)$$ is its phase

Example. Frequency response of a Capacitor

Consider the capacitor described by the equation

$$i = C\frac{dv}{dt}$$

where *v* is the input and *i* is the output. Determine the sinusoidal steady-state response of the capacitor.

Solution. The transfer function of the capacitor is

$$\frac{I(s)}{V(s)} = G(s) = Cs$$

So $$G(j\omega) = Cj\omega$$

Computing the magnitude and phase, we find that

$$\left|G(j\omega)\right| = \left|Cj\omega\right| = C\omega$$

$$\measuredangle G(j\omega) = \phi = 90^\circ$$

Gain characteristic: $|G(j\omega)| = |Cj\omega| = C\omega$

Phase characteristic: $\measuredangle G(j\omega) = \phi = 90^\circ$

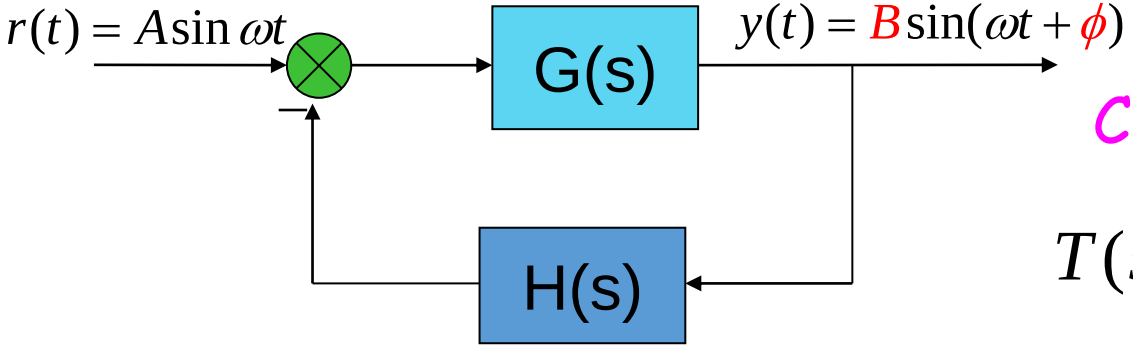
Output: $I(j\omega) = G(j\omega)V(j\omega)$

For a unit-amplitude sinusoidal input *v*, the output *i* will be a sinusoid with magnitude $C\omega$, and the phase of the output will lead the input by 90°.

Note that for this example the magnitude is proportional to the input frequency while the phase is independent of frequency.

Magnitude and Phase of a TF

Similar to OL systems, for a CL control system



CL transfer function

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

For frequency-domain analysis, we basically replace s by jω:

$$T(j\omega) = \frac{Y(j\omega)}{R(j\omega)} = \frac{G(j\omega)}{1 + G(j\omega)H(j\omega)}$$

The frequency-domain (sinusoidal) transfer function T(jω) may be expressed in terms of its magnitude and phase:

$$T(j\omega) = \underbrace{|T(j\omega)|}_{\text{magnitude}} \underbrace{\angle T(j\omega)}_{\text{phase}}$$

Magnitude and Phase of a TF

The magnitude of T(jω) is

$$|T(j\omega)| = \left|\frac{G(j\omega)}{1+G(j\omega)H(j\omega)}\right| = \frac{|G(j\omega)|}{|1+G(j\omega)H(j\omega)|}$$

⇐ Gain characteristics

The phase of T(jω) is

$$\measuredangle T(j\omega) = \phi_T(j\omega) = \measuredangle G(j\omega) - \measuredangle\left[1+G(j\omega)H(j\omega)\right]$$

⇐ Phase characteristics

Note that both the magnitude and the phase of a sinusoidal TF depend on the frequency ω of the input signal.

Therefore, the system might have different response to sinusoidal signals with different frequencies.

The magnitude and phase characteristics of the given TF can be represented in graphical form using Bode diagrams and Nyquist plots.

Bode Diagrams
(Logarithmic Plots)

Bode Diagram (Plot)

The Bode plot or diagram of the function $G(j\omega)$ consists of two plots:

-- the magnitude of $G(j\omega)$ in decibels (dB) versus $\log_{10}(\omega)$ or ω

-- the phase of $G(j\omega)$ in degrees as a function of $\log_{10}(\omega)$ or ω.

The magnitude of $G(j\omega)$ in decibels (dB) is obtained by multiplying the logarithm (base 10) of $|G(j\omega)|$ by 20

$$20\log_{10}|G(j\omega)|$$

The primary advantage of Bode plots is that the multiplicative factors in the magnitude are converted into additive factors.

In Bode plots, the contribution of each factor in the TF can be observed (in contrast to Nyquist plots).

Bode Diagram (Plot)

Consider the general TF to illustrate the construction of Bode plots

$$G(s)=\frac{K\prod_{i=1}^{Q}(1+T_i s)}{s^N\prod_{m=1}^{M}(1+\tau_m s)\prod_{k=1}^{R}\left(1+2\zeta_k\left(s/\omega_{n_k}\right)+\left(s/\omega_{n_k}\right)^2\right)}$$

where $T_i, i=1,\ldots,Q;\tau_m, m=1,\ldots,M;\zeta_k,\omega_{n_k}, k=1,\ldots,R$ are real constants.

Substituting $s=j\omega$ into $G(s)$ yields

$$G(j\omega)=\frac{K\prod_{i=1}^{Q}(1+j\omega T_i)}{(j\omega)^N\prod_{m=1}^{M}(1+j\omega\tau_m)\prod_{k=1}^{R}\left(1+2\zeta_k\left(j\omega/\omega_{n_k}\right)+\left(j\omega/\omega_{n_k}\right)^2\right)}$$

Bode Diagram (Plot)

The magnitude of $G(j\omega)$ in dB is

$$\left|G(j\omega)\right|_{dB} = 20\log_{10}\left|G(j\omega)\right|$$

$$= 20\log_{10}\left|K\right| + \sum_{i=1}^{Q} 20\log_{10}\left|1+jT_i\omega\right| - N\times 20\log_{10}\left|j\omega\right|$$

$$-\sum_{m=1}^{M} 20\log_{10}\left|1+j\tau_m\omega\right| - \sum_{k=1}^{R} 20\log_{10}\left|1+2\zeta_k\left(j\omega/\omega_{n_k}\right)-\left(\omega/\omega_{n_k}\right)^2\right|$$

The phase of $G(j\omega)$ is

$$\angle G(j\omega) = \angle K + \sum_{i=1}^{Q} \angle(1+jT_i\omega) - N\times(\angle j\omega)$$

$$-\sum_{m=1}^{M} \angle(1+j\tau_m\omega) - \sum_{k=1}^{R} \angle\left(1+2\zeta_k\left(j\omega/\omega_{n_k}\right)-\left(\omega/\omega_{n_k}\right)^2\right)$$

Bode Diagram (Plot)

Note that the contribution of each factor in the TF appears as additive term both in the magnitude and phase equations.

The Bode diagram of a general TF $G(j\omega)$ can be obtained by adding the bode diagrams of its factors.

Therefore, we will first consider the Bode diagrams of typical factors in TFs.

Typical factors which usually occur in TF's $G(j\omega)$ are:

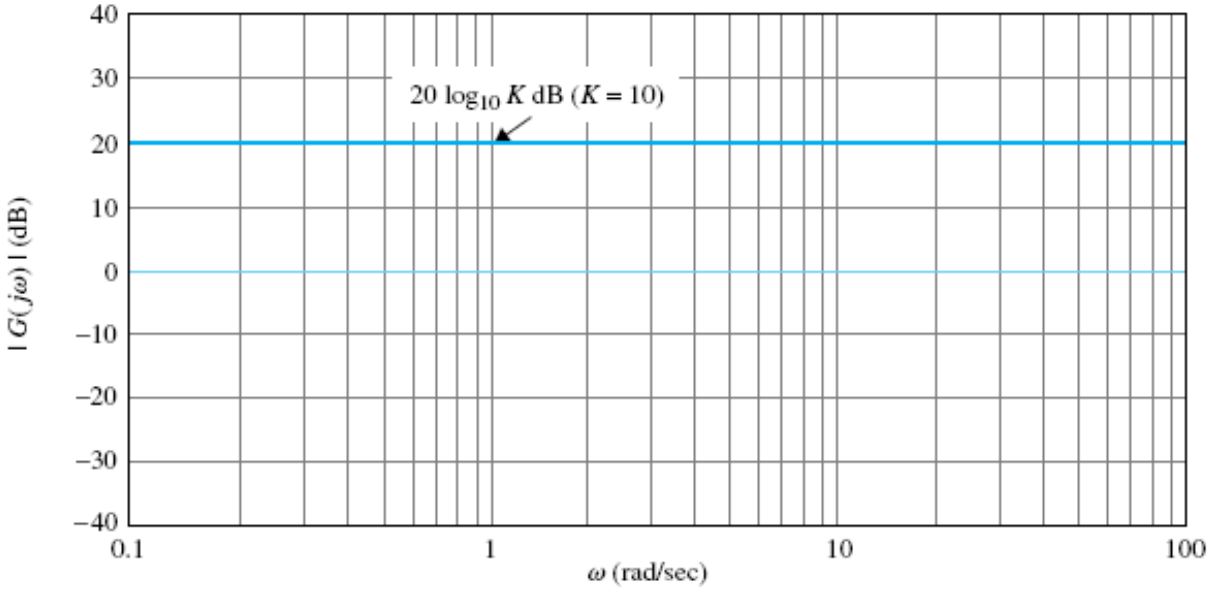
1. Constant gain: K
2. Poles or zeros at the origin: $(j\omega)$
3. Poles or zeros on the real axis: $(1+j\omega T)$
4. Complex conjugate poles or zeros:

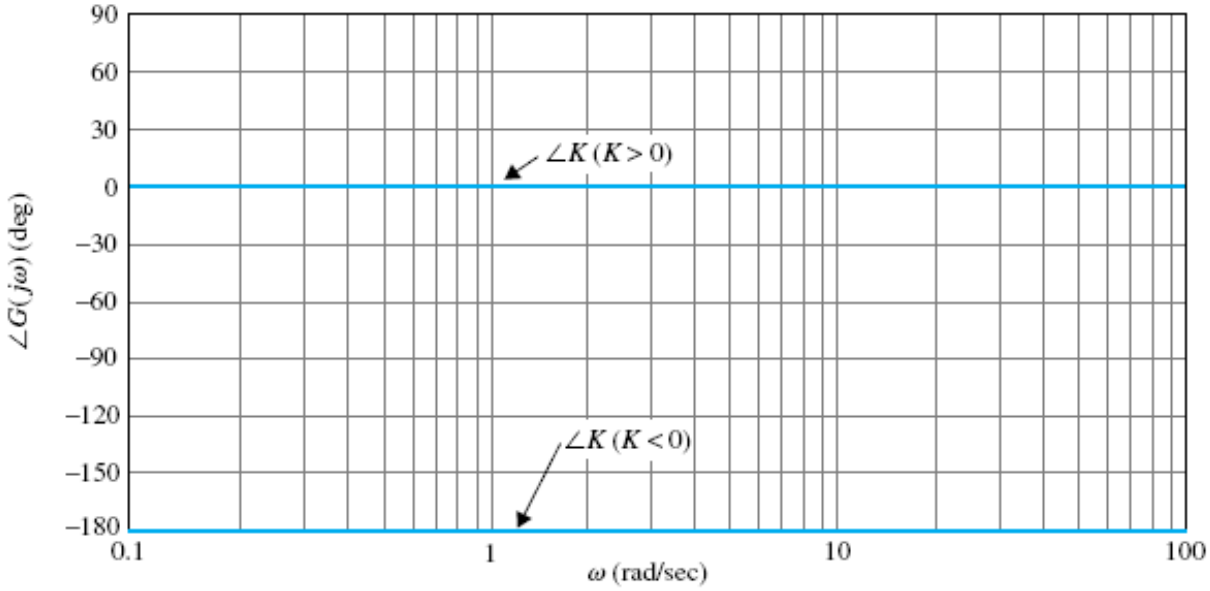
$$(1 + j2\zeta\omega/\omega_n - \omega^2/\omega_n^2)$$

1. Real constant K

$$K_{dB} = 20\log_{10}|K|$$
$$= \text{constant}$$

$$\measuredangle K = \begin{cases} 0^\circ & K > 0 \\ 180^\circ & K < 0 \end{cases}$$





2. Poles or zeros at the origin, $(j\omega)^{\pm N}$

Magnitude of $(j\omega)^{\pm N}$ in dB:

$$\left|(j\omega)^{\pm N}\right|_{dB} = 20\log_{10}\left|(j\omega)^{\pm N}\right| = \pm N \times 20\log_{10}\omega \quad \text{dB}$$

For a given N, it is a straight line with the slope:

$$\frac{d}{d\log_{10}\omega}\left(\pm N \times 20\log_{10}\omega\right) = \pm N \times 20 \quad \text{dB/decade}$$

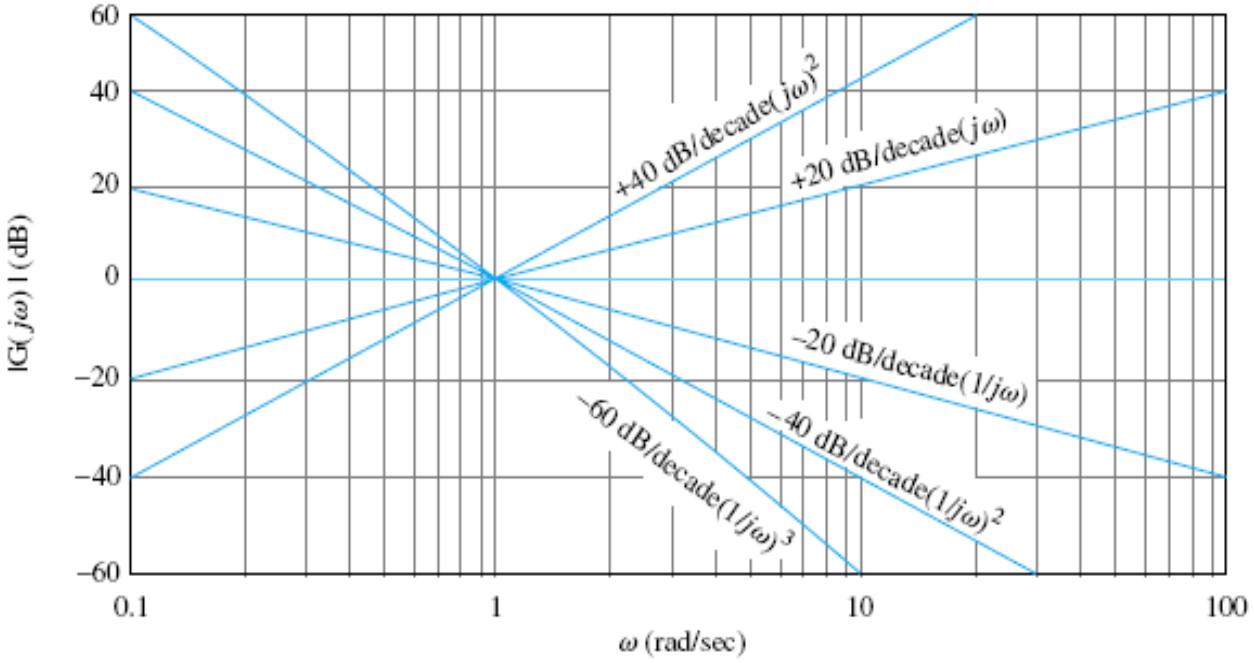
Thus, a unit change in $\log_{10}\omega$ corresponds to a change of $\pm$20 dB in the magnitude.

At ω=1, $\left|(j\omega)^{\pm N}\right|_{dB} = 0.$

So, these lines pass through the 0dB axis at ω=1.

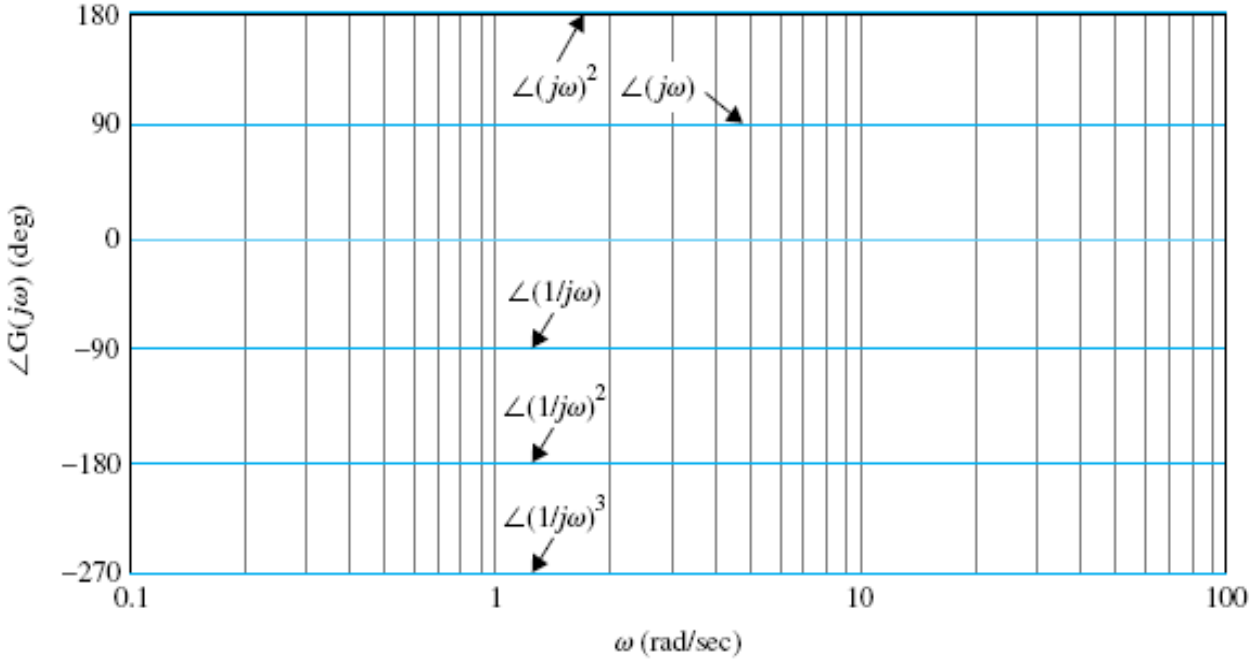
Magnitude of $(j\omega)^{\pm N}$:

$\pm N \times 20 \log_{10} \omega$ dB



Phase of $(j\omega)^{\pm N}$:

$\pm N \times 90^{\circ}$



3. (a) Simple zero on the real axis $(1+j\omega T)$

Consider the function $G(j\omega)=1+j\omega T$

where T is a positive real constant.

The magnitude of G(jω) in dB is

$$|G(j\omega)|_{dB}=20\log_{10}|G(j\omega)|=20\log_{10}\sqrt{1+\omega^2T^2}$$

At very low frequencies, $\omega T \ll 1$

$$|G(j\omega)|_{dB}\cong 20\log_{10}1=0 \ \text{dB}$$

(ω^2T^2 is neglected when compared with 1.)

At very high frequencies, $\omega T \gg 1$

$$|G(j\omega)|_{dB}\cong 20\log_{10}\sqrt{\omega^2T^2}=20\log_{10}\omega T \ \text{dB}$$

The two lines are called asymptotes.

The two lines intersect at:

$\boxed{\omega=1/T}$

(corner frequency)

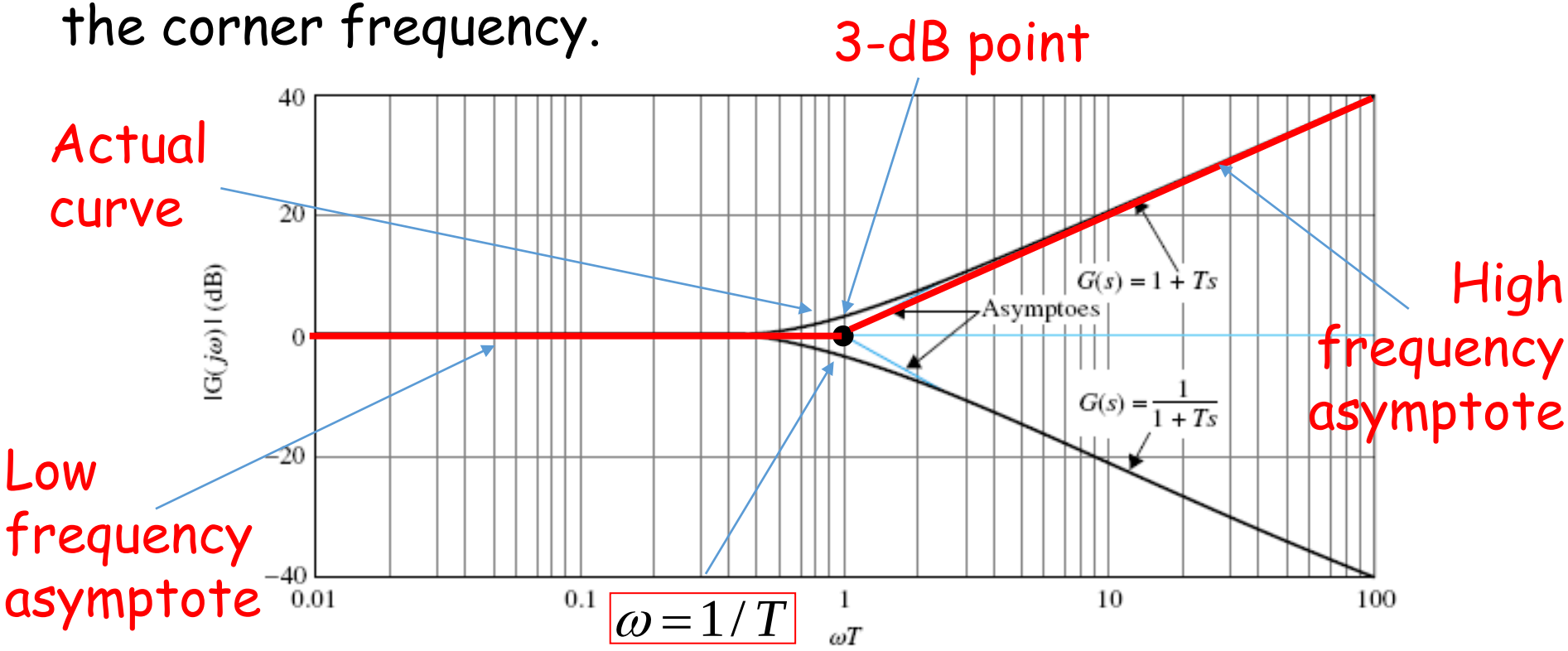
This represents a straight line with a slope of +20dB

The steps of making of sketch of $\left|1+j\omega T\right|_{dB}$

Step 1: Locate the corner frequency ω=1/T on the frequency axis;

Step 2: Draw the two asymptotes (i.e., the 20dB/decade line and the horizontal line at 0 dB) with the two lines intersecting at ω=1/T.

Step 3: Sketch a smooth curve by locating the 3-dB point at the corner frequency.

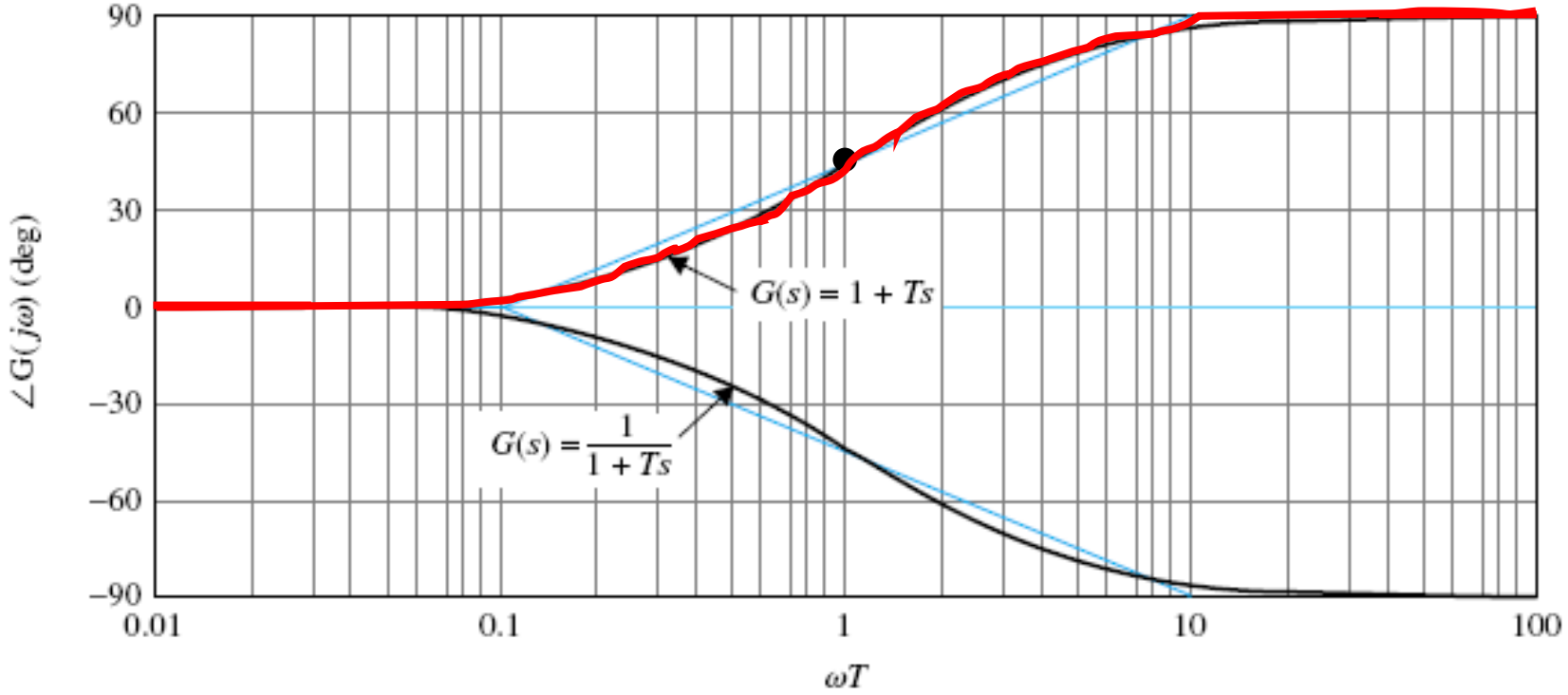


The phase of G(jω)=1+jωT is $\measuredangle G(j\omega) = \tan^{-1}\omega T$

At very low frequencies, $\measuredangle G(j\omega) \cong 0^\circ$

At very high frequencies, $\measuredangle G(j\omega) \cong 90^\circ$

At the corner frequency, $\measuredangle G(j\omega) = 45^\circ$



As ω varies from 0 to ∞, $\measuredangle G(j\omega)$ varies from 0° to +90°.

3. (b) Simple pole on the real axis $1/(1+j\omega T)$

Consider the function $G(j\omega) = \dfrac{1}{1+j\omega T}$

The magnitude of G(jω) in dB is

$$|G(j\omega)|_{dB} = 20\log_{10}|G(j\omega)| = -20\log_{10}\sqrt{1+\omega^2T^2}$$

At very low frequencies, $\omega T \ll 1$

$$|G(j\omega)|_{dB} \cong -20\log_{10} 1 = 0 \ \text{dB}$$

At very high frequencies, $\omega T \gg 1$

$$|G(j\omega)|_{dB} \cong -20\log_{10}\sqrt{\omega^2T^2} = -20\log_{10}\omega T \ \text{dB}$$

The two asymptotes intersect at:

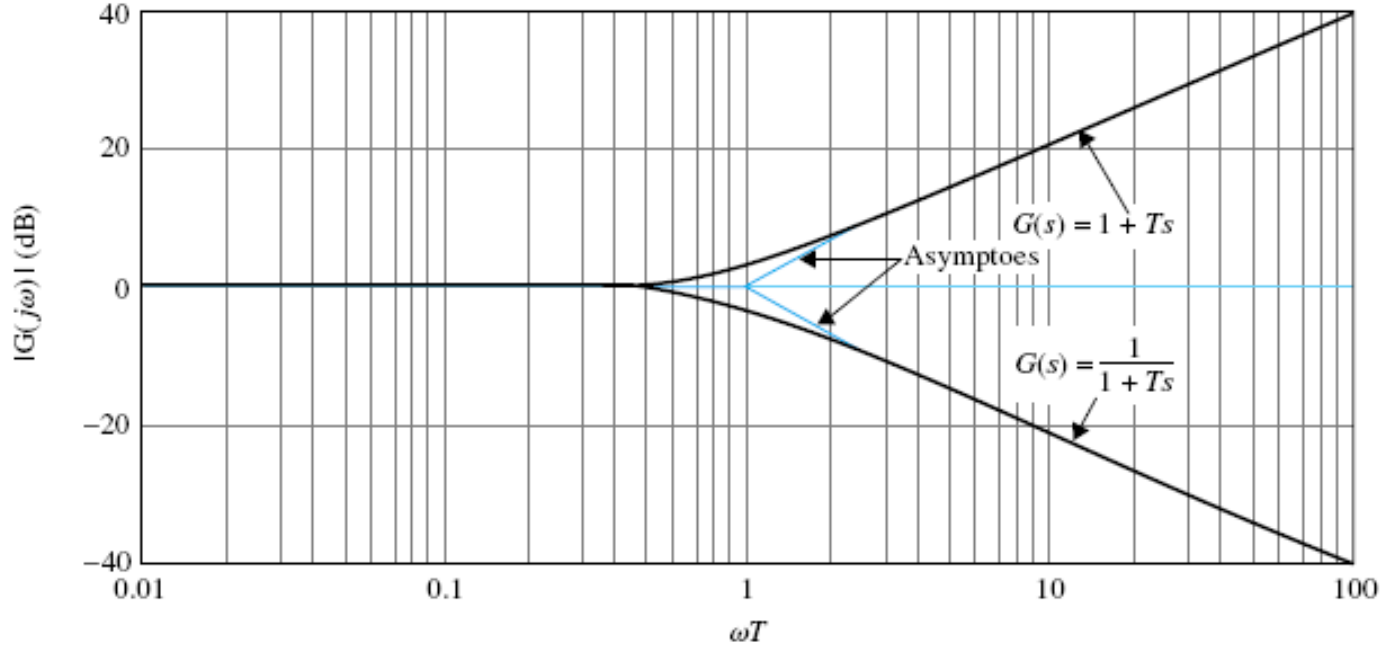
$\boxed{\omega = 1/T}$

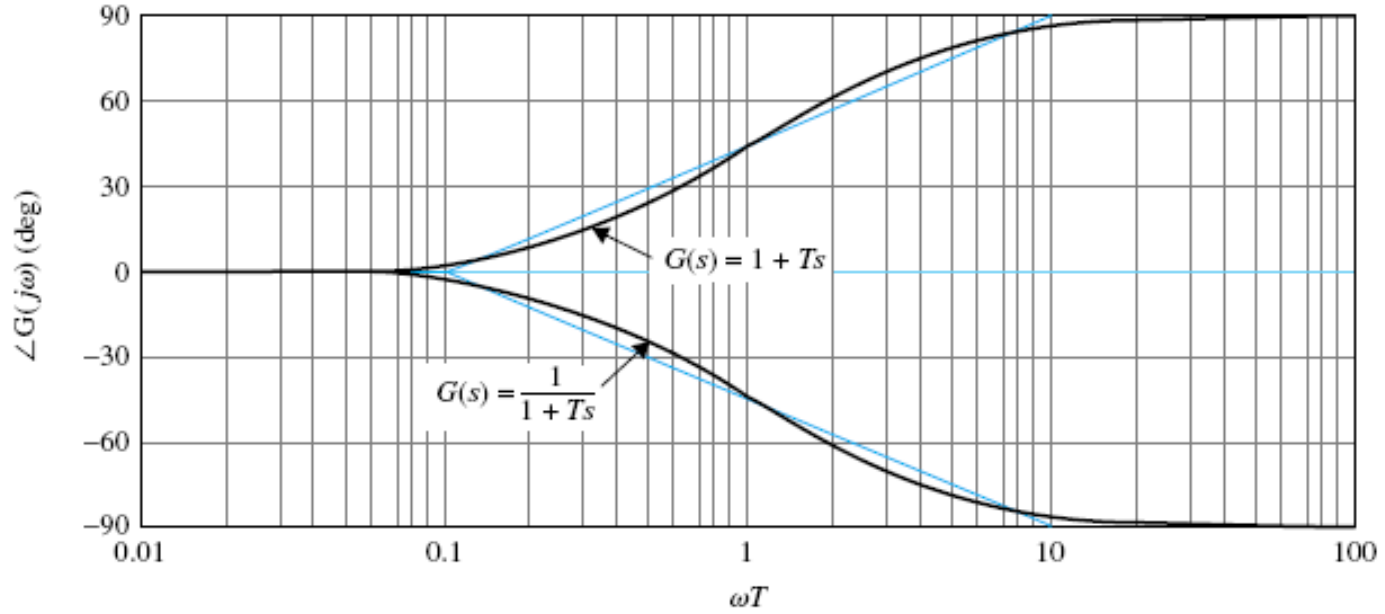
(corner frequency)

This represents a straight line with a slope of -20dB

The phase of G(jω): $\angle G(j\omega) = -\tan^{-1}\omega T$

As ω varies from 0 to ∞, $\angle G(j\omega)$ varies from 0° to -90°.


40
20
0
−20
−40
0.01
0.1
1
10
100
ωT
|G(jω)| (dB)
G(s) = 1 + Ts
Asymptoes
G(s) = 1/(1 + Ts)


90
60
30
0
−30
−60
−90
0.01
0.1
1
10
100
ωT
∠G(jω) (deg)
G(s) = 1 + Ts
G(s) = 1/(1 + Ts)

3. (c) Multi-order poles or zeros on the real axis $(1+j\omega T)^{\pm N}$

Corner frequency is still

$$\boxed{\omega = 1/T}$$

Slope of the high frequency asymptote is

$$\pm N \times 20 \text{ dB/decade}$$

The phase angle of G(jω) at each point is:

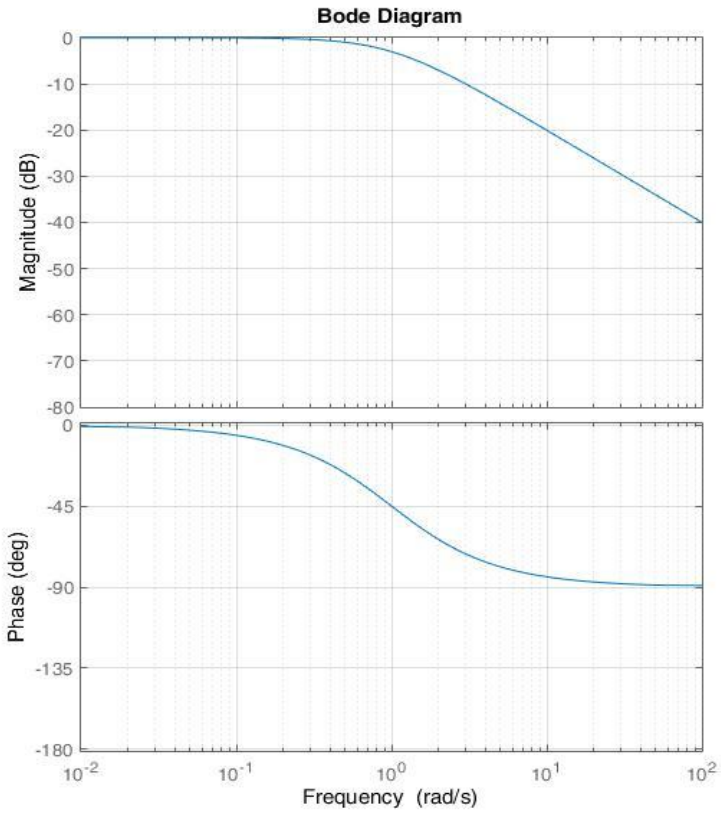
$$\pm N \times \phi$$

where ϕ is the angle for the simple pole/zero case.

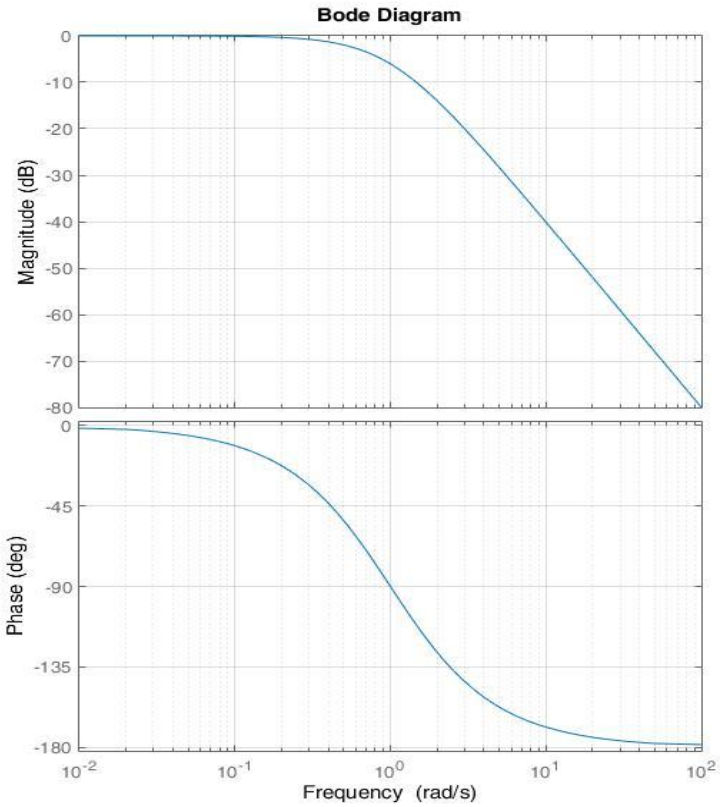
The deviation of the actual curve form the asymptotes at the corner frequency is also N times the deviation in the simple pole/zero case.

This case is equal to basically adding N times the Bode plot of the simple pole case.

Example



$$G(j\omega) = \frac{1}{1+ j\omega}$$



$$G(j\omega) = \frac{1}{(1+ j\omega)^2}$$

4. Complex poles and zeros $\left(1+2\zeta_k\left(j\omega/\omega_{n_k}\right)+\left(j\omega/\omega_{n_k}\right)^2\right)^{\mp 1}$

Consider the second-order transfer function

$$G(s)=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}=\frac{1}{1+(2\zeta/\omega_n)s+(1/\omega_n^2)s^2}$$

We are interested only in the case when $\zeta \leq 1$, since otherwise G(s) would have two unequal real poles, and the Bode plot can be obtained by considering G(s) as the product of two transfer functions with simple poles.

By letting s=jω, G(s) becomes

$$G(j\omega)=\frac{1}{[1-(\omega^2/\omega_n^2)]+j2\zeta(\omega/\omega_n)}$$

$$G(j\omega) = \frac{1}{[1-(\omega^2/\omega_n^2)] + j2\zeta(\omega/\omega_n)}$$

The magnitude of G(jω) in dB is

$$|G(j\omega)|_{dB} = 20\log_{10}|G(j\omega)|$$

$$= -20\log_{10}\sqrt{[1-(\omega^2/\omega_n^2)]^2 + 4\zeta^2(\omega/\omega_n)^2}$$

At very low frequencies, $\omega/\omega_n << 1$

$$|G(j\omega)|_{dB} \cong -20\log_{10} 1 = 0 \ \text{dB}$$

At very high frequencies, $\omega/\omega_n >> 1$

$$|G(j\omega)|_{dB} \cong -20\log_{10}\sqrt{(\omega/\omega_n)^4} = -40\log_{10}(\omega/\omega_n) \ \text{dB}$$

The two lines intersect at:

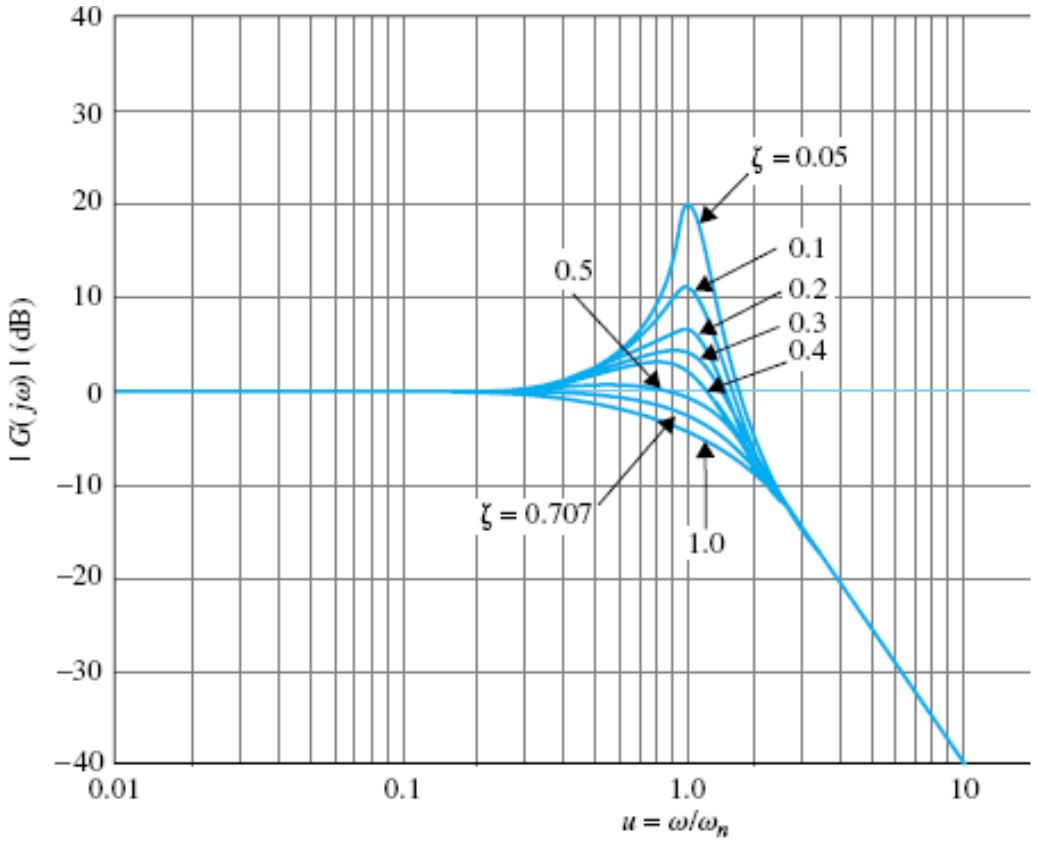
$$\boxed{\omega = \omega_n}$$

(corner frequency)

This equation represents a straight line with a slope of -40 dB/decade in the Bode plot coordinates.

The actual magnitude curve of G(jω) in this case may differ strikingly from the asymptotic curve.

The reason for this is that the amplitude and phase curves of the second-order G(jω) depend not only on the corner frequency ω_n, but also on the damping ratio ζ, which does not enter the asymptotic curve.



|G(jω)| has a peak which depends on ζ

Resonant peak

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

It occurs at resonant frequency

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}$$

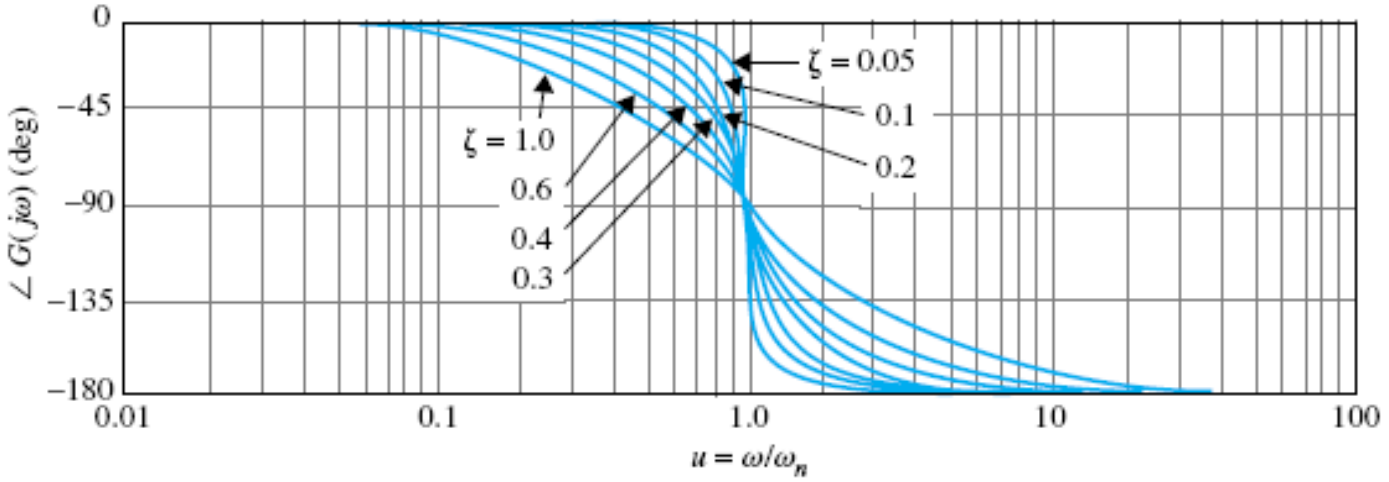
$$G(j\omega) = \frac{1}{[1-(\omega^2/\omega_n^2)] + j2\zeta(\omega/\omega_n)}$$

The phase of G(jω) is given by $\measuredangle G(j\omega) = -\tan^{-1}\left(\dfrac{2\zeta\,\omega/\omega_n}{1-\left(\omega/\omega_n\right)^2}\right)$

At very low frequencies, $\measuredangle G(j\omega) \cong 0^\circ$

At very high frequencies, $\measuredangle G(j\omega) \cong -180^\circ$

At the corner frequency, $\measuredangle G(j\omega) = -90^\circ$



Example Consider the following transfer function

$$G(s) = \frac{10(s+10)}{s(s+2)(s+5)}$$

Sketch its Bode Plot.

Solution. Letting s=jω, we have

$$G(j\omega) = \frac{10(j\omega+10)}{(j\omega)(j\omega+2)(j\omega+5)}$$

Reformulating it into the form for Bode Plot

$$G(j\omega) = \frac{K(1+jT_1\omega)}{(j\omega)(1+j\tau_1\omega)(1+j\tau_2\omega)} = \frac{10(1+j0.1\omega)}{(j\omega)(1+j0.5\omega)(1+j0.2\omega)}$$

where $K = 10, T_1 = 0.1, \tau_1 = 0.5, \tau_2 = 0.2$

So G(jω) has corner frequencies at ω = 10, 2, and 5 rad/sec.

Example

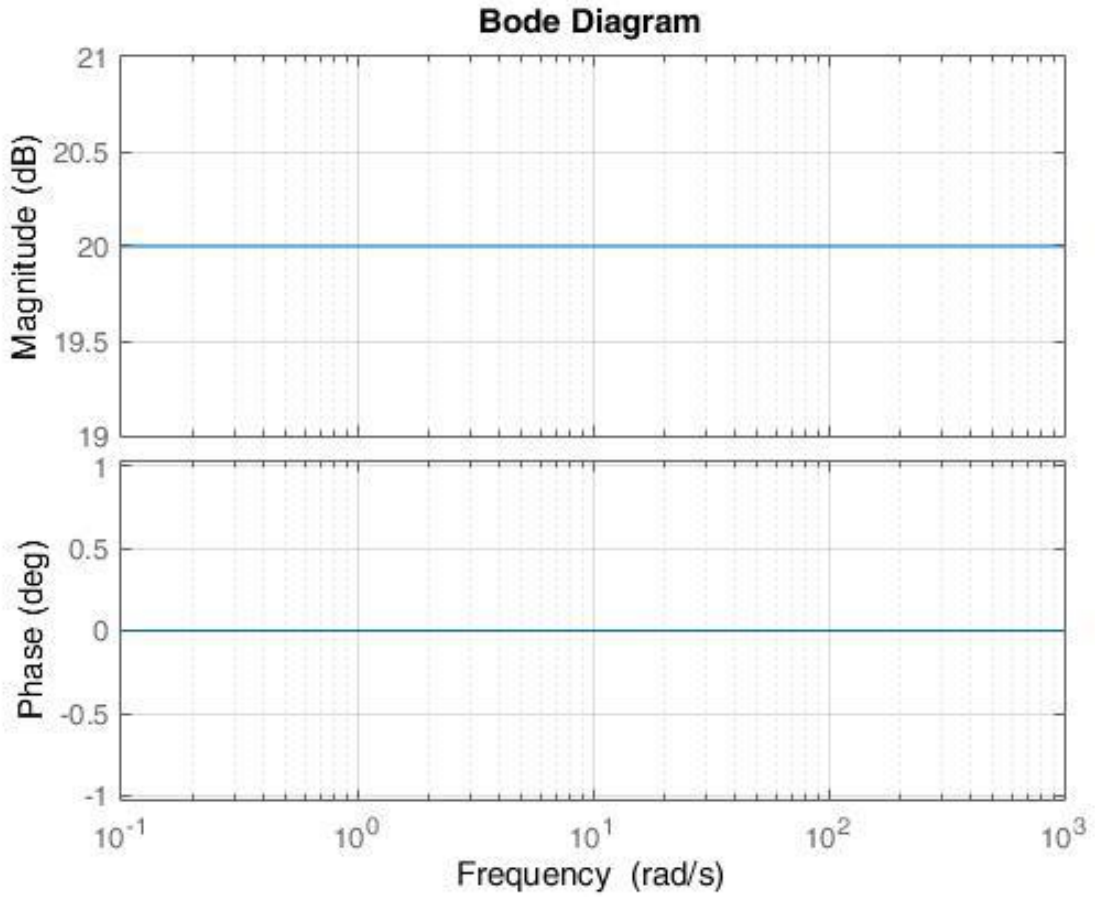
$$G(j\omega) = \frac{10(1 + j0.1\omega)}{(j\omega)(1 + j0.5\omega)(1 + j0.2\omega)}$$

The factors of G(jω) are

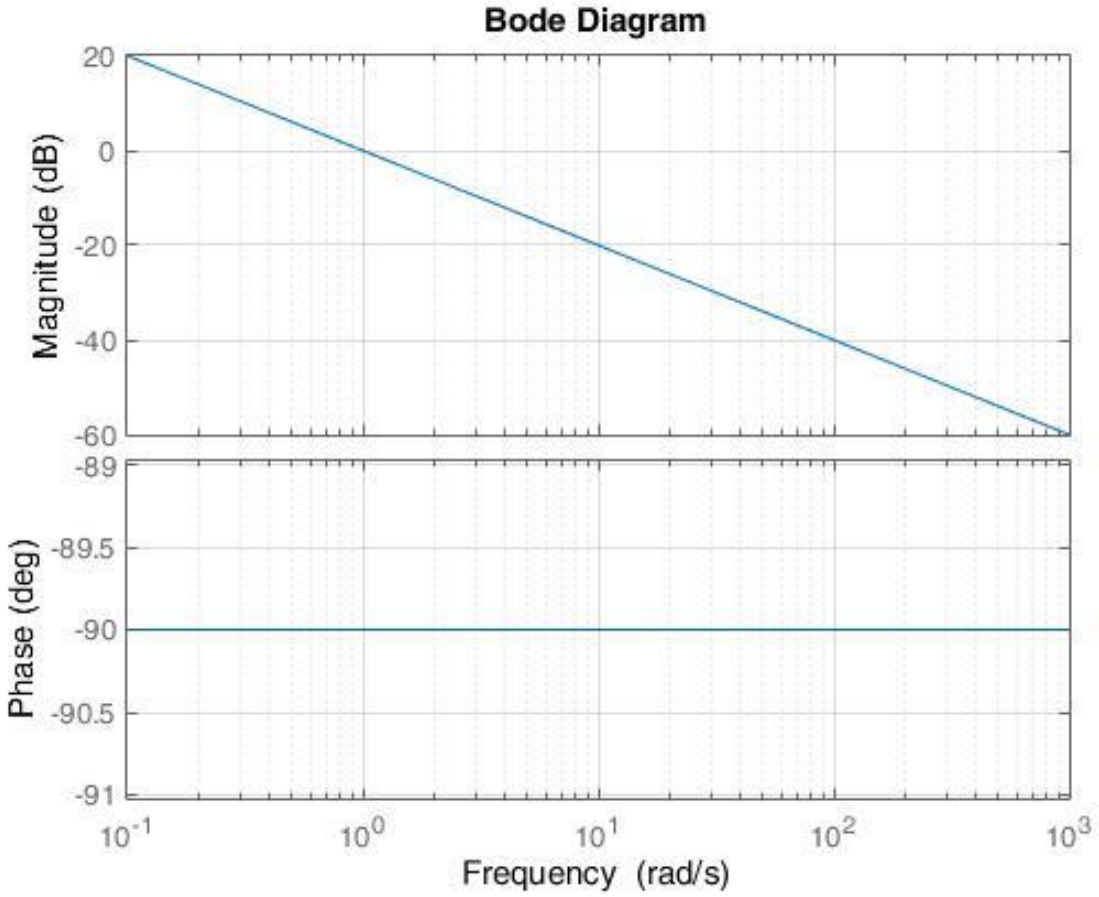
1. Constant gain: K=10
2. Real zero: (1+jω/10)
3. Pole at the origin: $(j\omega)^{-1}$
3. Real pole 1: $(1+j\omega/2)^{-1}$
4. Real pole 2: $(1+j\omega/5)^{-1}$

G(jω) has corner frequencies at ω = 10, 2, and 5 rad/sec.

1. Bode plot of the constant gain: K=10

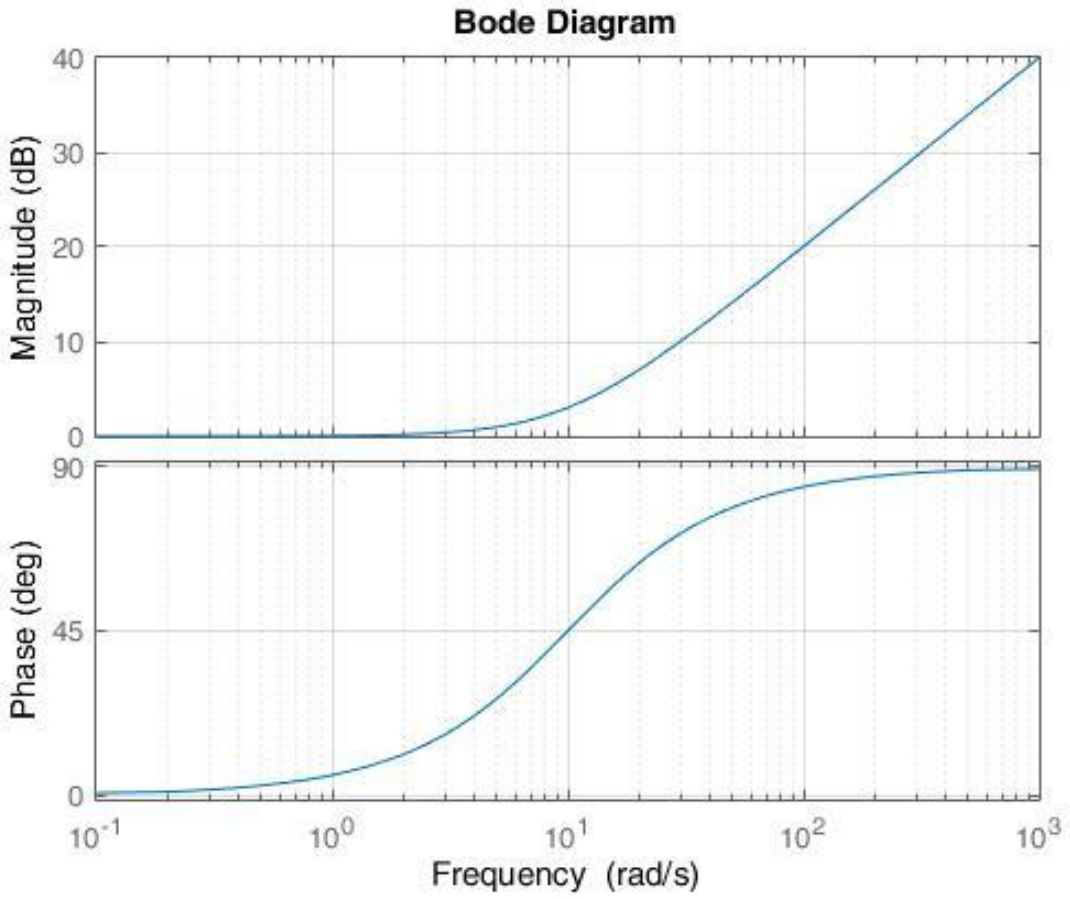


2. Bode plot of the pole at the origin factor: $(j\omega)^{-1}$



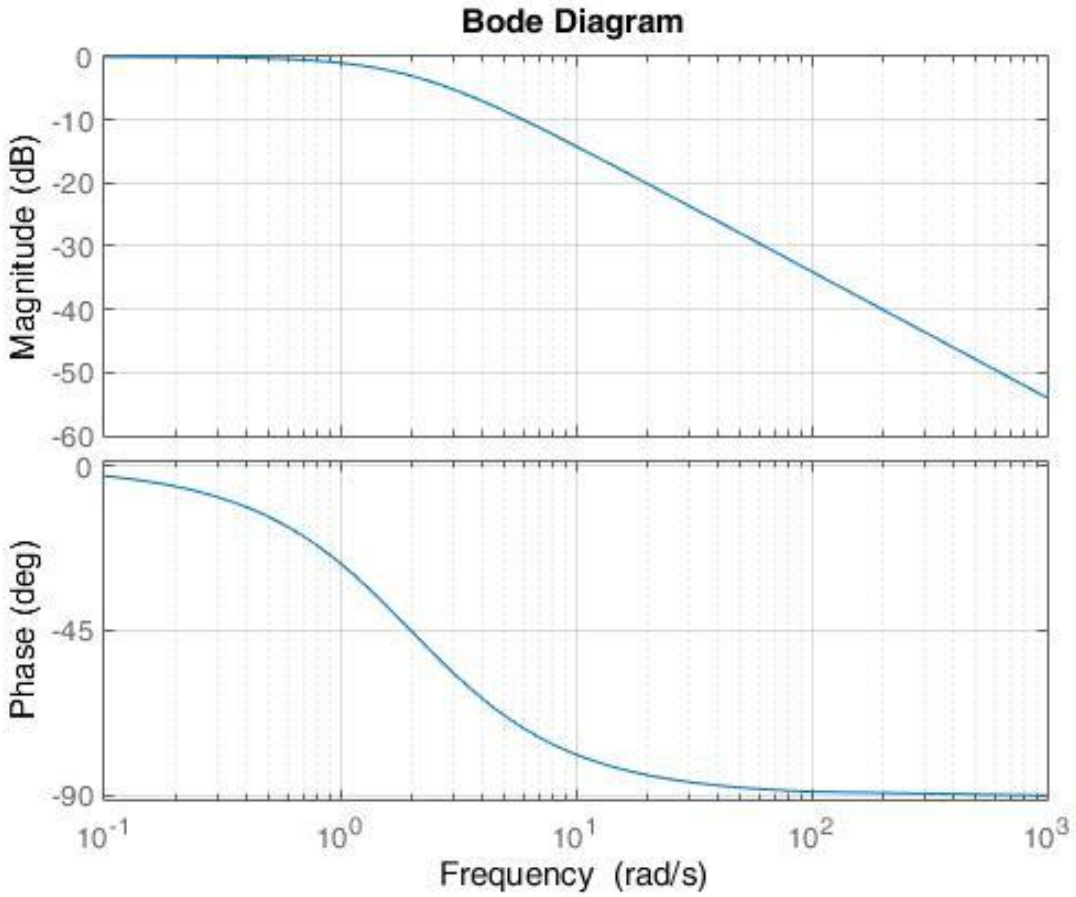
Magnitude curve: a straight line with slope of -20 dB/decade, passing through the ω=1 rad/sec point on the 0-dB axis.

3. Bode plot of the simple real zero factor: (1+j0.1ω)



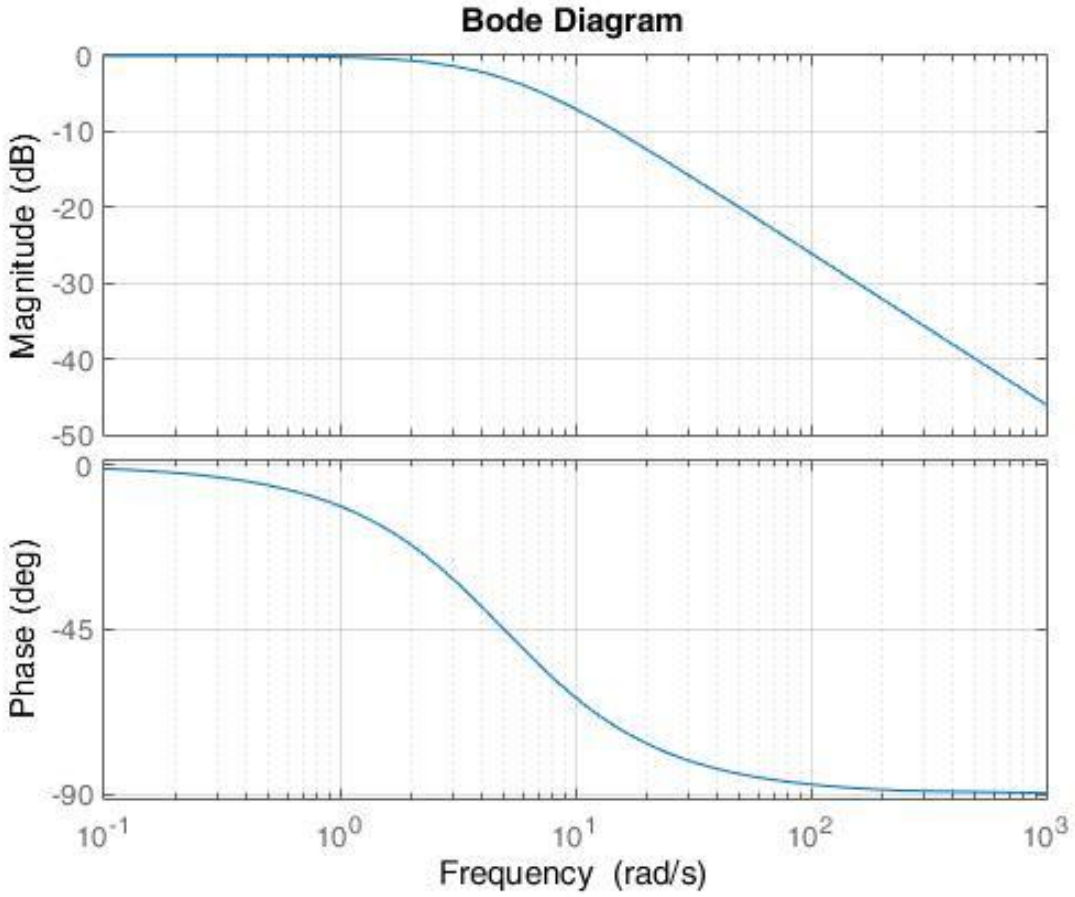
Corner frequency: ω=1/0.1=10 rad/sec

4. Bode plot of the simple real pole factor: 1/(1+j0.5ω)



Corner frequency: ω=1/0.5=2 rad/sec

5. Bode plot of the simple real pole factor: 1/(1+j0.2ω)

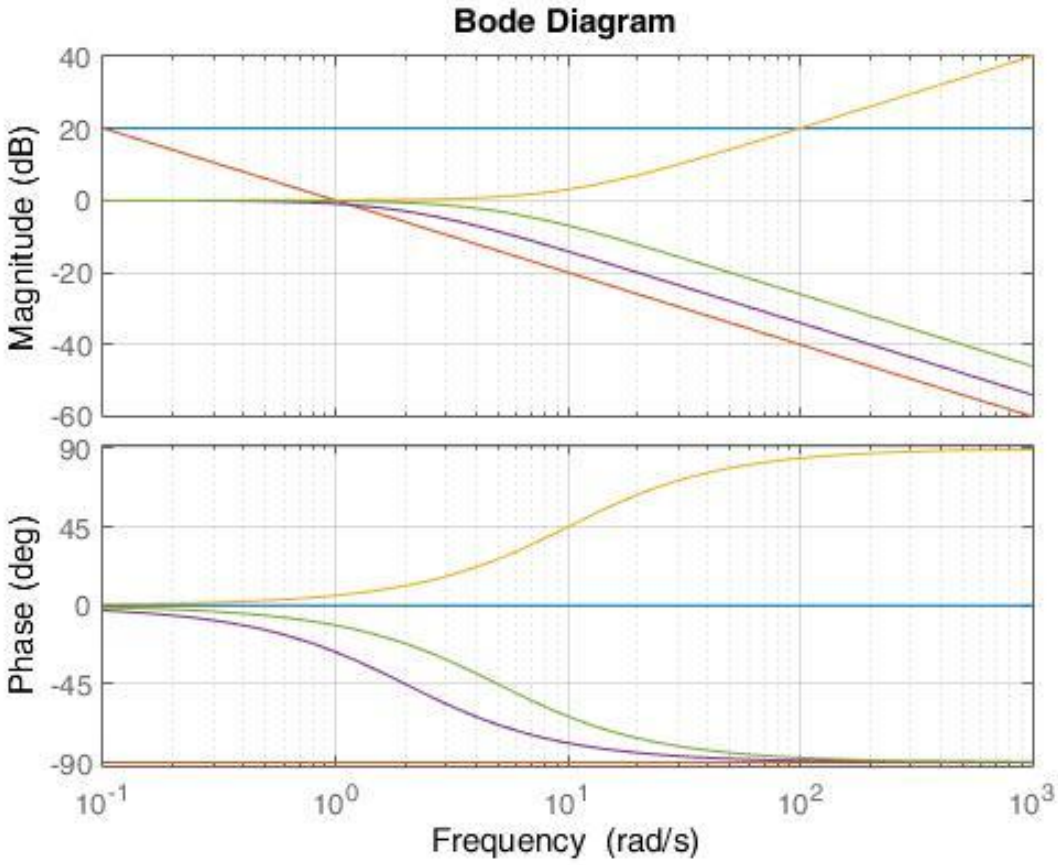


Corner frequency: ω=1/0.2=5 rad/sec

Bode plot of all components

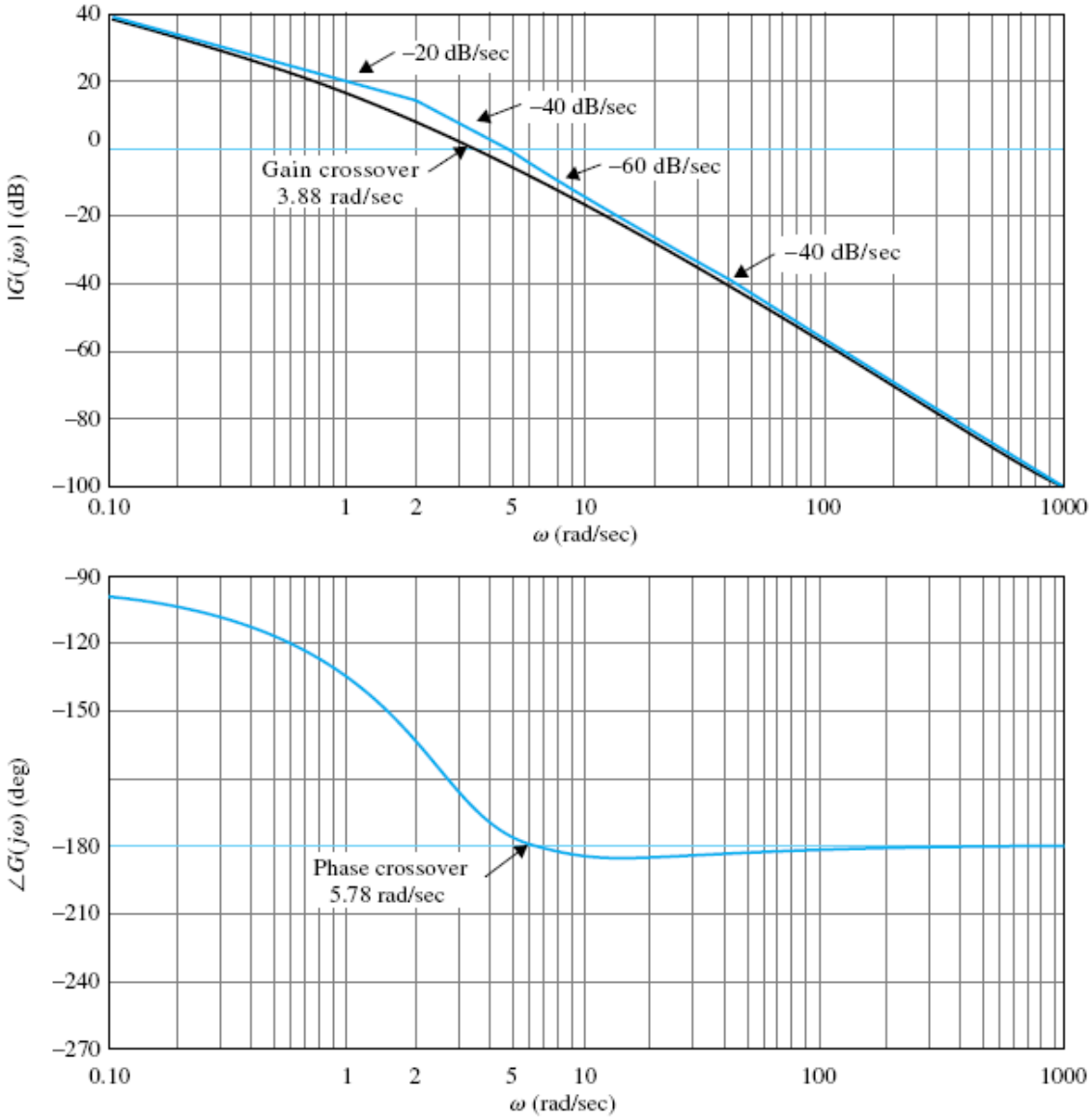
The gain $|G(j\omega)|_{dB}$ is obtained by adding the magnitude curves of the components together, point by point.

The phase $\measuredangle G(j\omega)$ is obtained by adding the phase curves of the components together, point by point.



Bode plot of the overall G(s)

Bode Plot:
Gain crossover point: $|G(j\omega)|_{dB}$ cross the 0-dB axis
Phase crossover point: where the phase curve cross the -180°axis.



Example Consider the following transfer function

$$G(s) = \frac{300(s+5)}{s(s+2)(s^2+10s+100)}$$

Sketch its Bode Plot.

Solution. Letting s=jω, we have

$$G(j\omega) = \frac{300(j\omega+5)}{(j\omega)(j\omega+2)\left((j\omega)^2+10(j\omega)+100\right)}$$

Reformulating it into the form for Bode Plot

$$G(j\omega) = \frac{7.5(1+j\omega/5)}{(j\omega)(1+j\omega/2)\left(1+j\omega/10+(j\omega/10)^2\right)}$$

G(jω) has corner frequencies at ω = 5, 2, and 10 rad/sec.

The damping ratio for the complex poles is ζ = 0.5.

Example

$$G(j\omega) = \frac{7.5(1 + j\omega/5)}{(j\omega)(1 + j\omega/2)\left(1 + j\omega/10 + (j\omega/10)^2\right)}$$

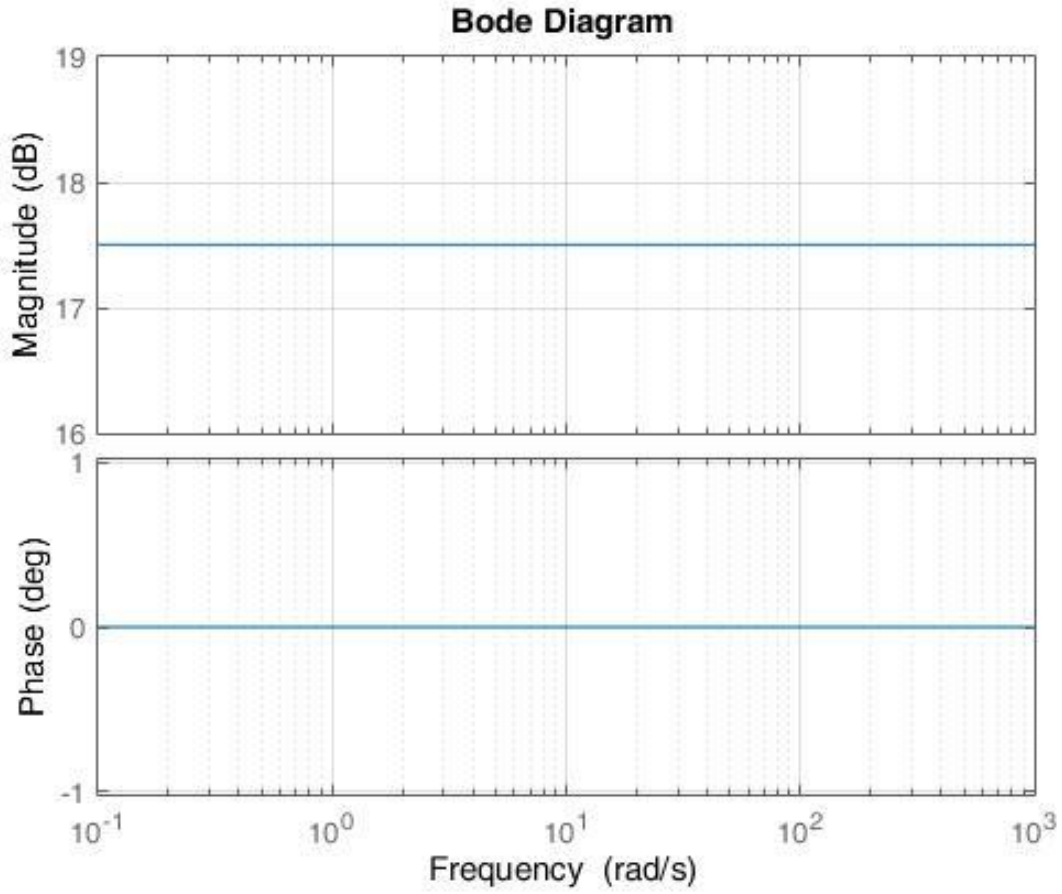
The factors of G(jω) are

1. Constant gain: K=7.5
2. Real zero: (1+jω/5)
3. Pole at the origin: $(j\omega)^{-1}$
3. Real pole: $(1+j\omega/2)^{-1}$
4. Complex conjugate poles: $(1 + j\omega/10 - \omega^2/10^2)^{-1}$

G(jω) has corner frequencies at ω = 5, 2, and 10 rad/sec.

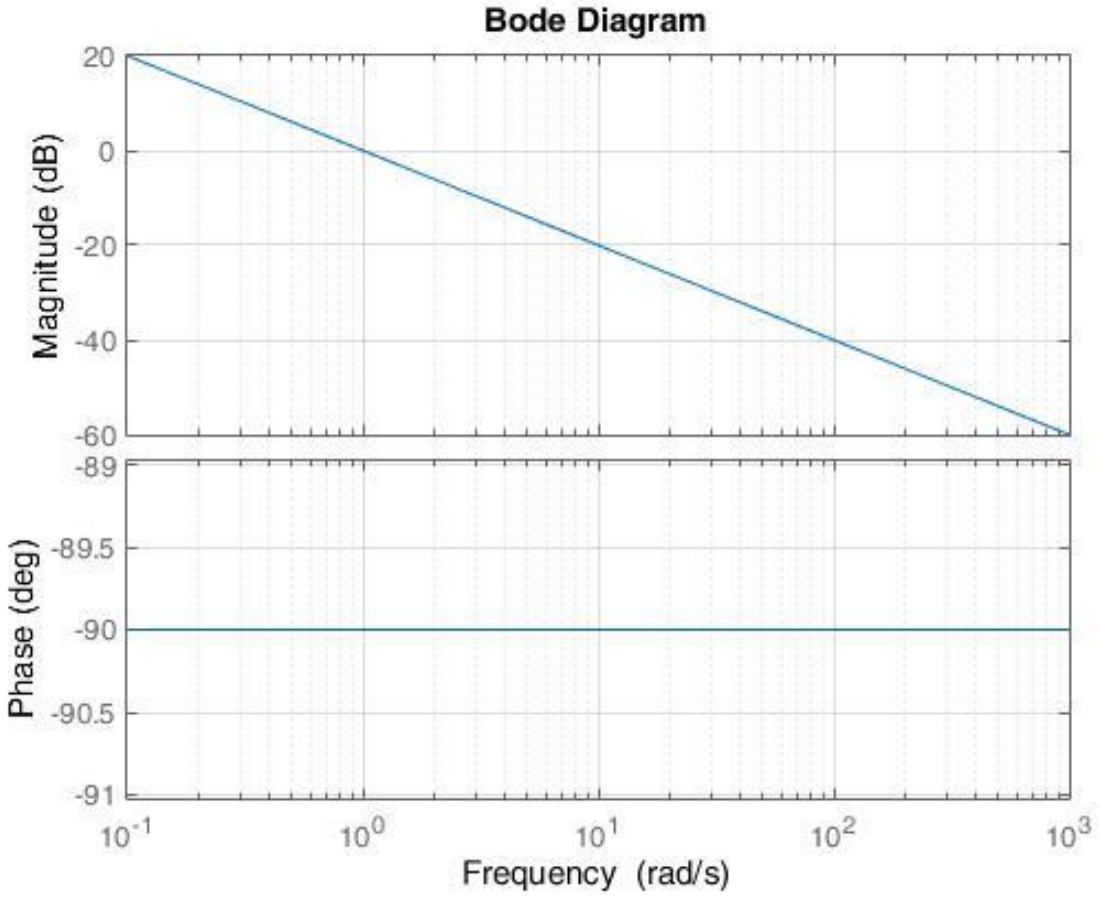
The damping ratio for the complex poles is ζ = 0.5.

1. Bode plot of the constant gain: K=7.5



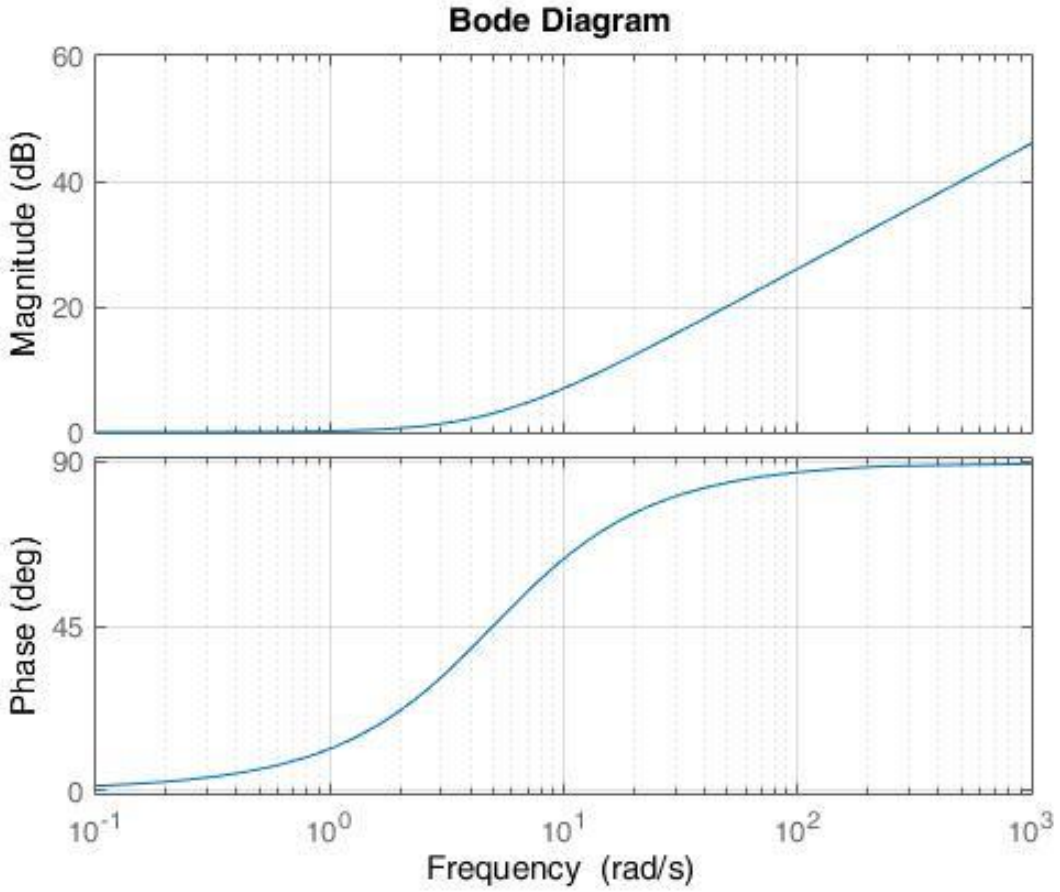
$$20\log_{10} 7.5 = 17.5012 \ \text{dB}$$

2. Bode plot of the pole at the origin factor: 1/(jω)



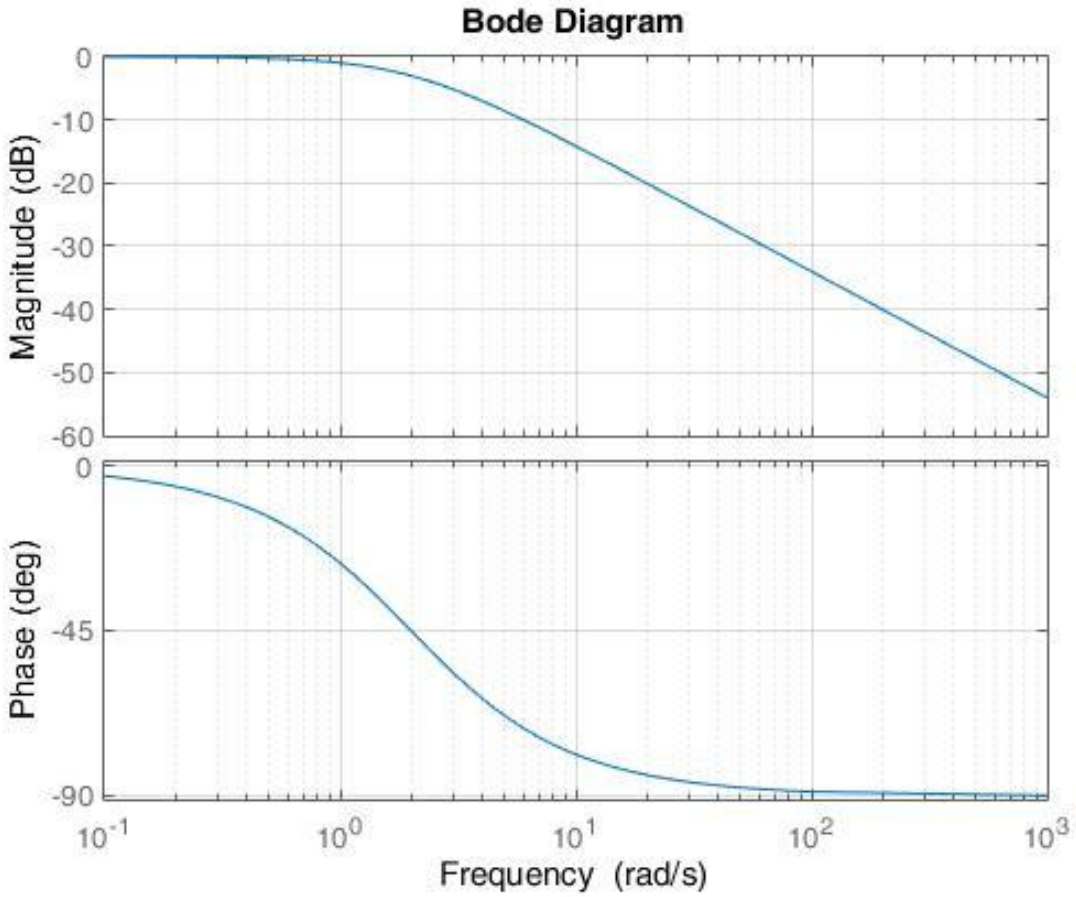
Magnitude curve: a straight line with slope of -20 dB/decade, passing through the ω=1 rad/sec point on the 0-dB axis.

3. Bode plot of the simple real zero factor: (1+jω/5)



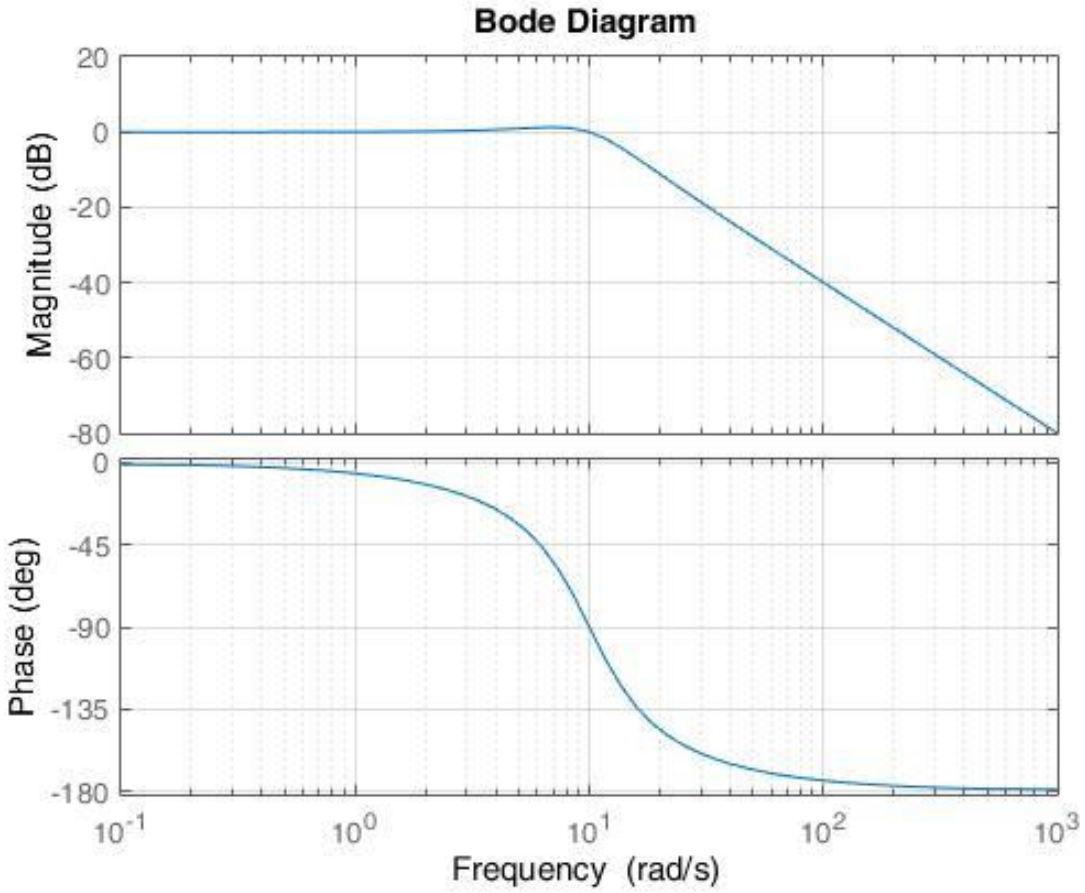
Corner frequency: ω=5 rad/sec

4. Bode plot of the simple pole factor: 1/(1+jω/2)



Corner frequency: ω=2 rad/sec

5. Bode plot of the complex poles factor: $1/(1+j\omega/10-\omega^2/10^2)$

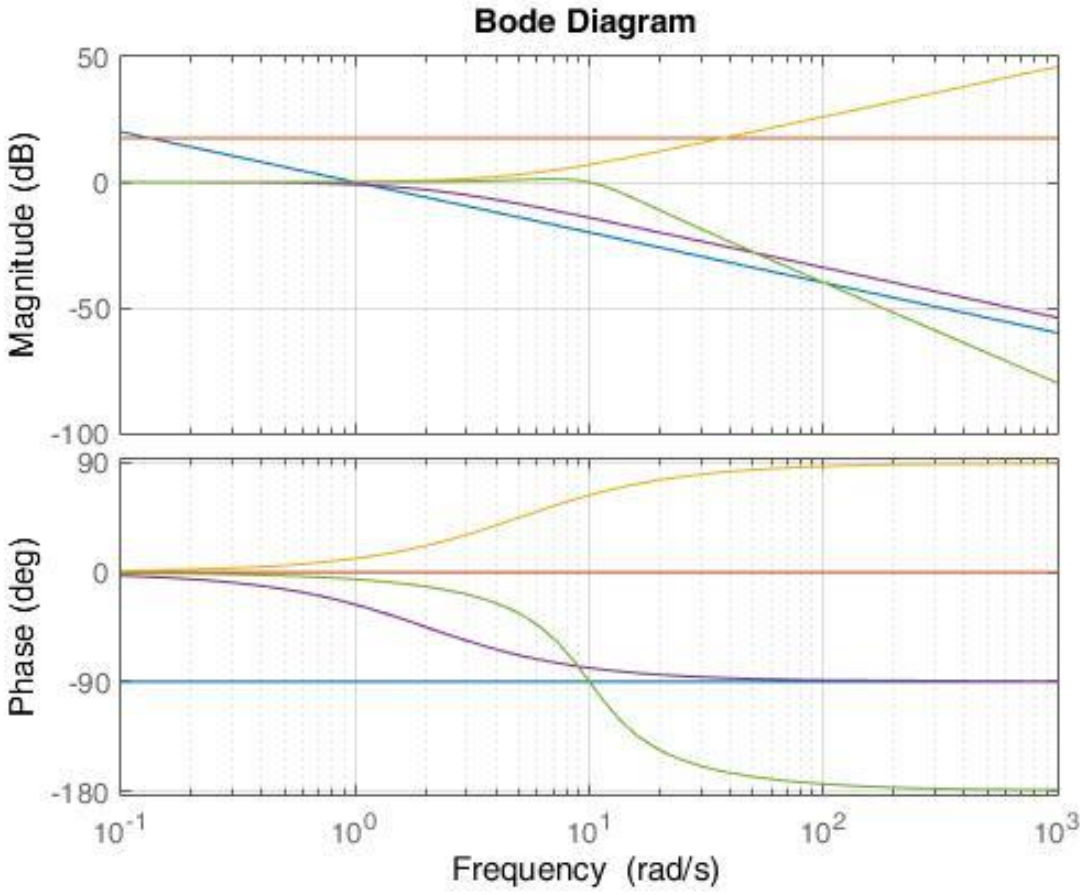


Corner frequency: $\omega=10$ rad/sec

Bode plot of all components

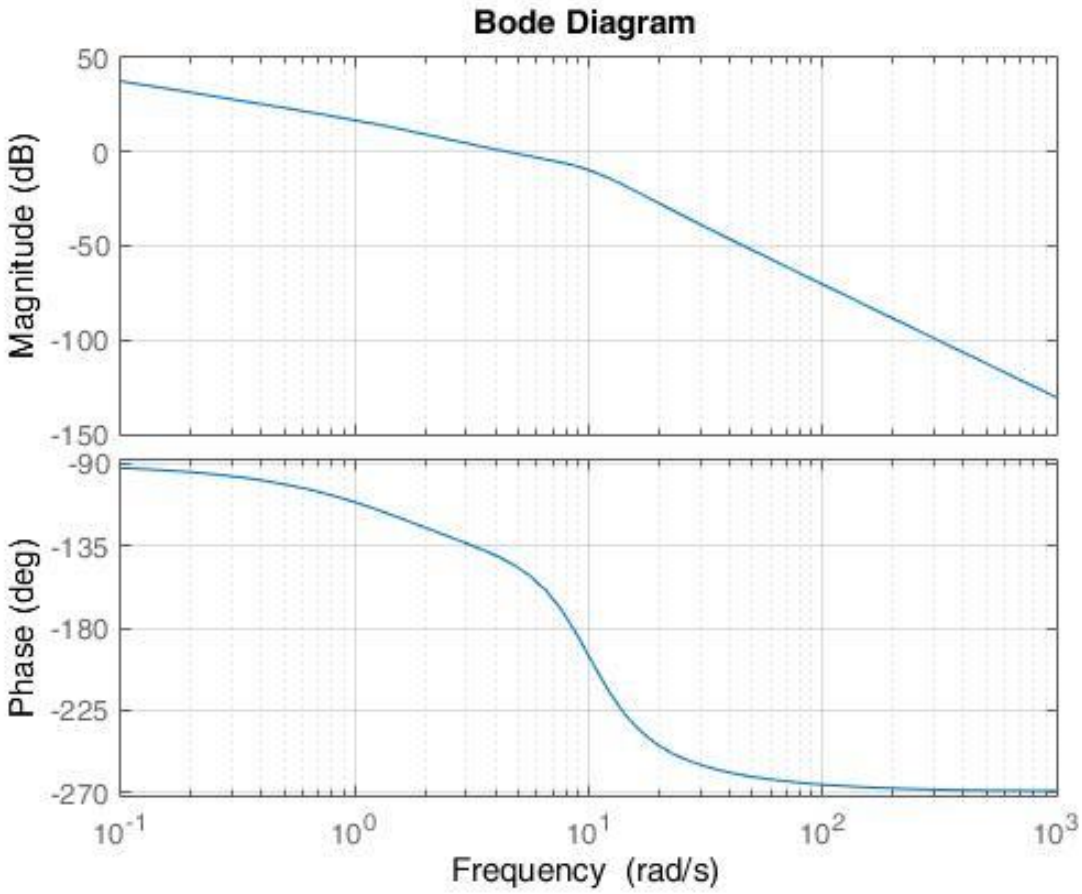
The gain $|G(j\omega)|_{dB}$ is obtained by adding the magnitude curves of the components together, point by point.

The phase $\measuredangle G(j\omega)$ is obtained by adding the phase curves of the components together, point by point.



Bode plot of the overall G(s)

Bode Plot:
Gain crossover point: $|G(j\omega)|_{dB}$ cross the 0-dB axis
Phase crossover point: where the phase curve cross the -180°axis.



Minimum Phase and Non-Minimum Phase Systems

A system which has all its poles and zeros in the left-half plane is called a **minimum phase system**.

A system is called a **non-minimum phase system** if it has either poles or zeros in the right-half plane.

Minimum phase systems are stable and have stable inverse.

Non-minimum phase systems are either unstable systems or have unstable inverse.

In other words, unstable systems and systems which have unstable inverse are non-minimum phase systems.

Minimum Phase and Non-Minimum Phase Systems

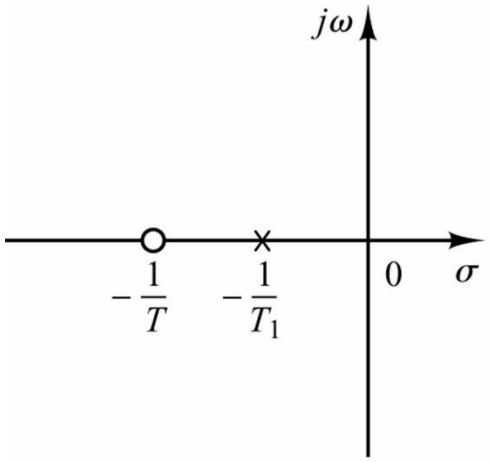
Among the stable TFs with the same magnitude curve but different phase angles, the minimum phase system has the least range of phase shift – hence the name minimum phase.

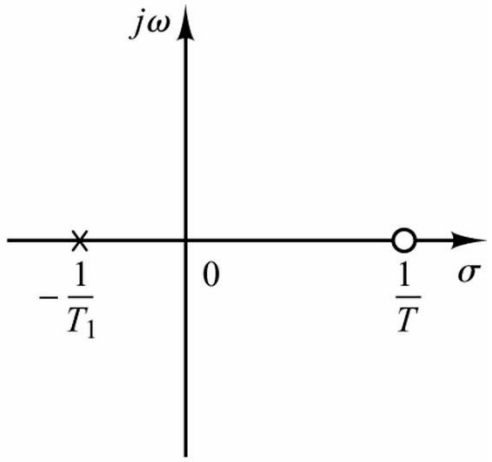
For minimum phase systems the phase and the magnitude curves are uniquely related and one of them is sufficient to uniquely determine the entire TF.

Non-minimum phase systems are usually not desired since they are slow in response. They create phase lag in the system and mostly in practical systems excessive phase lag should be avoided.

Minimum Phase and Non-Minimum Phase Systems

Consider the following TFs $G_1(s)$ and $G_2(s)$





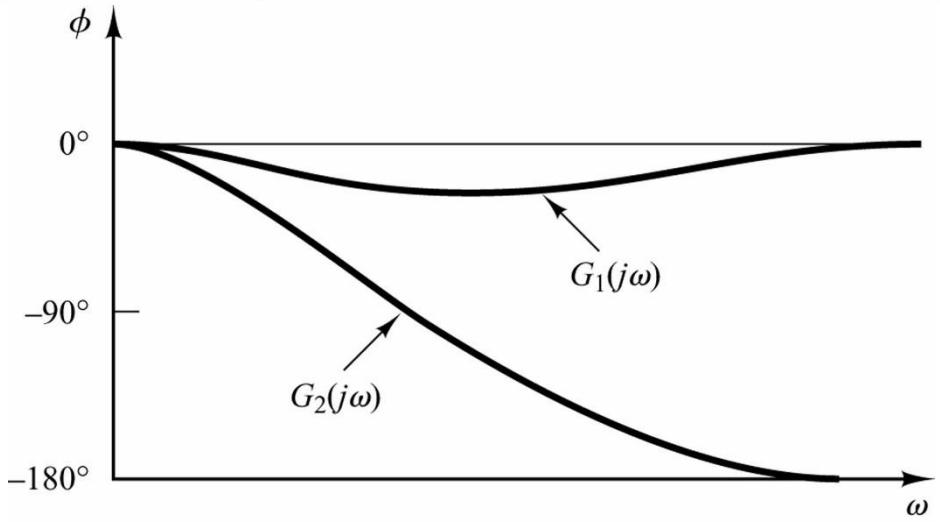
$$G_1(s) = \frac{1 + Ts}{1 + T_1 s}$$

$$G_2(s) = \frac{1 - Ts}{1 + T_1 s}$$

$G_1(s)$ - minimum phase system.

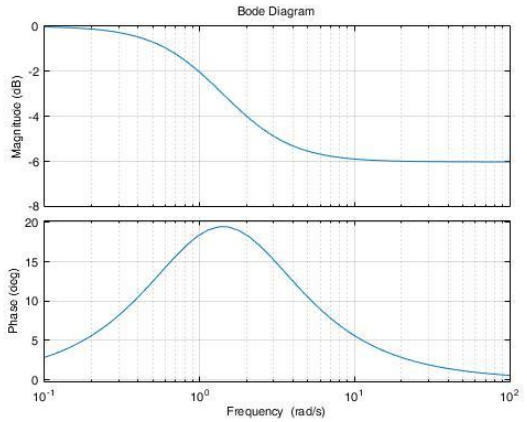
$G_2(s)$ - non-minimum phase system.

Their phase plots are

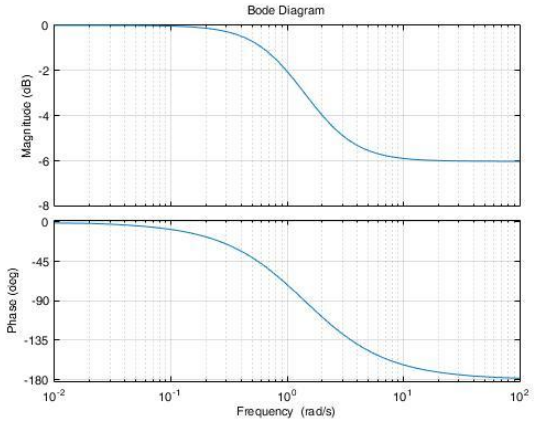


The range of phase shift in $G_2(s)$ is larger than the range of phase shift in $G_1(s)$.

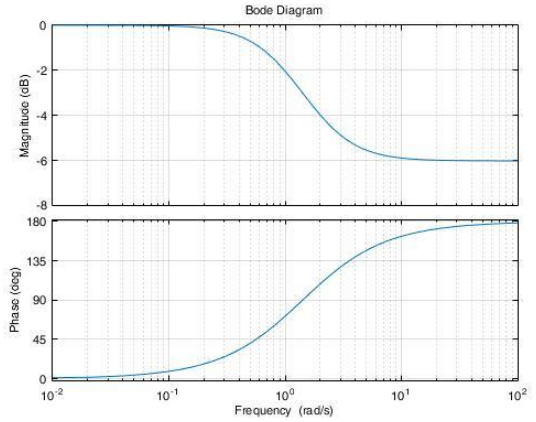
Example



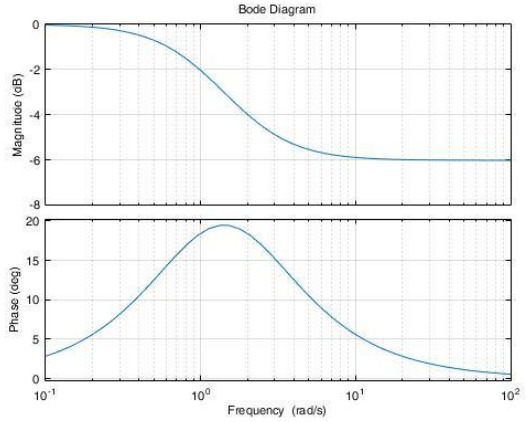
$$G_1(j\omega) = \frac{(1+ j0.5\omega)}{(1+ j\omega)}$$



$$G_2(j\omega) = \frac{(1- j0.5\omega)}{(1+ j\omega)}$$



$$G_3(j\omega) = \frac{(1+ j0.5\omega)}{(1- j\omega)}$$



$$G_4(j\omega) = \frac{(1- j0.5\omega)}{(1- j\omega)}$$

$G_1(s)$ – stable system with stable inverse

$G_2(s)$ – stable system with unstable inverse

$G_3(s)$, $G_4(s)$ – unstable systems



$G_1(s)$ - minimum phase system.

$G_2(s)$, $G_3(s)$, $G_4(s)$ - non-minimum phase systems.

Experimental Determination of TFs

In order to analyze and/or design control systems we need a mathematical model of the plant under consideration.

One method to derive a model of a given control system is to use physical laws and analytical methods.

However, in some cases it might be difficult to analytically derive a model.

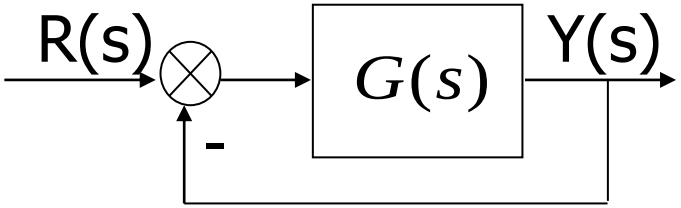
In such cases, one may use experimental identification methods.

By measuring the change in amplitude and the phase-shift of the response of the given LTI system to a sinusoidal inputs under different frequencies, one may plot the Bode diagram of the system.

Then, from the Bode diagram the TF may be determined.

System Type and The Log-Magnitude Curve

Consider the unity-feedback system



Open-loop TF:

$$G(s) = \frac{K\prod_{i=1}^{Q}(1+T_i s)}{s^N \prod_{m=1}^{M}(1+\tau_m s)\prod_{k=1}^{R}\left(1+2\zeta_k\left(s/\omega_{n_k}\right)+\left(s/\omega_{n_k}\right)^2\right)}$$

N=0 - Type-0 system

N=1 - Type-1 system

N=2 - Type-2 system

Can we determine the type of the system from its Bode diagram?

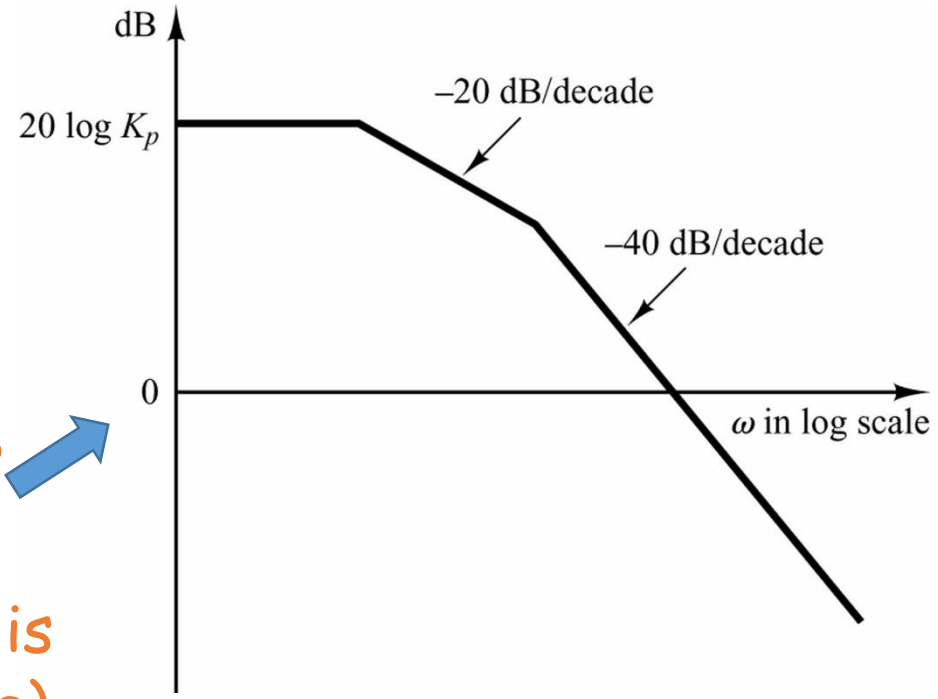
System Type and The Log-Magnitude Curve

Type-0 systems (N=0):

$$\lim_{s\to 0} G(s) = K_p = K$$

is the proportional (position) error constant.

The Bode magnitude curve for such systems has the form:



The low frequency asymptote is constant (i.e., has zero slope).

It is equal to $20\log_{10}(K_p)$

One can easily determine the value of K_p from the log-magnitude curve.

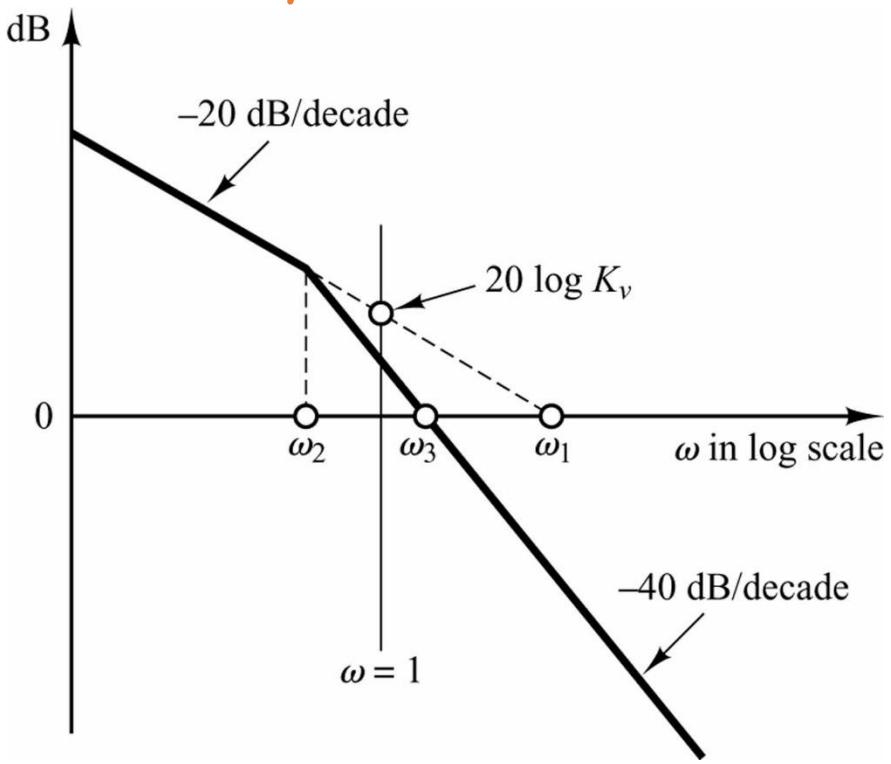
System Type and The Log-Magnitude Curve

Type-1 systems ($N=1$):

$$\lim_{s\to 0} sG(s) = K_v = K$$

is the velocity error constant.

The Bode magnitude curve for such systems has the form:



The low frequency asymptote has a slope of -20dB/dec.

It is given by $20\log_{10}\left|K_v / j\omega\right|$

Evaluating this at ω=1 we have

$$20\log_{10}\left|K_v / j\omega\right|_{|\omega=1} = 20\log_{10}\left(K_v\right)$$

Also, at the frequency ω_1 where the low frequency asymptote or its extension crosses the 0-dB line we have

$$20\log_{10}\left|K_v / j\omega_1\right| = 0$$

$$K_v / \omega_1 = 1 \quad \Rightarrow \quad K_v = \omega_1$$

The intersection of the first (-20dB/dec) segment with the 0-dB line is equal to K_v.

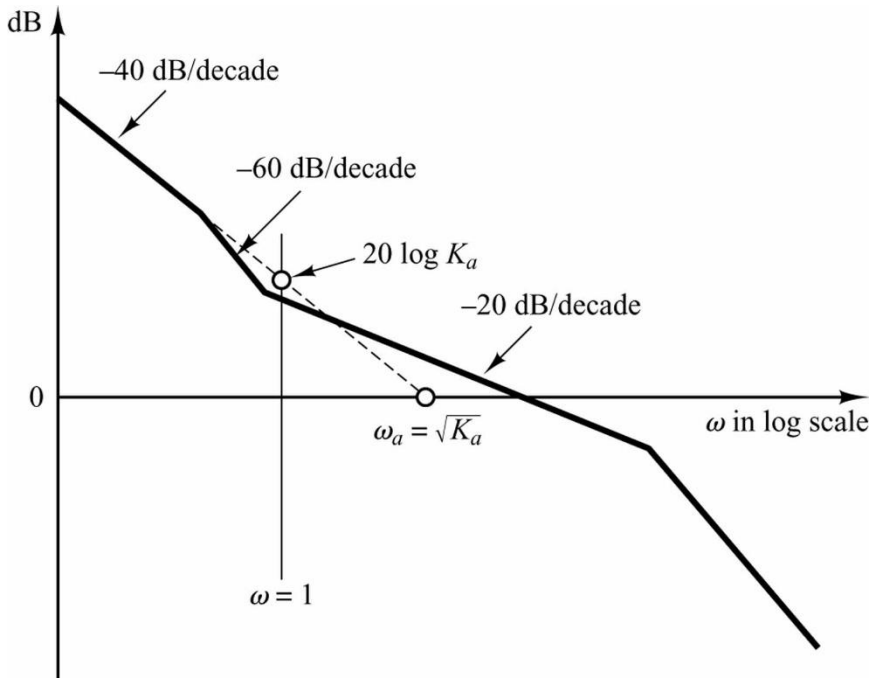
System Type and The Log-Magnitude Curve

Type-2 systems (N=2):

$$\lim_{s\to 0} s^2 G(s) = K_a = K$$

is the acceleration error constant.

The Bode magnitude curve for such systems has the form:



The low frequency asymptote has a slope of -40dB/dec.

It is given by $20\log_{10}\left|K_a/(j\omega)^2\right|$

Evaluating this at ω=1 we have

$$20\log_{10}\left|K_a/(j\omega)^2\right|_{|\omega=1} = 20\log_{10}\left(K_a\right)$$

Also, at the frequency ω_a where the low frequency asymptote or its extension crosses the 0-dB line we have

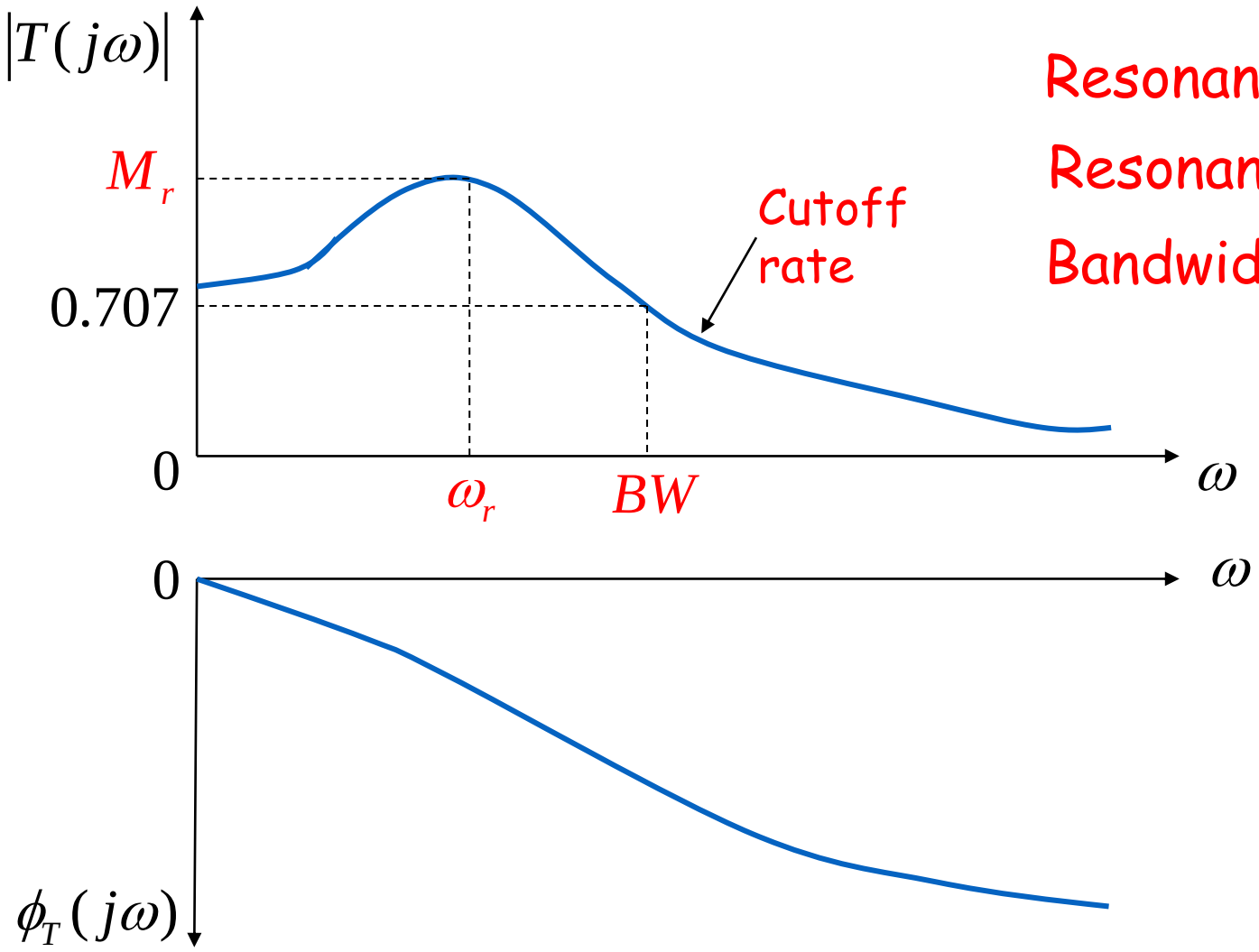
$$20\log_{10}\left|K_a/(j\omega_a)^2\right| = 0$$

$$K_a/(\omega_a)^2 = 1 \quad \Rightarrow \quad \omega_a = \sqrt{K_a}$$

The intersection of the first (-40dB/dec) segment with the 0-dB line is equal to $\sqrt{K_a}$.

Frequency Domain Specifications

Frequency-Domain Specifications



Resonant peak M_r

Resonant frequency ω_r

Bandwidth BW

$$\frac{d|T(j\omega)|}{d\omega} = 0$$

$$\Downarrow$$

$$\omega_r$$

Typical gain-phase characteristic of a control system

Frequency response of a prototype second-order system

Closed-loop transfer function:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Its frequency-domain transfer function:

$$T(j\omega) = \frac{Y(j\omega)}{R(j\omega)} = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n (j\omega) + \omega_n^2}$$

Define $u = \omega/\omega_n$

$$T(ju) = \frac{1}{1 + j2\zeta u - u^2}$$

The magnitude of $T(ju)$ is

$$|T(ju)| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}$$

The phase of $T(ju)$ is

$$\measuredangle T(j\omega) = \phi_T(j\omega) = -\tan^{-1}\frac{2\zeta u}{1-u^2}$$

The resonant frequency of $T(ju)$ is

$$\frac{d|T(ju)|}{du} = 0 \quad \Rightarrow \quad u_r = \sqrt{1-2\zeta^2}$$

Resonant peak

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

With $u_r = \omega_r/\omega_n$, we have $\omega_r = \omega_n\sqrt{1-2\zeta^2}$

Since frequency is a real quantity, it requires $1-2\zeta^2 \geq 0$

So $\zeta \leq \frac{1}{\sqrt{2}} \cong 0.707$

According to the definition of Bandwidth

$$\left|T(ju)\right| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}} = \frac{1}{\sqrt{2}} \cong 0.707$$

$$\Rightarrow \quad u^2 = (1-2\zeta^2) \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2}$$

With $u = \omega/\omega_n$, we have

$$BW = \omega_n \left[(1-2\zeta^2) \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2} \right]^{1/2}$$

$$= \omega_n \sqrt{(1-2\zeta^2) \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

For a prototype second-order system ($\zeta \le 0.707$)

Resonant peak

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

M_r depends on ζ only.

For $\zeta \le 0$, the system is unstable;

For $0<\zeta \le 0.707$, $\zeta \uparrow$ $M_r \downarrow$;

For $\zeta \ge 0.707$, $M_r = 1$

Resonant frequency

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}$$

ω_r depends on both ζ and ω_n.

For $0<\zeta \le 0.707$, ω_n fixed, $\zeta \uparrow$ $\omega_r \downarrow$;

For $\zeta \ge 0.707$, $\omega_r = 0$.

Bandwidth

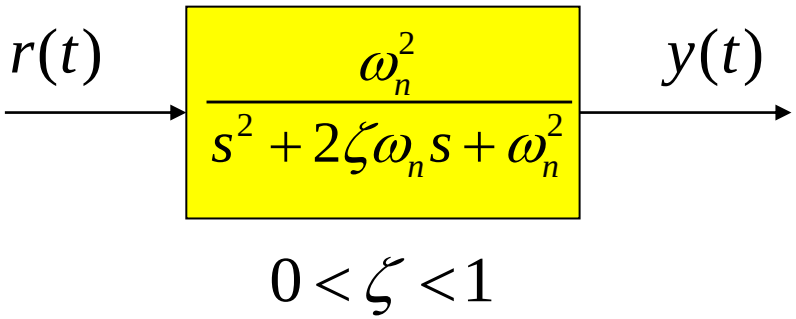
$$BW = \omega_n\left[(1-2\zeta^2) \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2}\right]^{1/2}$$

BW is directly proportional to ω_n, $\omega_n \uparrow$ $BW \uparrow$

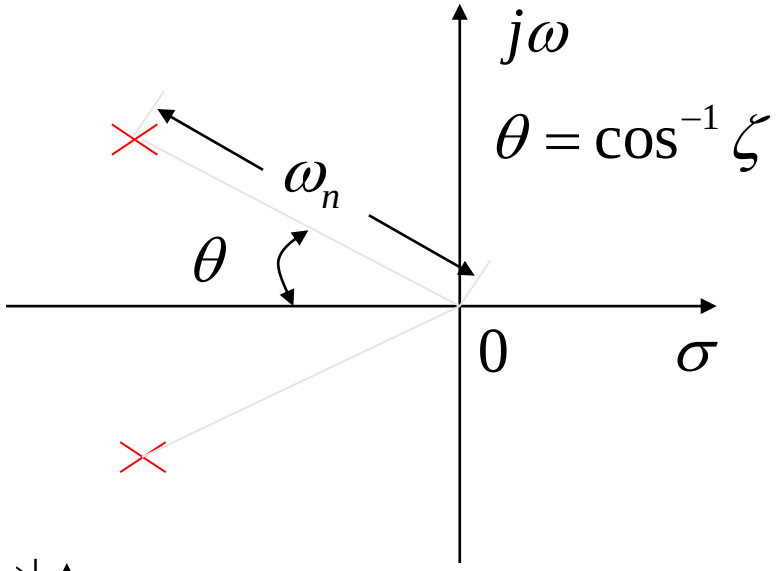
$\zeta \uparrow$ $BW/\omega_n \downarrow$

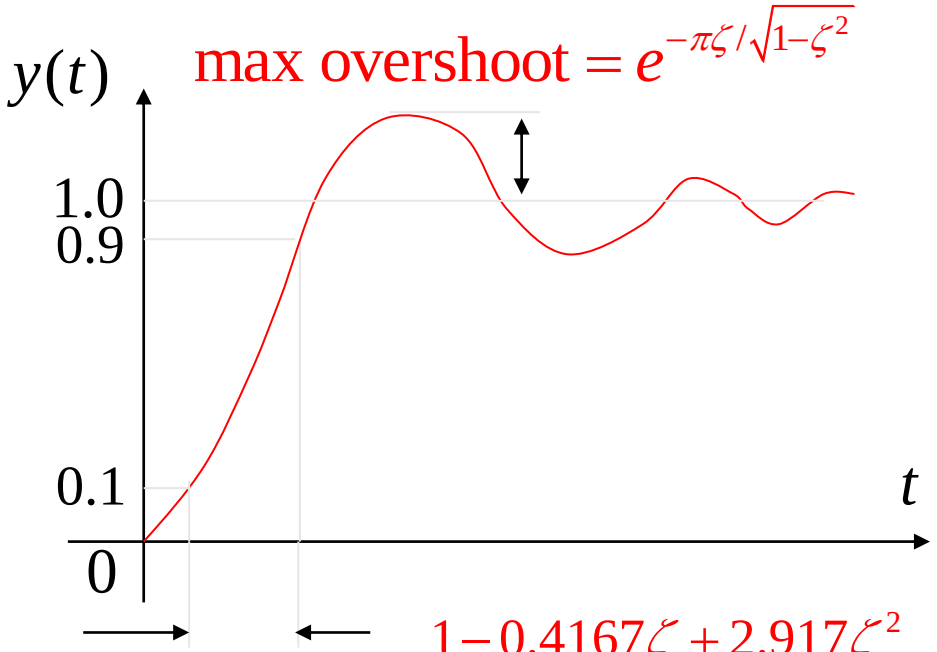
For $0 \le \zeta \le 0.707$, ω_n fixed, $\zeta \uparrow$ $BW \downarrow$;

Correlation between pole locations, unit-step response and the magnitude of the frequency response

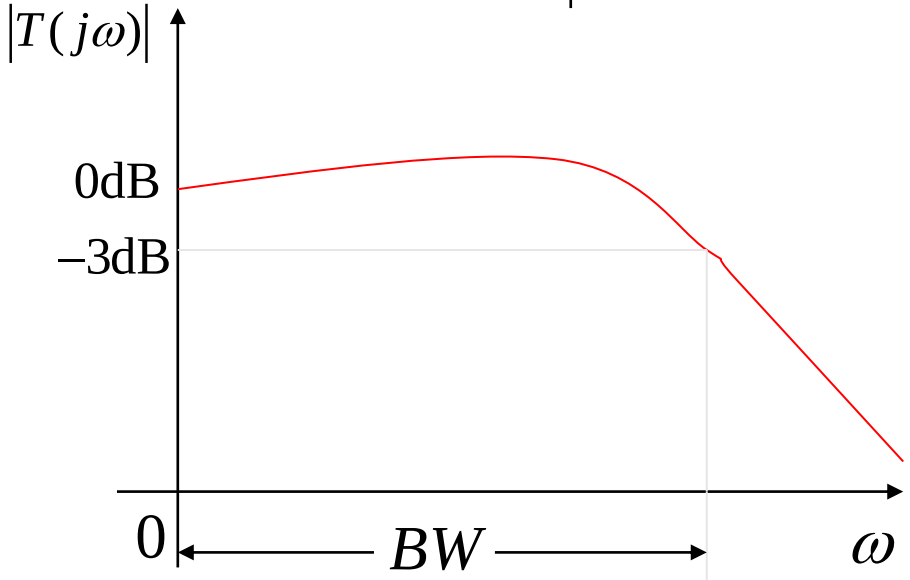


$0<\zeta<1$





$$t_r \approx \frac{1-0.4167\zeta+2.917\zeta^2}{\omega_n}$$



$$BW=\omega_n[(1-2\zeta^2)+\sqrt{4\zeta^4-4\zeta^2+2}]^{1/2}$$

Example. The specifications on a second-order unity-feedback control system with the closed-loop transfer function

$$T(s)=\frac{Y(s)}{R(s)}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

are that the maximum overshoot must not exceed 10 percent, and the rise time be less than 0.1 sec. Find the corresponding limiting values of M_r and BW analytically.

Solution. Maximum overshoot:

$$M_p\%=e^{-\zeta\pi/\sqrt{1-\zeta^2}}\times 100\%\le 10\% \quad\Rightarrow\quad \boxed{\zeta\ge 0.6}$$

Rise time: $$t_r\approx\frac{1-0.4167\zeta+2.917\zeta^2}{\omega_n}\le 0.1 \quad (0<\zeta<1)$$

$$\Rightarrow \quad 2.917\zeta^2-0.4167\zeta+1\le 0.1\omega_n$$

$$\Rightarrow \quad \omega_n\ge 29.17\zeta^2-4.167\zeta+10 \quad\Rightarrow\quad \boxed{\omega_n\ge 18}$$

Based on time-domain analysis, we obtain $\zeta \geq 0.6$ and $\omega_n \geq 18$

Frequency-domain specifications:

Resonant peak $M_r = \dfrac{1}{2\zeta\sqrt{1-\zeta^2}} \quad (\zeta \leq 0.707)$

For $0<\zeta \leq 0.707$, $\zeta \uparrow$ $M_r \downarrow$;

For $\zeta \geq 0.707$, $M_r = 1$

$\zeta \geq 0.6$ ⇒ $1 \leq M_r \leq 1.04$

Bandwidth $BW = \omega_n[(1-2\zeta^2)+\sqrt{4\zeta^4-4\zeta^2+2}]^{1/2}$

BW is directly proportional to ω_n, $\omega_n \uparrow BW \uparrow$

$\zeta \uparrow$ $BW/\omega_n \downarrow$

For $0 \leq \zeta \leq 0.707$, ω_n fixed, $\zeta \uparrow BW \downarrow$;

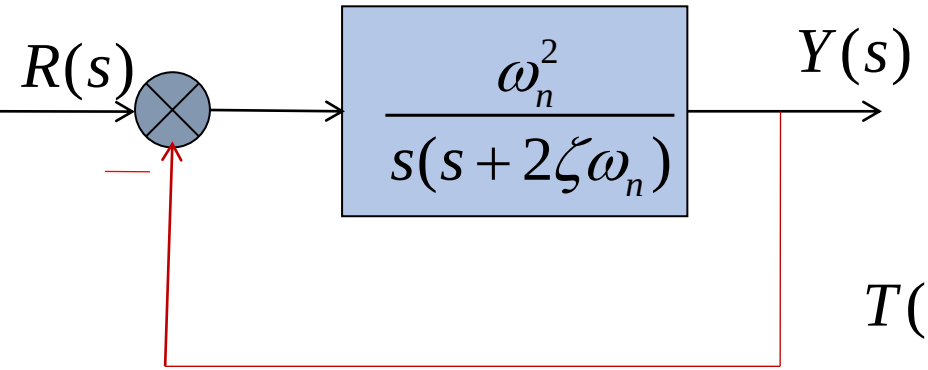
$0.6 \leq \zeta \leq 0.707$ ⇒ $1 \leq BW/\omega_n \leq 1.15$

⇒ $\omega_n \leq BW \leq 1.15\omega_n$

$\omega_n \geq 18$

⇒ $BW \geq 18$

Effects of adding a zero to the OL TF

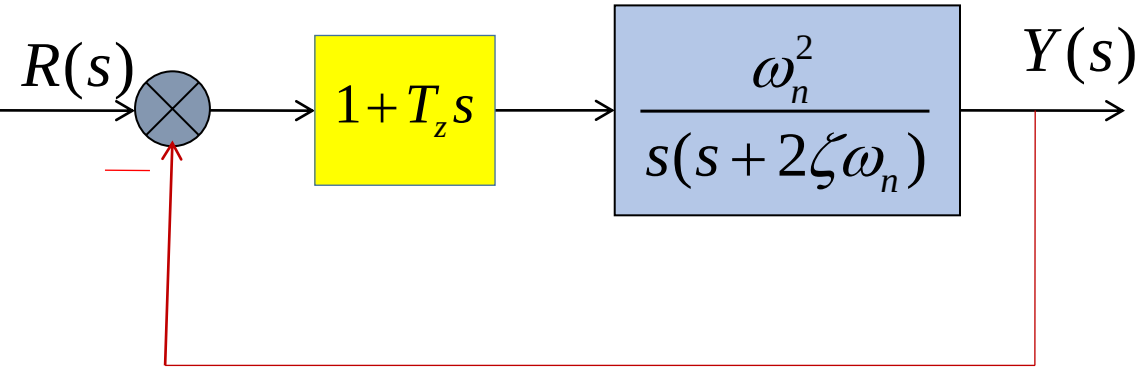


Open-loop TF: $G(s) = \dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$

Closed-loop TF:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

Adding a zero at $s = -1/T_z$



Open-loop TF:

$$G(s) = \frac{(1+T_z s)\omega_n^2}{s(s+2\zeta\omega_n)}$$

Closed-loop TF: $T(s) = \dfrac{\omega_n^2(1+T_z s)}{s^2+(2\zeta\omega_n+T_z\omega_n^2)s+\omega_n^2}$

The additional zero changes both numerator and denominator.

As analyzing the prototype second-order system, using similar but more complicate calculation, we obtain

Bandwidth $BW = (-b + 1/2\sqrt{b^2 + 4\omega_n^4})^{1/2}$

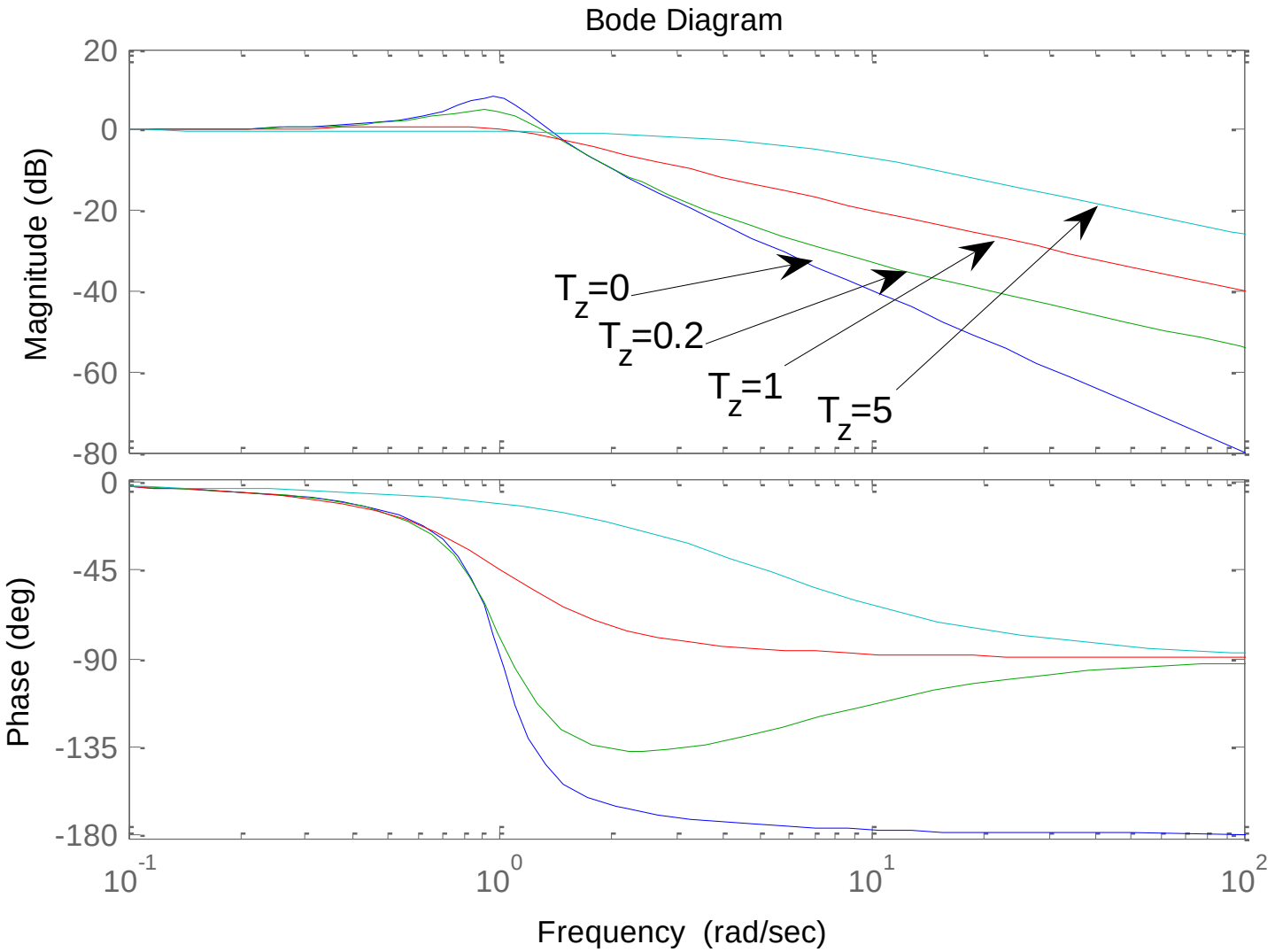
where $b = 4\zeta^2\omega_n^2 + 4\zeta\omega_n^3 T_z - 2\omega_n^2 - \omega_n^4 T_z^2$

For fixed ω_n and ζ, we analyze the effect of T_z .

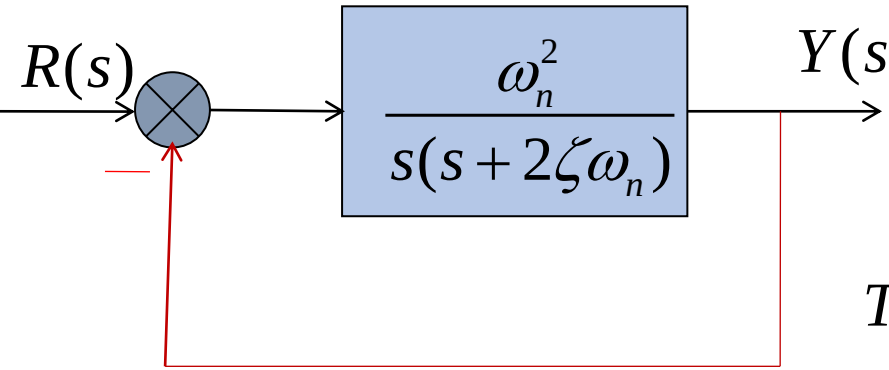
The general effect of adding a zero the open-loop transfer function is to increase the bandwidth of the closed-loop system.

$$\omega_n = 1$$

$$\zeta = 0.2$$



Effects of adding a pole to the OL TF

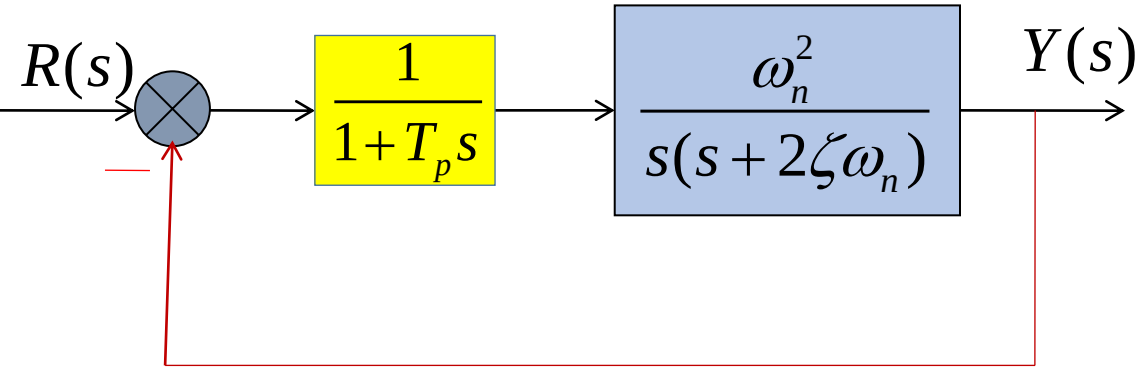


Open-loop TF: $G(s) = \dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$

Closed-loop TF:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

Adding a pole at $s = -1/T_p$



Open-loop TF:

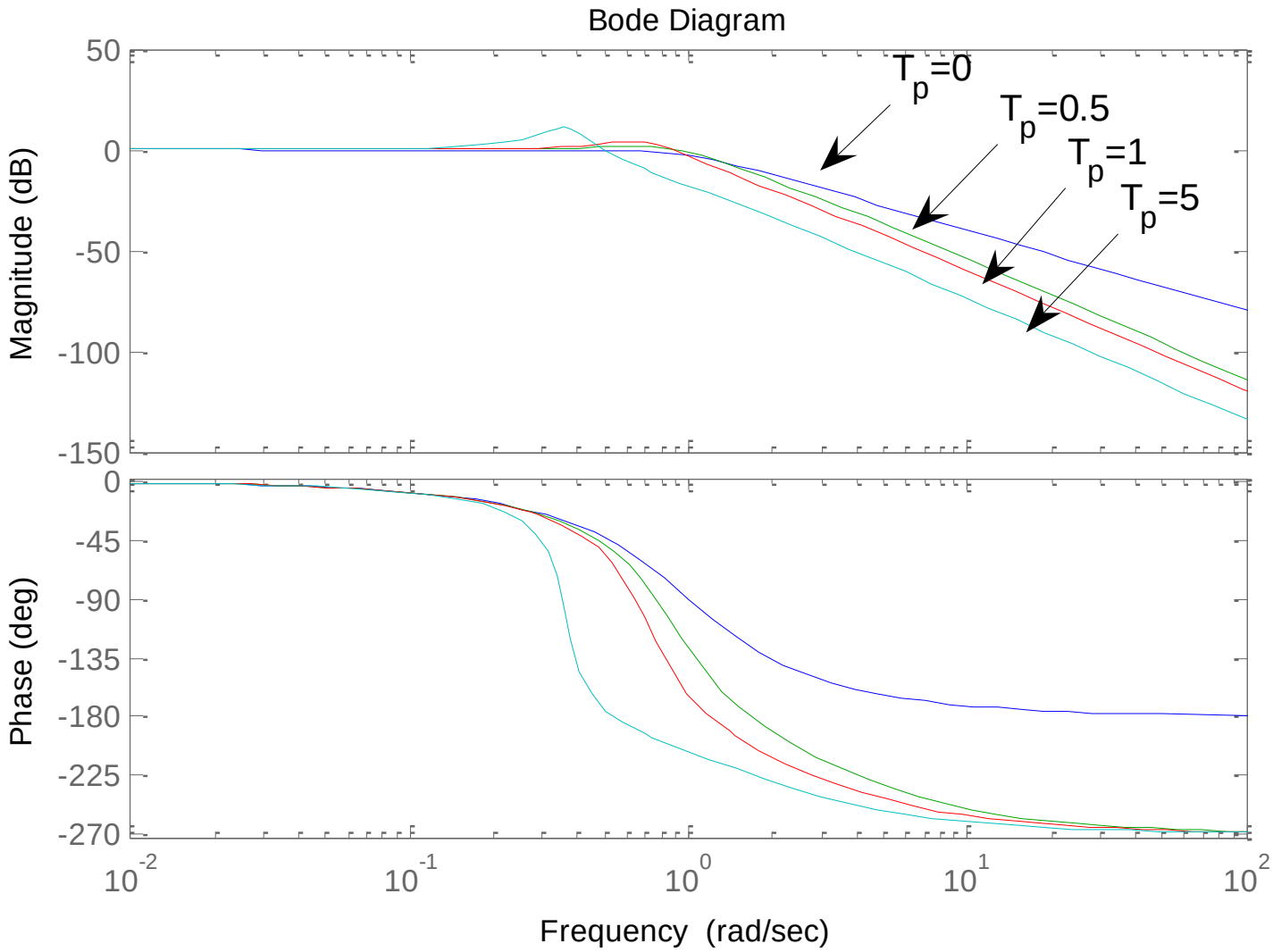
$$G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)(1+T_p s)}$$

Closed-loop TF: $T(s) = \dfrac{\omega_n^2}{T_p s^3 + (1+2\zeta\omega_n T_p)s^2 + 2\zeta\omega_n s + \omega_n^2}$

The effect of adding a pole the open-loop transfer function is to make the closed-loop system less stable, while decreasing the bandwidth.

$\omega_n = 1$

$\zeta = 0.707$



Nyquist (Polar) Plots

Nyquist (Polar) Plots

In the Bode diagram representation of sinusoidal TFs we plot the magnitude and the phase angle of the TF separately.

It is possible also to plot $G(j\omega) = |G(j\omega)| \angle G(j\omega)$ as ω varies from 0 to ∞ on a single graph/plot in the complex plane using polar coordinates. Such plots are called polar plots or Nyquist plots.

Polar plots have the advantage of depicting the frequency response characteristics on a single plot and the disadvantage of not showing the contribution of each factor of the TF.

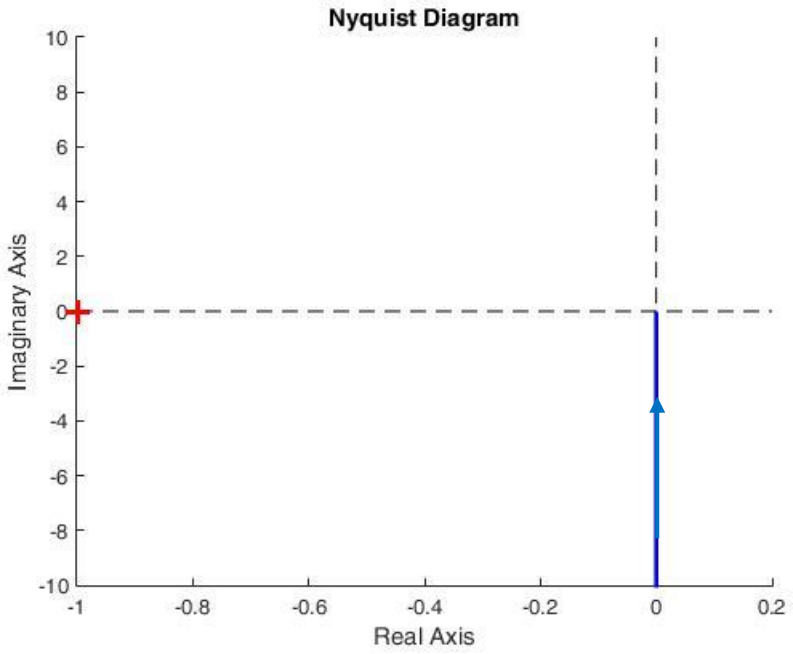
Example

Consider the TF

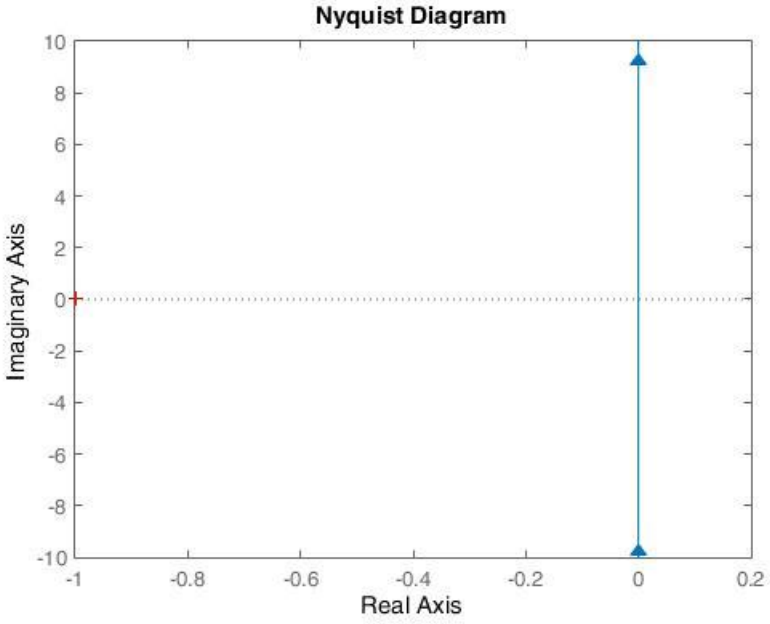
$$G(j\omega) = \frac{1}{j\omega} = -j\frac{1}{\omega}$$

$$= \frac{1}{\omega} \angle(-90^{o})$$

Its Nyquist plot is (for $\omega \in (0, \infty)$)



If ω is allowed to vary from $-\infty$ to $+\infty$ then the Nyquist plot becomes

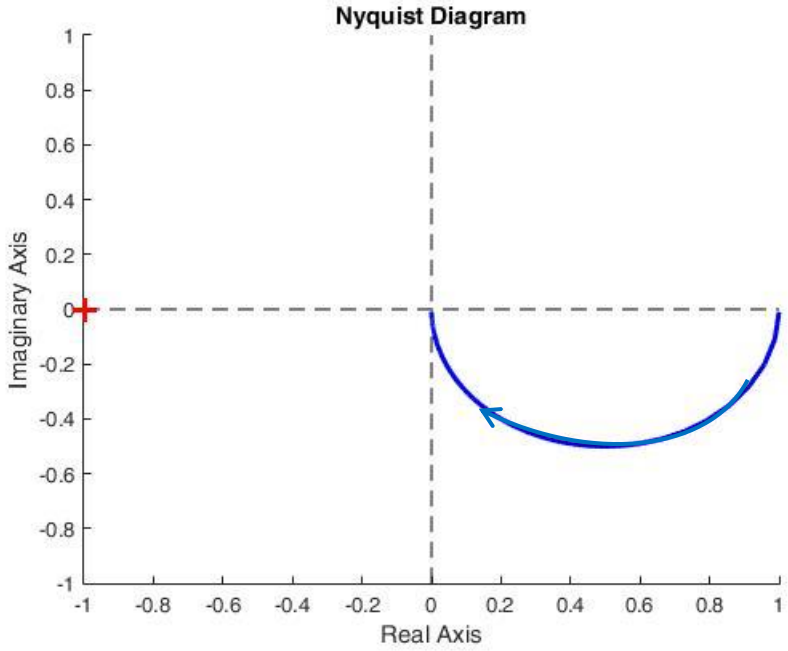


Example

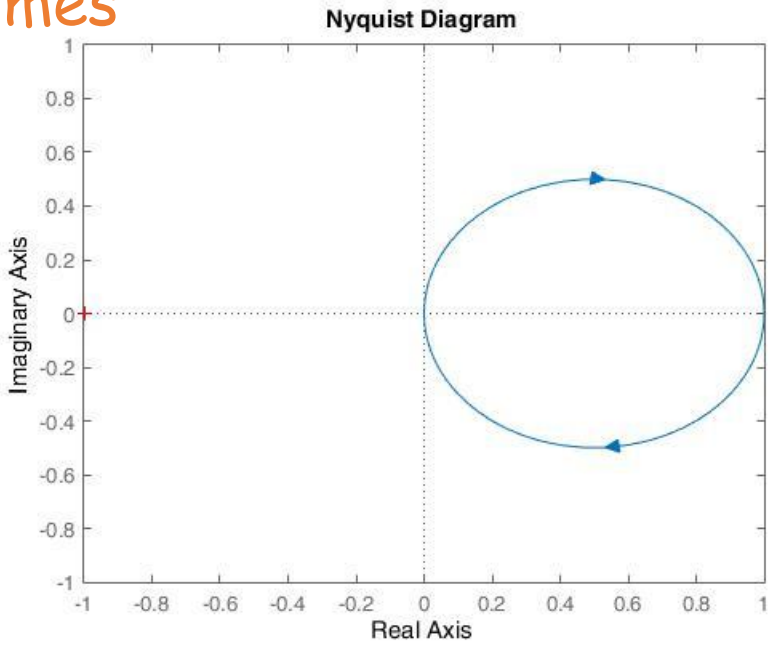
Consider the TF $G(j\omega) = \dfrac{1}{1+j\tau\omega} = \dfrac{1}{\sqrt{1+\tau^2\omega^2}} \measuredangle\left(-\tan^{-1}(\tau\omega)\right)$

ω	0	$1/\tau$	∞
$\lvert G(j\omega)\rvert$	1	$1/\sqrt{2}$	0
$\measuredangle G(j\omega)$	0	-45°	-90°

Its Nyquist plot is (for $\tau = 1, \omega \in (0, \infty)$)



If ω is allowed to vary from $-\infty$ to $+\infty$ then the Nyquist plot becomes



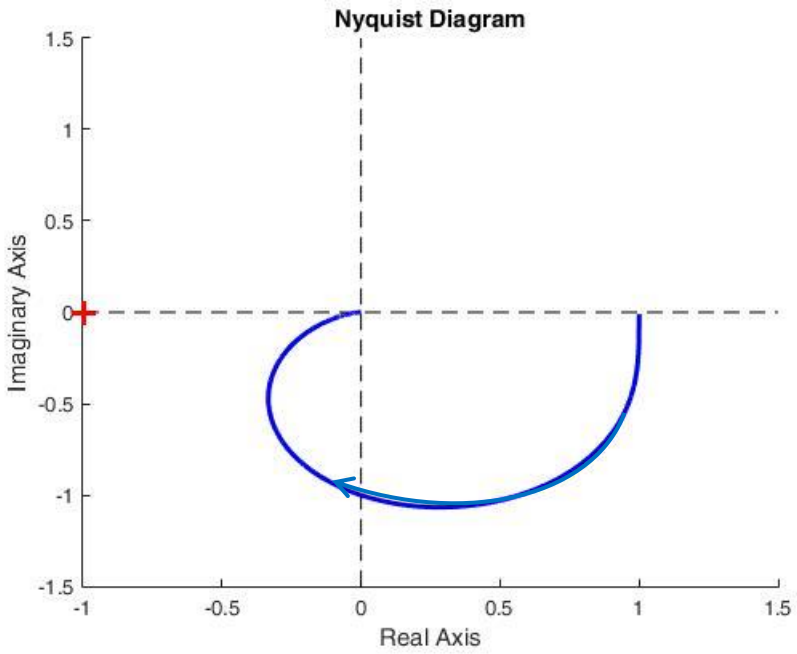
Example

Consider the TF $G(j\omega) = \dfrac{1}{1+2\zeta\left(j\omega/\omega_n\right)+\left(j\omega/\omega_n\right)^2}, \zeta > 0$

$$G(j\omega) = \frac{1}{\sqrt{\left(1-\left(\omega/\omega_n\right)^2\right)^2+\left(2\zeta\omega/\omega_n\right)^2}} \measuredangle\left(-\tan^{-1}\left(\frac{2\zeta\omega/\omega_n}{1-\left(\omega/\omega_n\right)^2}\right)\right)$$

ω	0	ω_n	∞
$\lvert G(j\omega)\rvert$	1	$1/2\zeta$	0
$\measuredangle G(j\omega)$	0	-90°	-180°

Its Nyquist plot is
(for $\omega_n = 1, \zeta = 0.5, \omega \in (0,\infty)$)



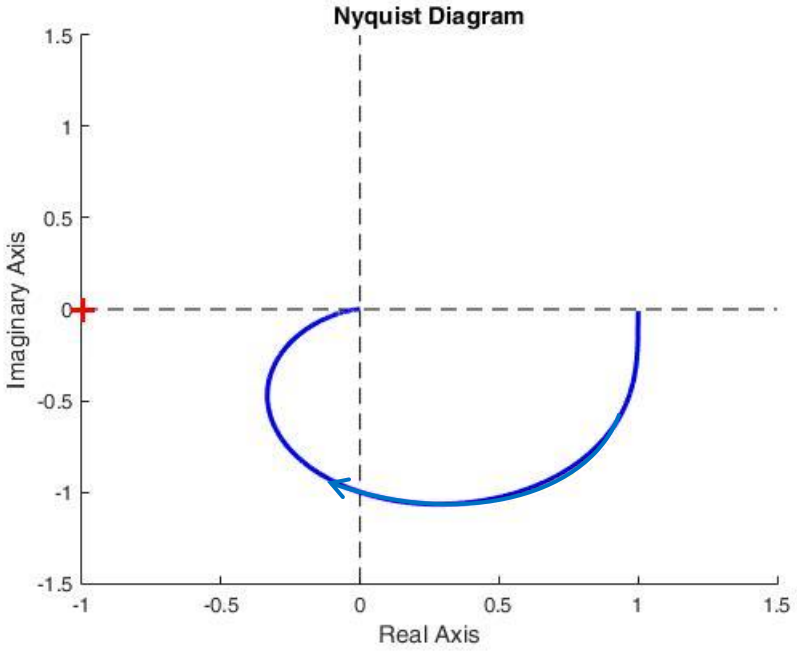
Example

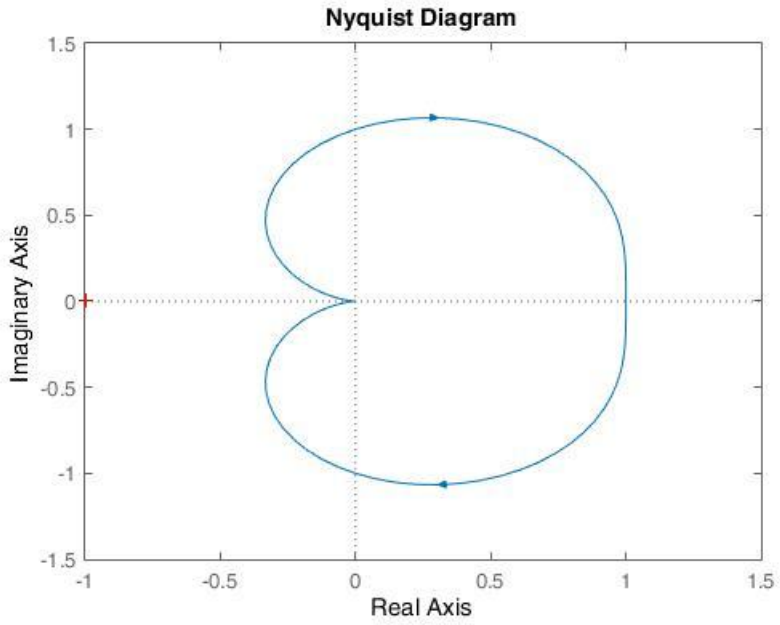
$$G(j\omega)=\frac{1}{\sqrt{\left(1-\left(\omega/\omega_n\right)^2\right)^2+\left(2\zeta\omega/\omega_n\right)^2}}\angle\left(-\tan^{-1}\left(\frac{2\zeta\omega/\omega_n}{1-\left(\omega/\omega_n\right)^2}\right)\right)$$

ω	0	ω_n	∞
$\lvert G(j\omega)\rvert$	1	$1/2\zeta$	0
$\angle G(j\omega)$	0	-90°	-180°

If ω is allowed to vary from $-\infty$ to $+\infty$ then the Nyquist plot becomes

Its Nyquist plot is (for $\omega_n=1,\zeta=0.5,\omega\in(0,\infty)$)





Example Consider the transfer function

$$G(s)=\frac{10}{s(s+1)}$$

Draw its Nyquist plot as ω varies from 0 to ∞.

Solution. Substituting s=jω into G(s) yields: $G(j\omega)=\dfrac{10}{j\omega(j\omega+1)}$

The magnitude and phase of G(jω) at ω=0 and ω=∞ are computed as follows.

$$\lim_{\omega\to 0}\left|G(j\omega)\right|=\lim_{\omega\to 0}\left|\frac{10}{j\omega(j\omega+1)}\right|=\lim_{\omega\to 0}\frac{10}{\omega}=\infty$$

$$\lim_{\omega\to 0}\angle G(j\omega)=\lim_{\omega\to 0}\angle\frac{10}{j\omega(j\omega+1)}=\lim_{\omega\to 0}\angle\frac{10}{j\omega}=-90^\circ$$

Thus, the properties of the Nyquist plot of G(jω) at ω=0 and ω=∞ are ascertained.

$$\lim_{\omega\to \infty}\left|G(j\omega)\right|=\lim_{\omega\to \infty}\left|\frac{10}{j\omega(j\omega+1)}\right|=0$$

$$\lim_{\omega\to \infty}\angle G(j\omega)=\lim_{\omega\to \infty}\angle\frac{10}{j\omega(j\omega+1)}=\lim_{\omega\to \infty}\angle\frac{10}{-\omega^2}=-180^\circ$$

Next, we determine the intersection.

If the Nyquist plot of G(jω) intersects with the real axis, we have

$$\mathrm{Im}[G(j\omega)] = 0$$

$$G(j\omega) = \frac{10}{j\omega(j\omega+1)} = -\frac{10}{\omega^2+1} - j\frac{10}{\omega(\omega^2+1)} \quad \Rightarrow \quad \frac{10}{\omega(\omega^2+1)} = 0 \quad \Rightarrow \quad \omega = \infty$$

This means that the G(jω) plot intersects with the real axis of the G(jω)-plane only at the origin.

Similarly, intersection of G(jω) with the imaginary axis:

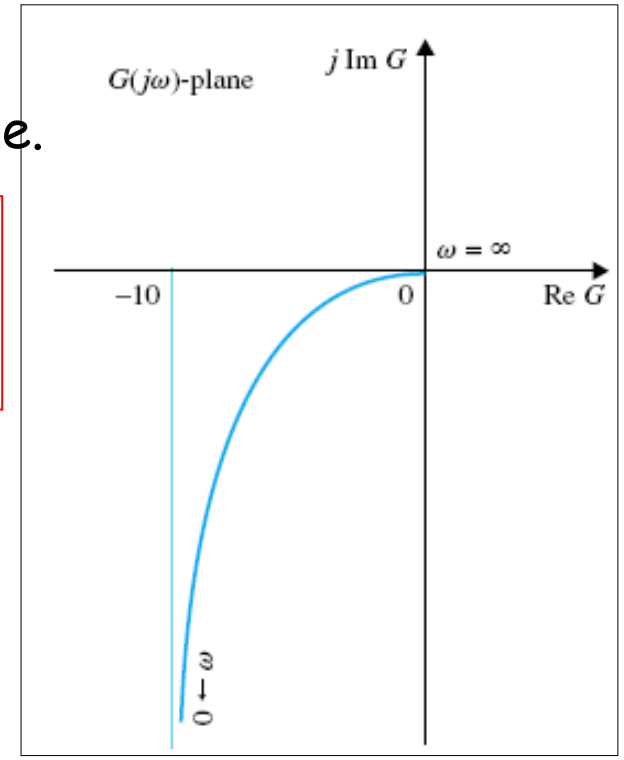
$$\mathrm{Re}[G(j\omega)] = 0 \quad \Rightarrow \quad \omega = \infty$$

which corresponds to the origin of the G(jω)-plane.

The conclusion is that the Nyquist plot of G(jω) *does not intersect any one of the axes* at any finite nonzero frequency.

At ω=∞, $\mathrm{Re}[G(j\omega)] = 0$

At ω=0, $\mathrm{Re}[G(j\omega)] = -10$



Example Consider again the transfer function

$$G(s)=\frac{300(s+5)}{s(s+2)(s^2+10s+100)}$$

We sketched the Bode diagram for this system.

Sketch its Nyquist Plot.

Solution. Letting s=jω, we have

$$G(j\omega)=\frac{300(j\omega+5)}{(j\omega)(j\omega+2)\left((j\omega)^2+10(j\omega)+100\right)}$$

ω	0	0.1	2	5	10	∞
$\lvert G(j\omega)\rvert$	∞	7.5	3.8	0.6	0.3	0
$\measuredangle G(j\omega)$	-90°	$\cong -90^o$	<-90°	$\cong -150^o$	$\cong -180^o$	-270°

These values can be extracted from the Bode diagram.

Otherwise, we would need to express the magnitude and phase angle in terms of ω and calculate their values for different specific values of ω.

Using these values one can roughly sketch the Nyquist plot.

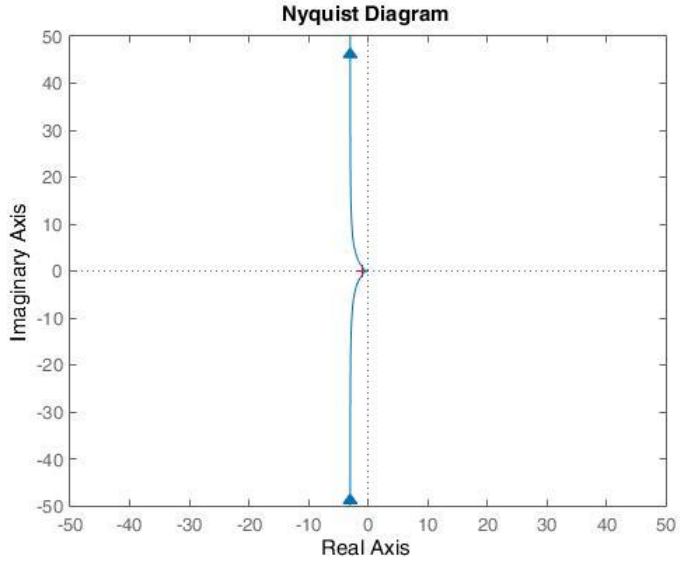
Example

$$G(j\omega)=\frac{300(j\omega+5)}{(j\omega)(j\omega+2)\left((j\omega)^2+10(j\omega)+100\right)}$$

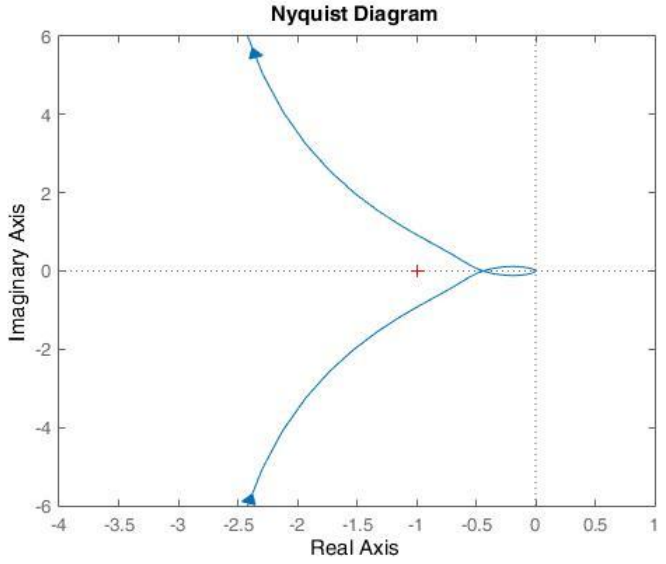
We sketched the Bode diagram for this system.

ω	0	0.1	2	5	10	∞
$\lvert G(j\omega)\rvert$	∞	7.5	3.8	0.6	0.3	0
$\measuredangle G(j\omega)$	-90°	$\cong -90^o$	<-90°	$\cong -150^o$	$\cong -180^o$	-270°

These values can be extracted from the Bode diagram.



The Nyquist plot.



Nyquist plot zoomed in the vicinity of the origin.

System Type and the Nyquist Plot

Consider the general TF

$$G(s) = \frac{K\prod_{i=1}^{Q}(1+T_i s)}{s^N \prod_{m=1}^{M}(1+\tau_m s)\prod_{k=1}^{R}\left(1+2\zeta_k\left(s/\omega_{n_k}\right)+\left(s/\omega_{n_k}\right)^2\right)}$$

where $m=Q$, $n=N+M+2R$, $K>0$, $m<n$

$m=Q$ – number of zeros $n=N+M+2R$ – number of poles

N=0 - Type-0 system

$$G(0) = K\angle(0^o) \quad and \quad G(\infty) = 0\angle(-(n-m)\times 90^o)$$

N=1 - Type-1 system

$$G(0) = \infty\angle(-90^o) \quad and \quad G(\infty) = 0\angle(-(n-m)\times 90^o)$$

N=2 - Type-2 system

$$G(0) = \infty\angle(-180^o) \quad and \quad G(\infty) = 0\angle(-(n-m)\times 90^o)$$

System Type and the Nyquist Plot

N=0 - Type-0 system

$$G(0) = K\angle(0^o)$$

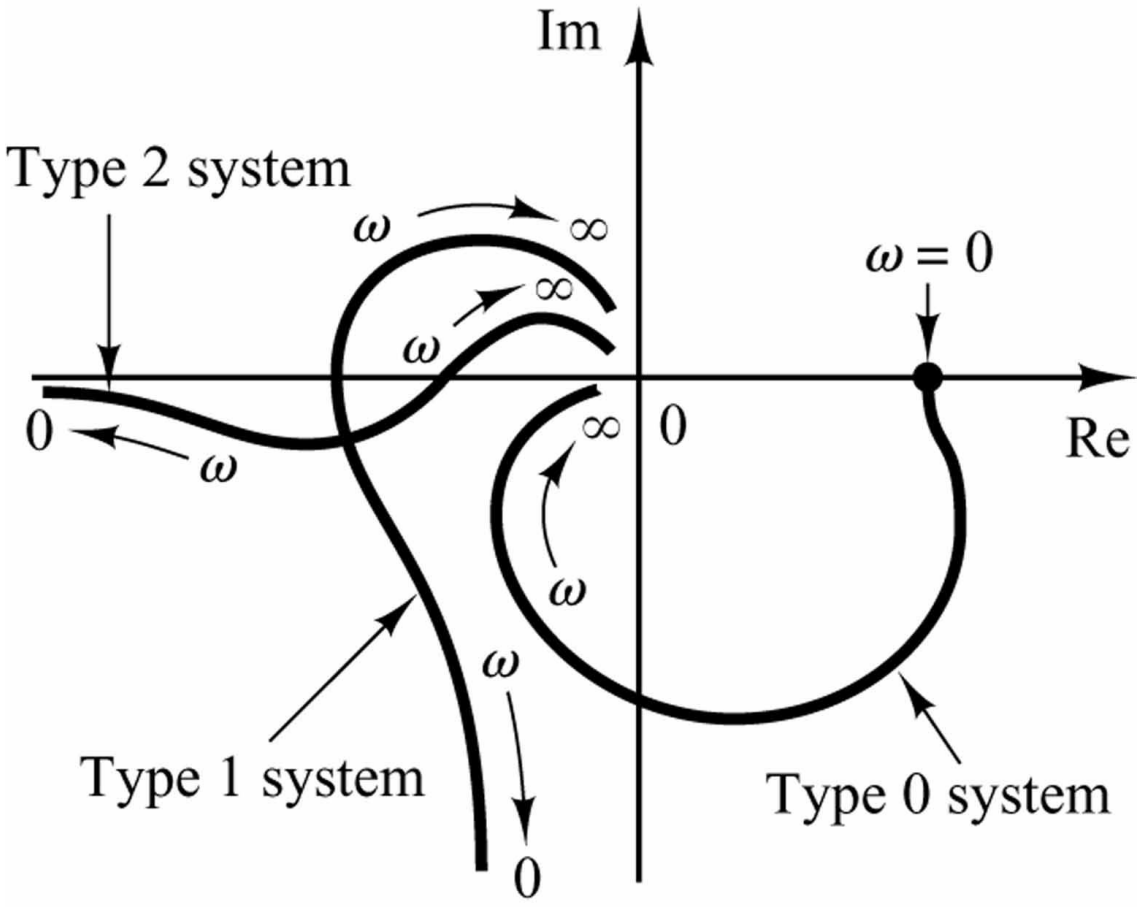
N=1 - Type-1 system

$$G(0) = \infty\angle(-90^o)$$

N=2 - Type-2 system

$$G(0) = \infty\angle(-180^o)$$

Starting points of the Nyquist plots of type-0,1,2 systems are as shown in the figure.



Nyquist Plots in the High Frequency Range

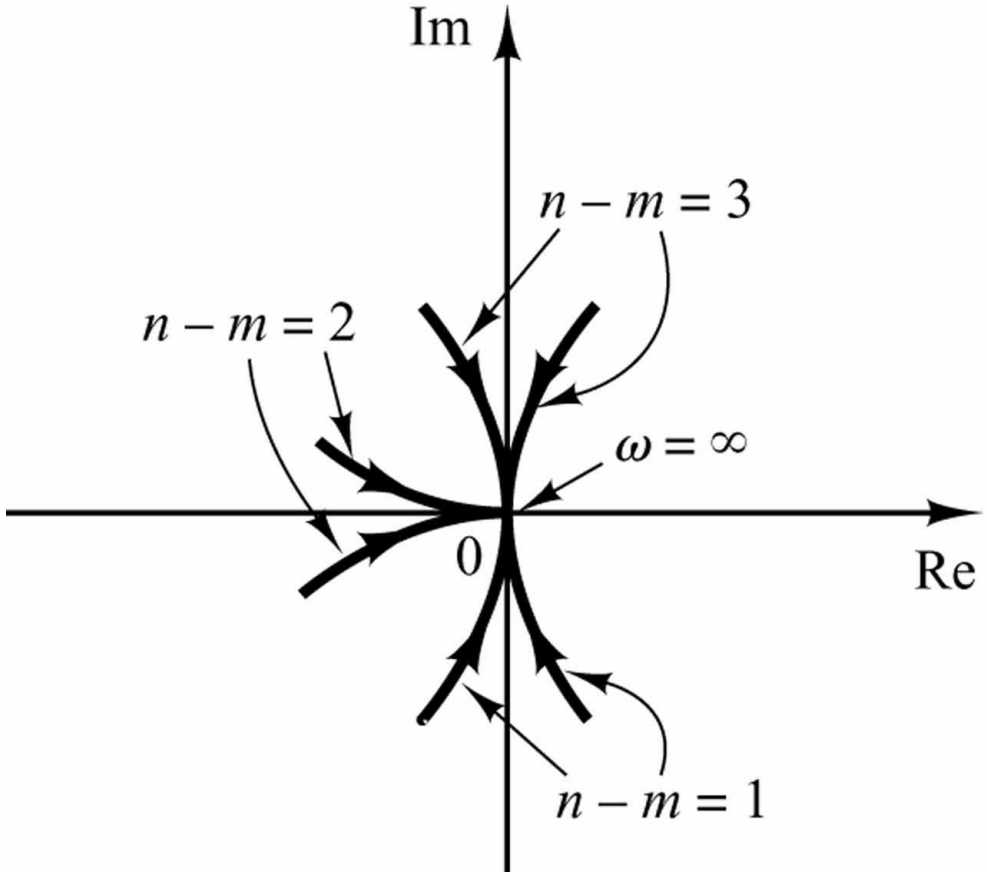
$$G(s)=\frac{K\prod_{i=1}^{Q}(1+T_i s)}{s^N\prod_{m=1}^{M}(1+\tau_m s)\prod_{k=1}^{R}\left(1+2\zeta_k\left(s/\omega_{n_k}\right)+\left(s/\omega_{n_k}\right)^2\right)}$$

n=number of poles
m=number of zeros
$m < n$

Nyquist plots in the high frequency range can be expressed by

$$G(\infty)=0\angle(-(n-m)\times 90^{o})$$

They converge to the origin as shown in the figure.



Nyquist Stability Criterion

Nyquist Stability Criterion

As we discussed before stability is an important desired property of control systems.

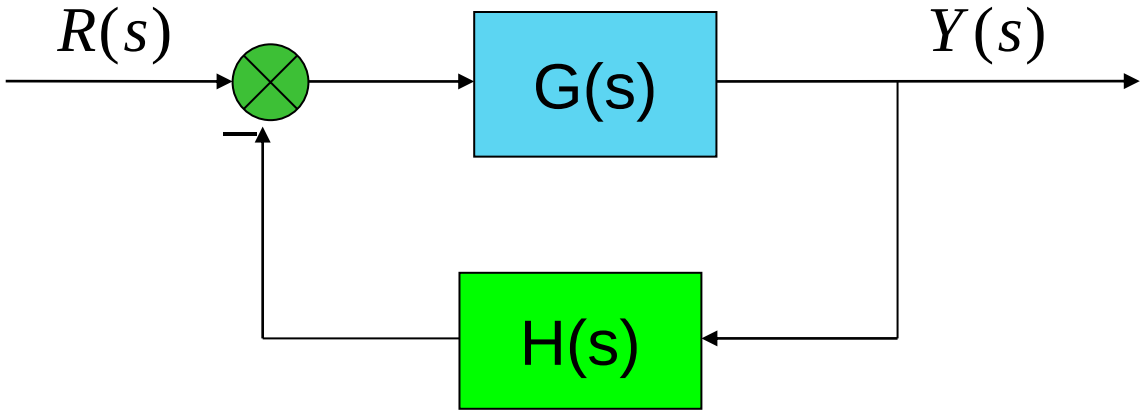
There are various methods for determining the stability of LTI systems – recall the Routh-Hurwitz and the root-locus methods.

There is also a method developed by Nyquist and called Nyquist Stability Criterion, which can be used to determine the stability and relative stability of an LTI system from its sinusoidal OL TF $G(j\omega)H(j\omega)$.

It is based on a theorem from the theory of complex variables and uses the mapping of contours in the complex plane.

Nyquist Stability Criterion

What is Nyquist stability criterion used for?



The Nyquist stability criterion is a semigraphical method that can be used to determine the stability of a closed-loop system;

The Nyquist stability criterion allows us to determine the stability of a closed-loop system from the frequency-response of the loop function G(jω)H(jω).

Review about stability

Closed-loop TF: $T(s) = \dfrac{G(s)}{1 + G(s)H(s)}$

Characteristic equation (CE):

$$\Delta(s) = 1 + G(s)H(s) = 0$$

Stability conditions:

Open-loop stability: poles of the loop TF G(s)H(s) are all in the left-half s-plane.

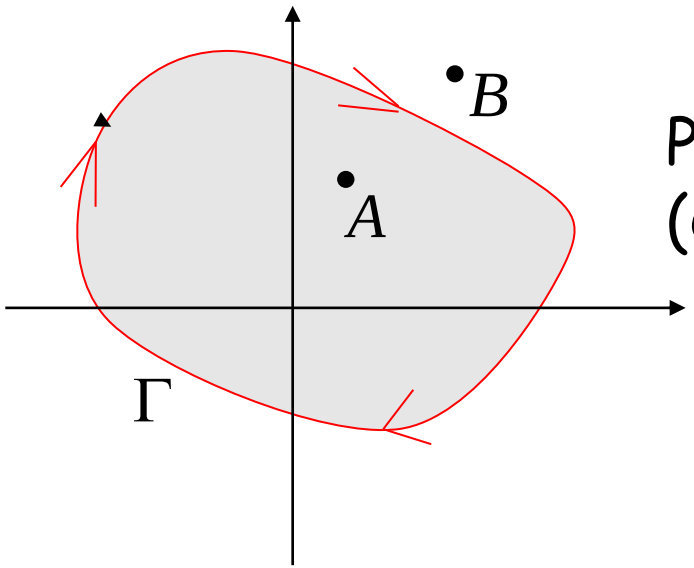
Closed-loop stability: poles of the closed-loop TF T(s) or roots of the CE Δ(s) are all in the left-half s-plane.

Definition of Encircled and Enclosed

Encircled: A point or region in a complex function plane is said to be encircled by a closed path if it is found inside the path.

Traversal Direction: We use the convention that the clockwise (CW) direction is positive.

Enclosed: A point or region in a complex function plane is said to be enclosed by a closed path if it is encircled by the path.

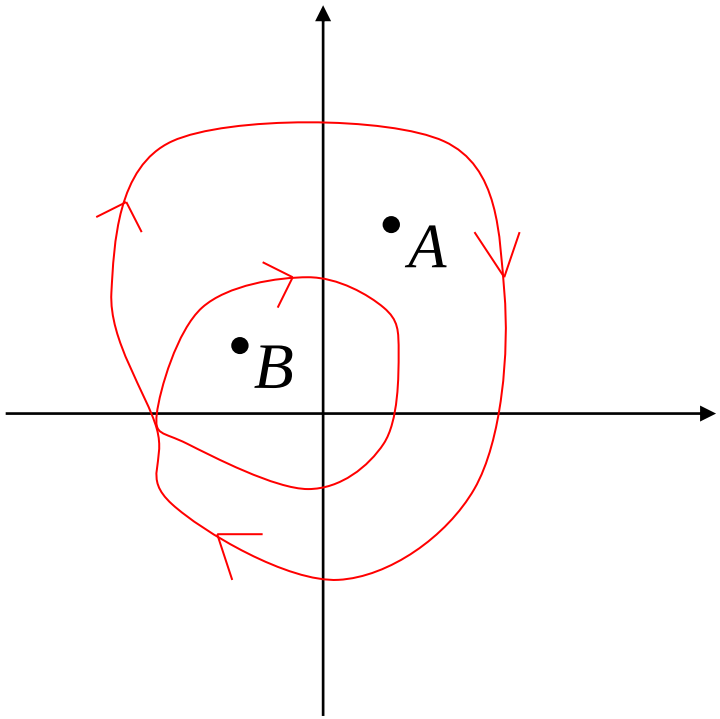


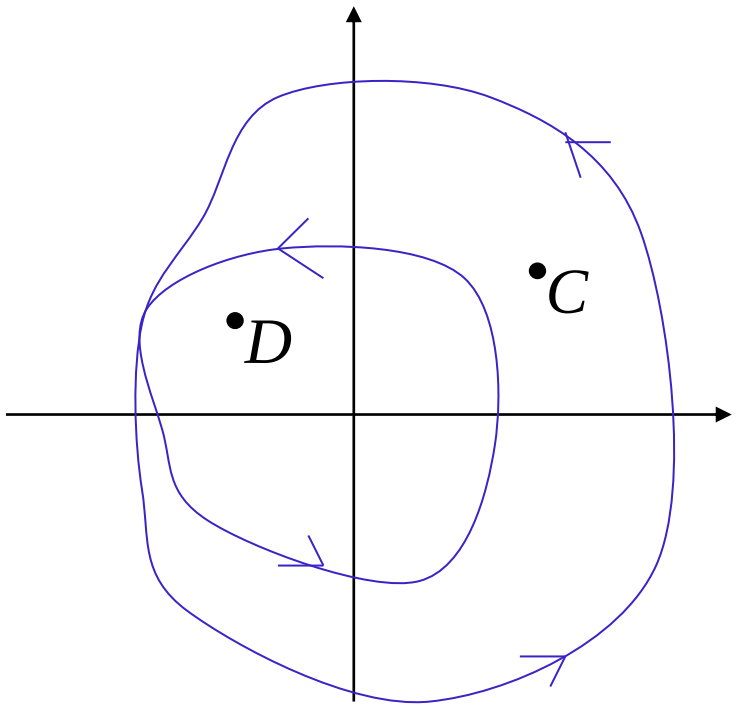
Point A is encircled in the positive (CW) direction by the closed path.

Point A is also enclosed in the positive (CW) direction by the closed path.

For positive (i.e., CW) traversal the area to the right is enclosed.
For negative (i.e., CCW) traversal the area to the left is enclosed.

Number of Encirclements and Enclosures





Point A is encircled/enclosed once in CW direction.
Point B is encircled/enclosed twice in CW direction.

Point C is encircled/enclosed once in CCW direction.
Point D is encircled/enclosed twice in CCW direction.

Mapping Contours From The s-plane to The Δ(s)-plane

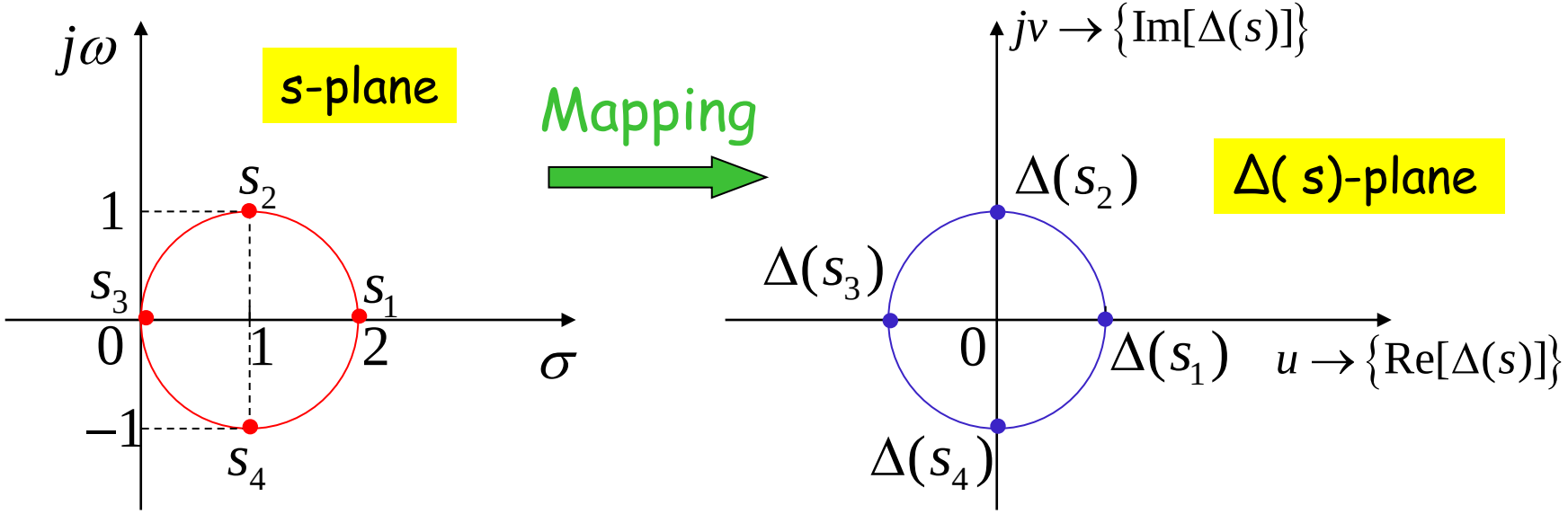
A contour map is a contour or trajectory in one plane (the s-plane) mapped or translated into another plane (the Δ(s)-plane) by a function or relation Δ(s).

Since $s=\sigma+j\omega$ is a complex variable, Δ(s) is also a complex variable, i.e., $\Delta(s)=u+jv$, and can be represented in the complex plane by its coordinates u and v.

A closed contour in the s-plane which does not pass through the singularities (i.e., the poles and zeros) of Δ(s) results in a closed curve in the Δ(s)-plane.

Mapping Contours From The s-plane to The Δ(s)-plane

Example: Consider a function Δ(s) =s-1. Map a circle with a radius 1 centered at 1 from s-plane to the Δ(s)-plane.



$s_1 = 2; \quad s_2 = 1 + j$

$s_3 = 0; \quad s_4 = 1 - j$

$\Delta(s_1) = 1; \quad \Delta(s_2) = j$

$\Delta(s_3) = -1; \quad \Delta(s_4) = -j$

Mapping Contours From The s-plane to The Δ(s)-plane

Example: Consider a function Δ(s) =(s+2)/s. $s=\sigma+j\omega$

$$\Delta(s)=\Delta(\sigma+j\omega)=\frac{\sigma+j\omega+2}{\sigma+j\omega}$$

$$=\underbrace{\left(1+\frac{2\sigma}{\sigma^2+\omega^2}\right)}_{u}+j\underbrace{\left(-\frac{2\omega}{\sigma^2+\omega^2}\right)}_{v}$$

$$=u+jv$$

$$u=1+\frac{2\sigma}{\sigma^2+\omega^2}$$

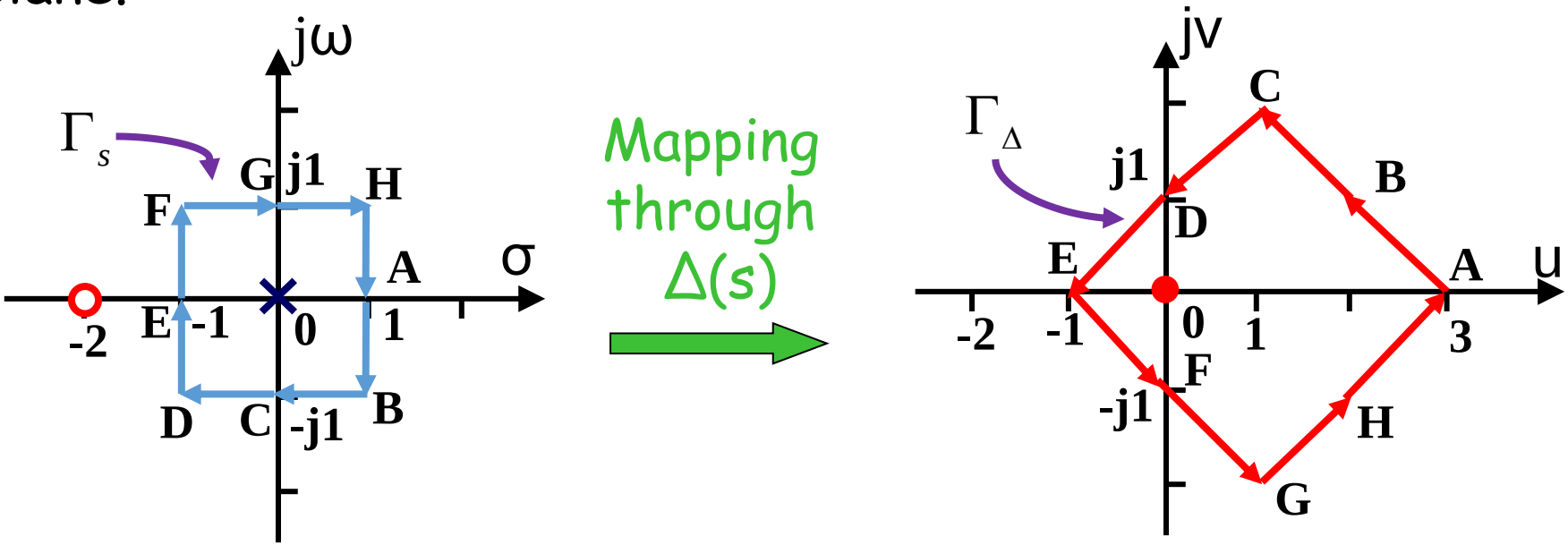
$$v=\left(-\frac{2\omega}{\sigma^2+\omega^2}\right)$$

For every (σ,ω) in the s-plane there is only one corresponding (u,v) in the Δ(s)-plane.

The opposite is in general not true, i.e., given (u,v) in the Δ(s)-plane, there may be multiple (σ,ω) in the s-plane corresponding to (resulting in) that (u,v).

Example: (Continued) $\Delta(s) = (s+2)/s$.

Consider arbitrarily the following closed contour in the s-plane.



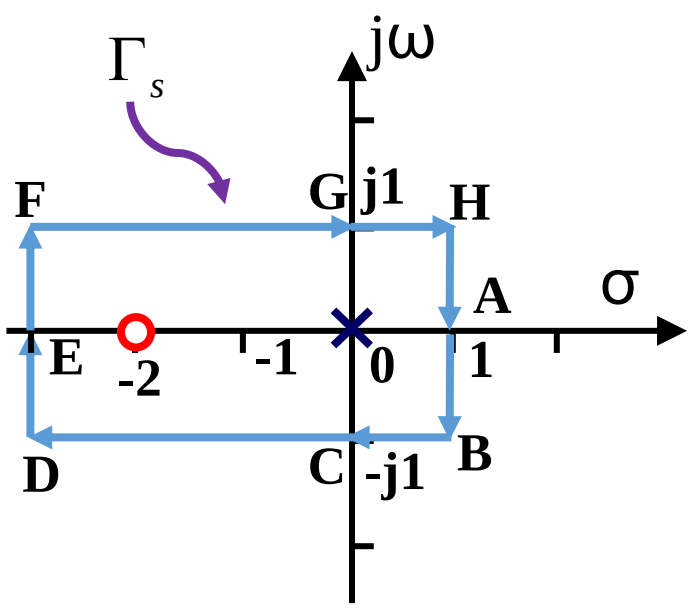
s	1	1-j1	-j1	-1-j1	-1	-1+j1	j1	1+j1
Δ(s)	3	2+j1	1+j2	j1	-1	-j1	1-j2	2-j1

The contour Γ_s encircles one pole of Δ(s) in the s-plane (in CW direction).

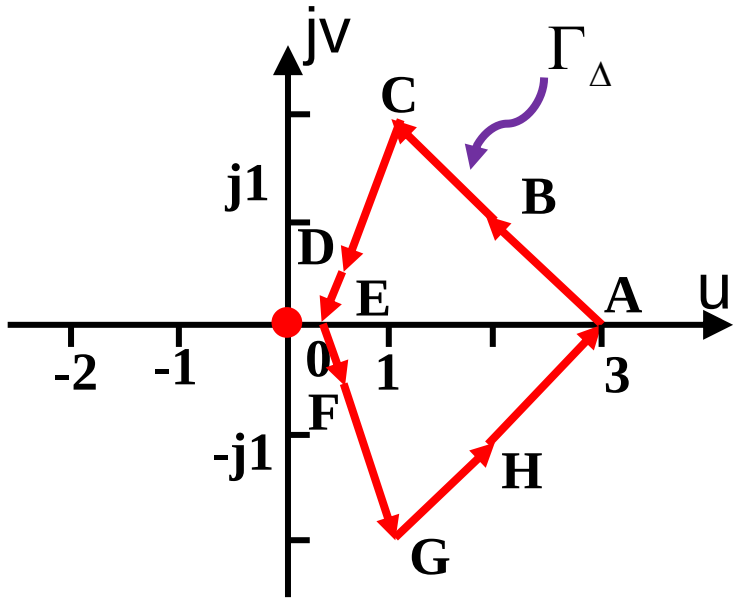
The contour Γ_Δ encircles the origin of Δ(s)-plane once in CCW direction.

Example: (Continued) $\Delta(s) = (s+2)/s$.

Consider another closed contour in the s-plane.



Mapping through $\Delta(s)$



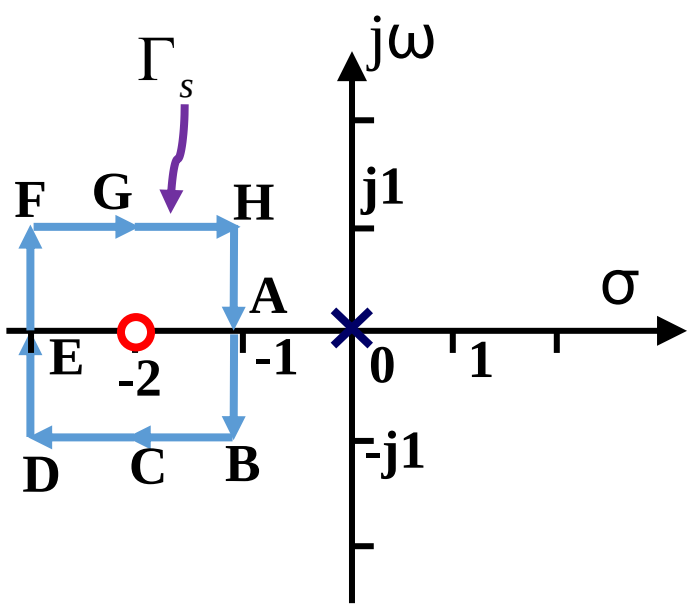
s	1	1-j1	-j1	-3-j1	-3	-3+j1	j1	1+j1
Δ(s)	3	2+j1	1+j2	2/5+j1/5	1/3	2/5-j1/5	1-j2	2-j1

The contour Γ_s encircles one zero and one pole of $\Delta(s)$ in the s-plane (in CW direction).

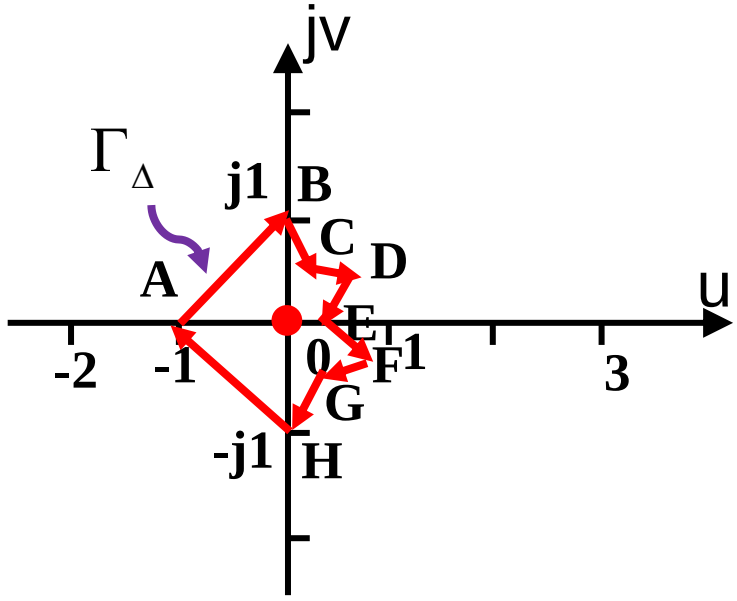
The contour Γ_Δ does not encircle the origin of $\Delta(s)$-plane.

Example: (Continued) $\Delta(s) = (s+2)/s$.

Consider another closed contour in the s-plane.



Mapping through $\Delta(s)$



s	-1	-1-j1	-2-j1	-3-j1	-3	-3+j1	-2+j1	-1+j1
$\Delta(s)$	-1	j1	1/5+j2/5	2/5+j1/5	1/3	2/5-j1/5	1/5-j2/5	-j1

The contour Γ_s encircles one zero of $\Delta(s)$ in the s-plane (in CW direction).

The contour Γ_Δ encircles the origin of $\Delta(s)$-plane once in CW direction.

Mapping Contours From The s-plane to The $\Delta(s)$-plane

As can be seen form the previous example, the number of encirclements of the origin in the $\Delta(s)$-plane depends on the number of poles and zeros of $\Delta(s)$ encircled by the contour Γ_s in the s-plane.

Consider a contour Γ_s in the s-plane with traversal in the positive (CW) direction. If it encircles Z zeros and P poles of the mapping $\Delta(s)$ and does not pass through any poles or zeros of $\Delta(s)$, then the corresponding contour Γ_Δ in the $\Delta(s)$-plane encircles the origin of the $\Delta(s)$-plane **N=Z-P times in the CW direction**.

We saw this in the previous example, but it is also a general property.

Can this be used for determining the stability of control systems?

Principle Argument

Let $\Delta(s)$ be a single-valued function that has a finite number of poles in the s-plane.

Suppose that an arbitrary closed path Γ_s is chosen in the s-plane so that the path does not go through any one of the poles or zeros of $\Delta(s)$;

The corresponding Γ_Δ contour mapped in the $\Delta(s)$ -plane will encircle the origin as many times as the difference between the number of zeros (Z) and poles (P) of $\Delta(s)$ that are encircled by the s-plane contour Γ_s .

In equation form: $N = Z - P$

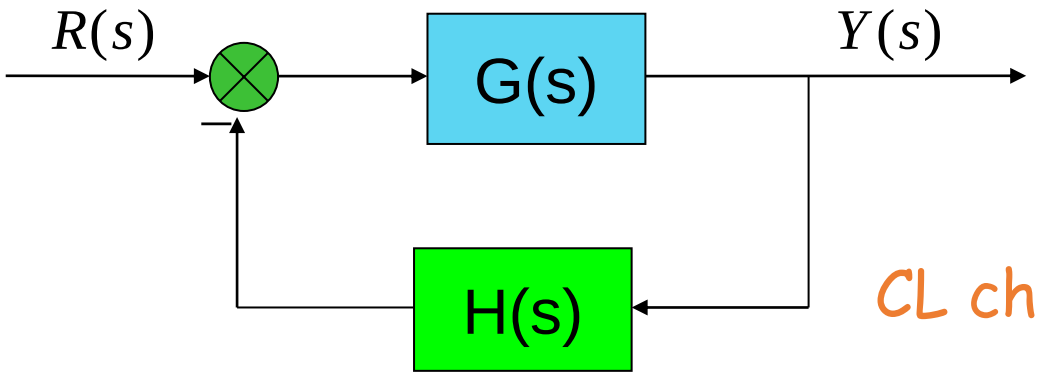
N - number of encirclements of the origin by the $\Delta(s)$-plane contour

Z - number of zeros of $\Delta(s)$ encircled by the s-plane contour

P - number of poles of $\Delta(s)$ encircled by the s-plane contour

Nyquist Stability Criterion

Consider the negative feedback CL control system



Closed-loop TF:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)}$$

CL characteristic equation (CE):

$$\Delta(s) = 1 + G(s)H(s) = 0$$

Note that the zeros of $\Delta(s)$ are poles of the CL TF.

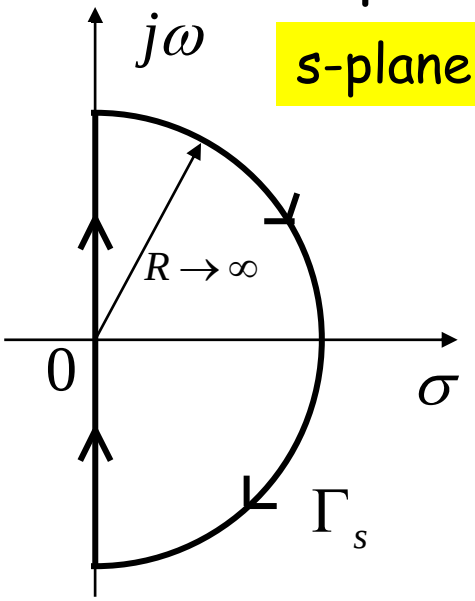
For the CL system to be stable they must be located in the LHP.

Also, recall that even though the poles and zeros of the OL TF G(s)H(s) may be located in the RHP, the CL system is still stable if the poles of the CL TF T(s) are in the LHP.

Nyquist Path Γ_s

Consider the CL CE $\Delta(s) = 1 + G(s)H(s) = 0$ and map with it the closed contour Γ_s which encircles the entire RH s-plane.

Γ_s is a curve composed of the imaginary axis and an arc of infinite radius such that the curve completely encloses the right half of the s-plane .



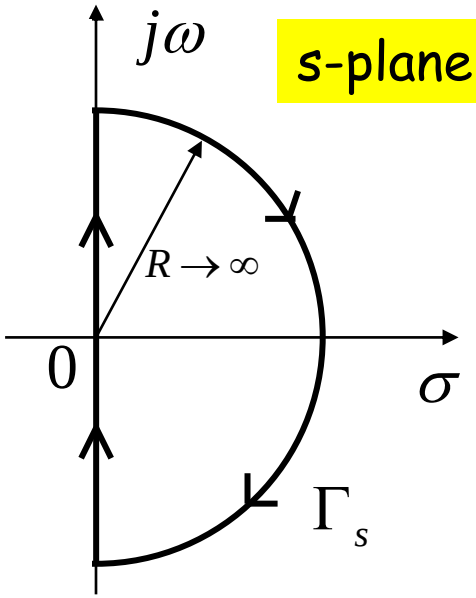
The Nyquist path Γ_s is in the positive CW direction

We can map Γ_s using $\Delta(s)$ into the $\Delta(s)$-plane.

Γ_s does not pass through any poles or zeros of $\Delta(s)$.

If $\Delta(s)$ has any poles or zeros in the right-half plane, they will be encircled by Γ_s .

Nyquist Criterion



The closed contour Γ_s encircles the entire RH s-plane.

We map Γ_s using $\Delta(s)$ into the $\Delta(s)$-plane.

Let P be the number of poles of the OL TF $G(s)H(s)$ in the RHP.

Note that these poles are also poles of $\Delta(s)=1+G(s)H(s)=0$.

The zeros of $\Delta(s)=1+G(s)H(s)=0$ are poles of the CL TF.

From before we know that $Z=N+P$, where N is the number of encirclements of the origin in the $\Delta(s)$-plane and Z is the number of CL poles encircled by Γ_s (i.e., CL poles in the RHP).

From the OL TF $G(s)H(s)$ we know P. ➡ By looking at the number N of encirclements of the origin in the $\Delta(s)$-plane, we can determine the number unstable CL poles (i.e., Z).

Nyquist Criterion

Let P be the number of poles of the OL TF G(s)H(s) in the RHP.

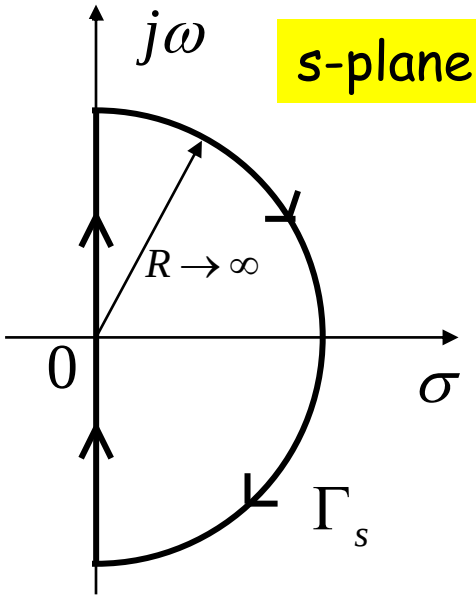
These poles are also poles of Δ(s)=1+G(s)H(s)=0.

$$G(s)H(s)=\frac{K\prod_{i=1}^{m}(s-z_i)}{\prod_{j=1}^{n}(s-p_i)}$$

$$\Delta(s)=1+G(s)H(s)=\frac{\prod_{j=1}^{n}(s-p_i)+K\prod_{i=1}^{m}(s-z_i)}{\prod_{j=1}^{n}(s-p_i)}$$

Δ(s) has the same poles as G(s)H(s), so P can be obtained by counting the number of poles of G(s)H(s) in the RHP.

Nyquist Criterion



Let m be the number of zeros and n the number of poles of G(s)H(s).

For practically realizable systems we have $m \le n$.

Therefore, for practically realizable systems we have

$$\lim_{s\to\infty} \Delta(s) = \lim_{s\to\infty} \left(1 + G(s)H(s)\right) = \text{constant}$$

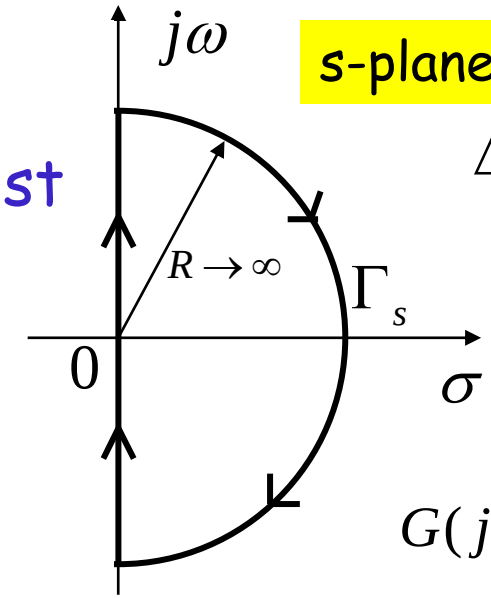
The entire semicircle with the infinite radius in the s-plane is mapped into a single point in the Δ(s)-plane.

Therefore, we can consider only the part of the contour Γ_s on the jω-axis, i.e., we can consider only s=jω or basically Δ(jω)=1+G(jω)H(jω) on $-\infty < \omega < +\infty$.

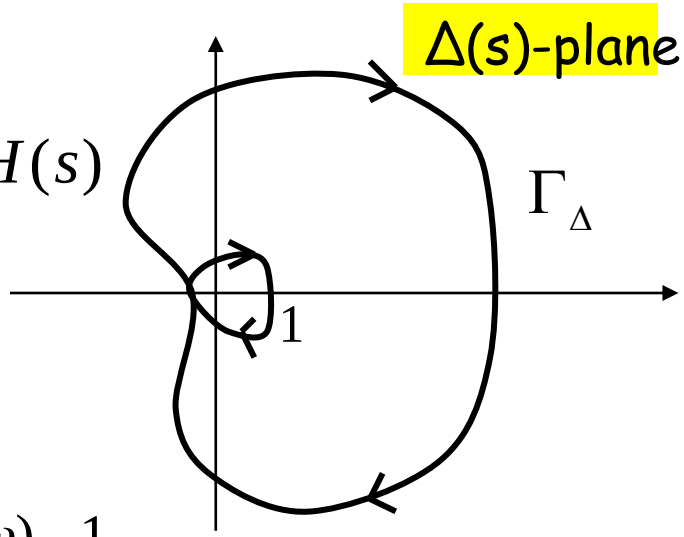
In other words, encirclement of the origin in the Δ(s)-plane (if there is any) occurs while ω varies from $-\infty$ to $+\infty$. (Assuming no poles or zeros are located on the jω-axis.)

Nyquist Criterion and Nyquist Diagram

Nyquist Path



$$\Delta(s) = 1 + G(s)H(s)$$

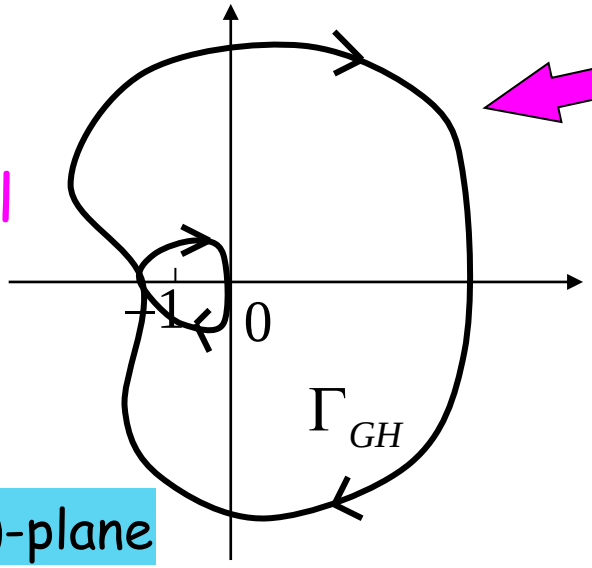


Since

$$G(j\omega)H(j\omega) = \Delta(j\omega) - 1$$

G(jω)H(jω) is only shifted to left by 1 version of Δ(jω).

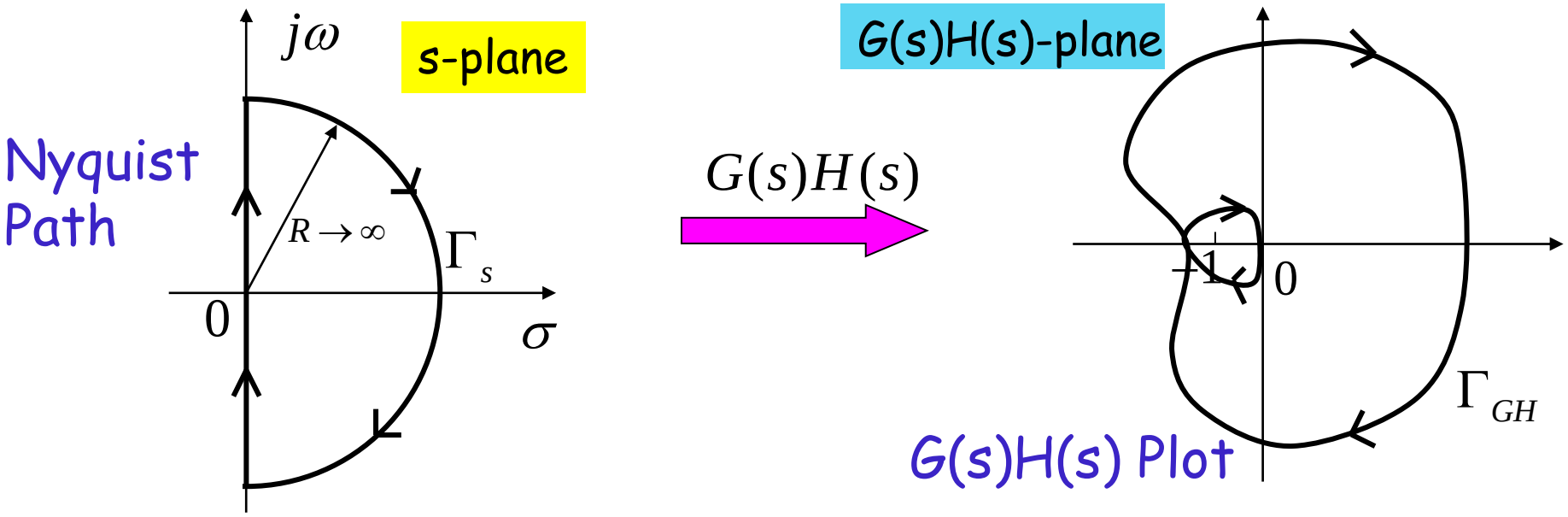
Critical point: (-1+j0)



G(s)H(s)-plane

Instead of mapping into the Δ(s)-plane and checking the encirclement of the origin, we can map into the G(s)H(s)-plane and check the encirclement of the point -1.

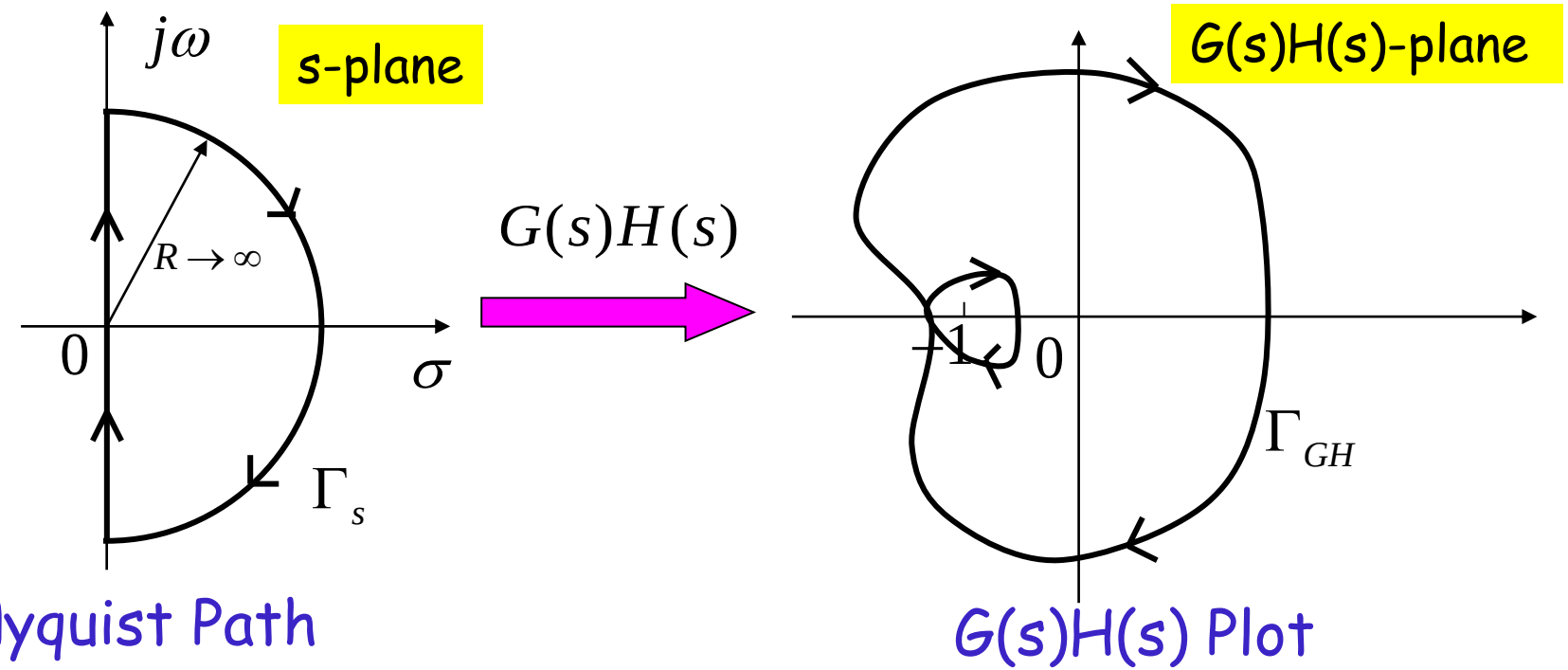
Nyquist Criterion and Nyquist Plot



Nyquist Diagram: Plot the Nyquist diagram of the loop function G(jω)H(jω) and check the encirclement of the point (-1+j0) to determine the closed-loop stability.

The Nyquist Path is shown in the left figure. This path is mapped through the loop transfer function G(s)H(s) to the G(s)H(s) plot in the right figure.

Nyquist Criterion and Nyquist Plot

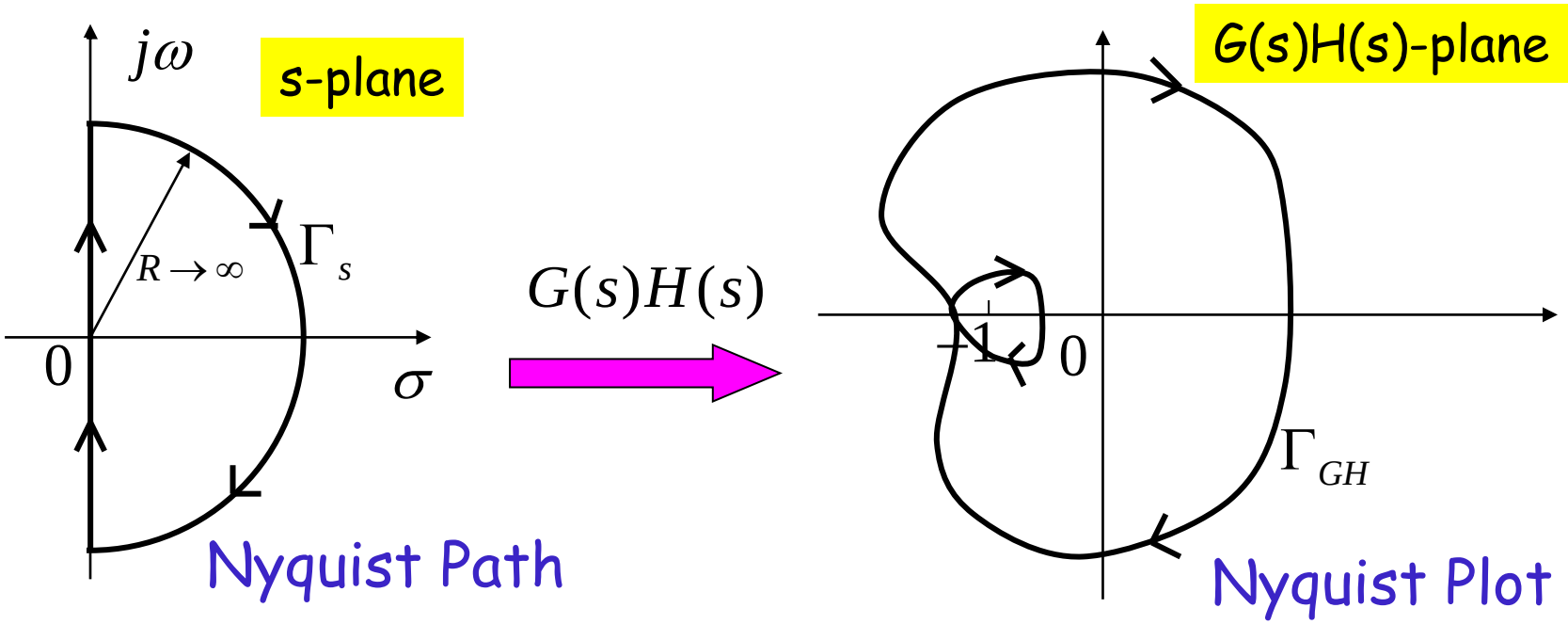


Nyquist Path

G(s)H(s) Plot

The Nyquist Stability Criterion can be stated as follows: Let P be the number of poles of the OL TF G(s)H(s) in the RHP and N the number of encirclements of -1 in the G(s)H(s)-plane in the CW direction. Then, the number of CL poles in the RHP is given by Z=N+P. For the system to be stable we need Z=N+P=0 and therefore N = -P.

Nyquist Criterion and Nyquist Plot



N - number of encirclements of (-1,j0) by the G(s)H(s) plot

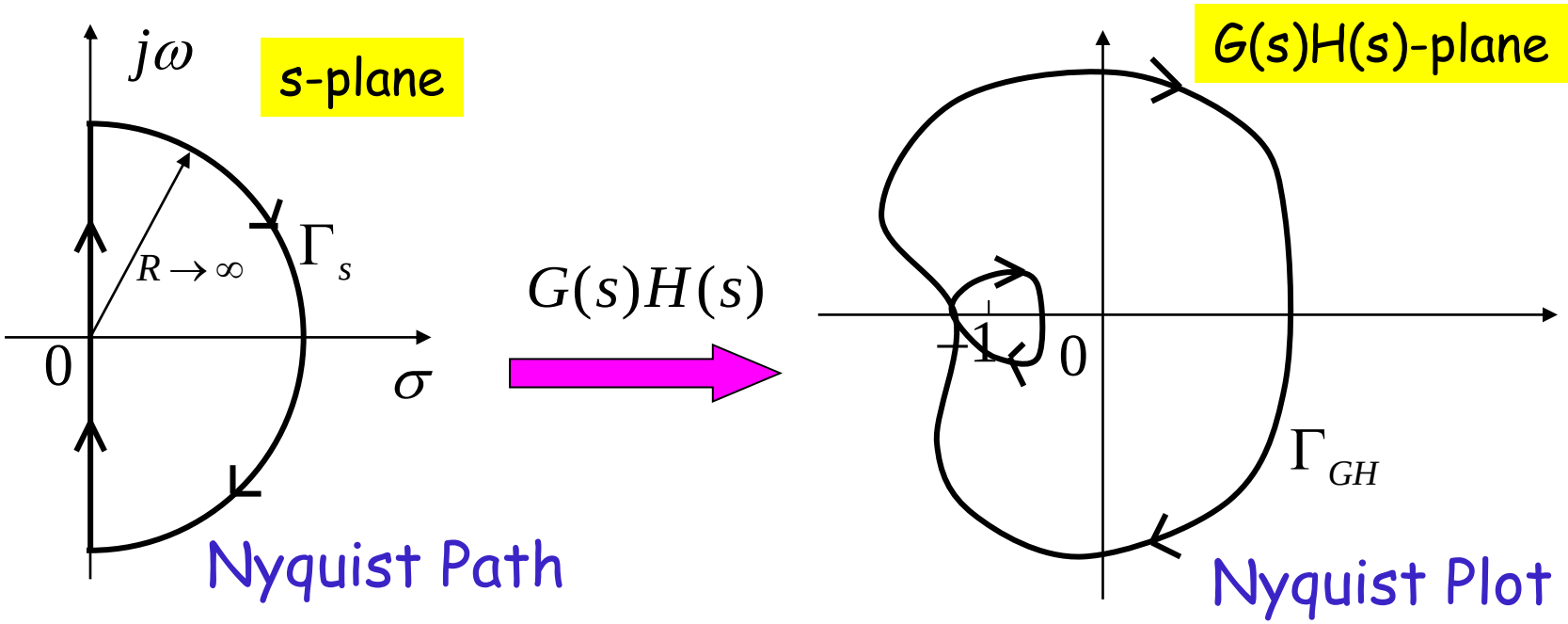
Z - number of CL poles (i.e., zeros of Δ(s)) that are inside the RHP

P - number of poles of G(s)H(s) that are inside the RHP

The condition of closed-loop stability according to the Nyquist Stability Criterion is:

$$N = -P$$

Nyquist Criterion and Nyquist Plot



The closed-loop system is stable if and only if the contour Γ_{GH} encircles -1 P times in the CCW direction, where P is the number of poles of the OL TF G(s)H(s) in the RHP.

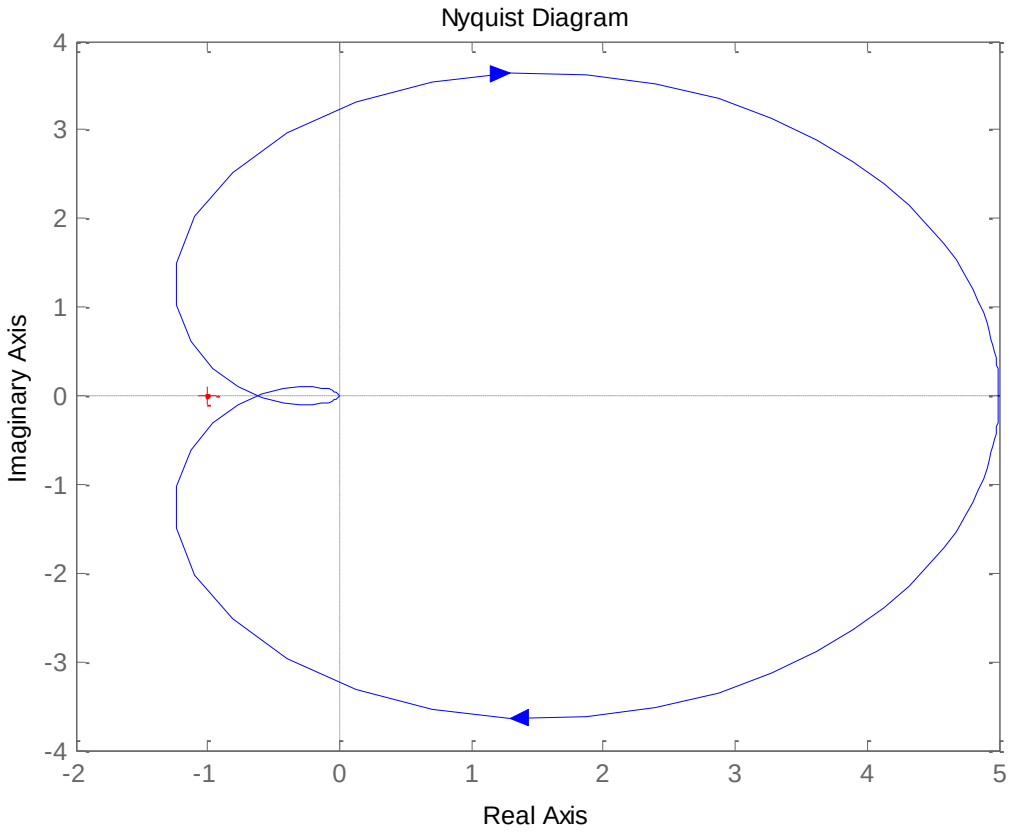
Example

Consider the system with the loop function

$$G(s)H(s) = \frac{5}{(s+1)^3}$$

Matlab program for Nyquist plot (G(s)H(s) plot)

```
>>num=5;
>>den=[1 3 3 1];
>>nyquist(num,den);
```



N=0, P=0,
N=-P, Z = 0 => stable

Question 1: is the closed-loop system stable?

Question 2: what if

$$G(s)H(s) = \frac{5K}{(s+1)^3}?$$

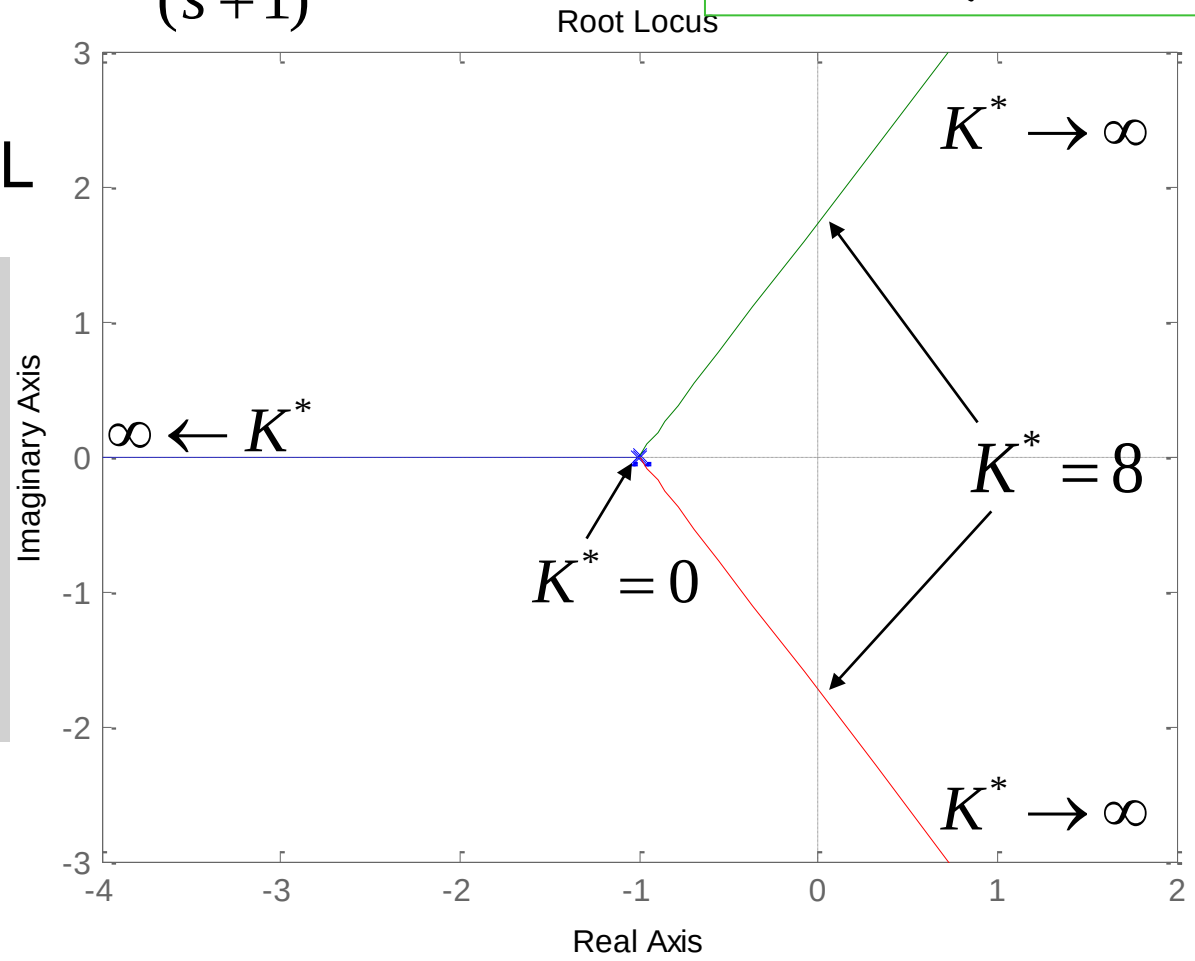
1. With root locus technique:

$$G(s)H(s)=\frac{5K}{(s+1)^3}=K^*\frac{1}{(s+1)^3}$$

```
>>num=1;
>>den=[1 3 3 1];
>>rlocus(num,den);
```

A K* varies from 0 to ∞, we draw the RL

When K*=8 (K=1.6), the RL cross the jω-axis, the closed-loop system is marginally stable.



For K*>8 (K>1.6), the closed-loop system has two roots in the RHP and is unstable.

2. With Nyquist plot and Nyquist criterion:

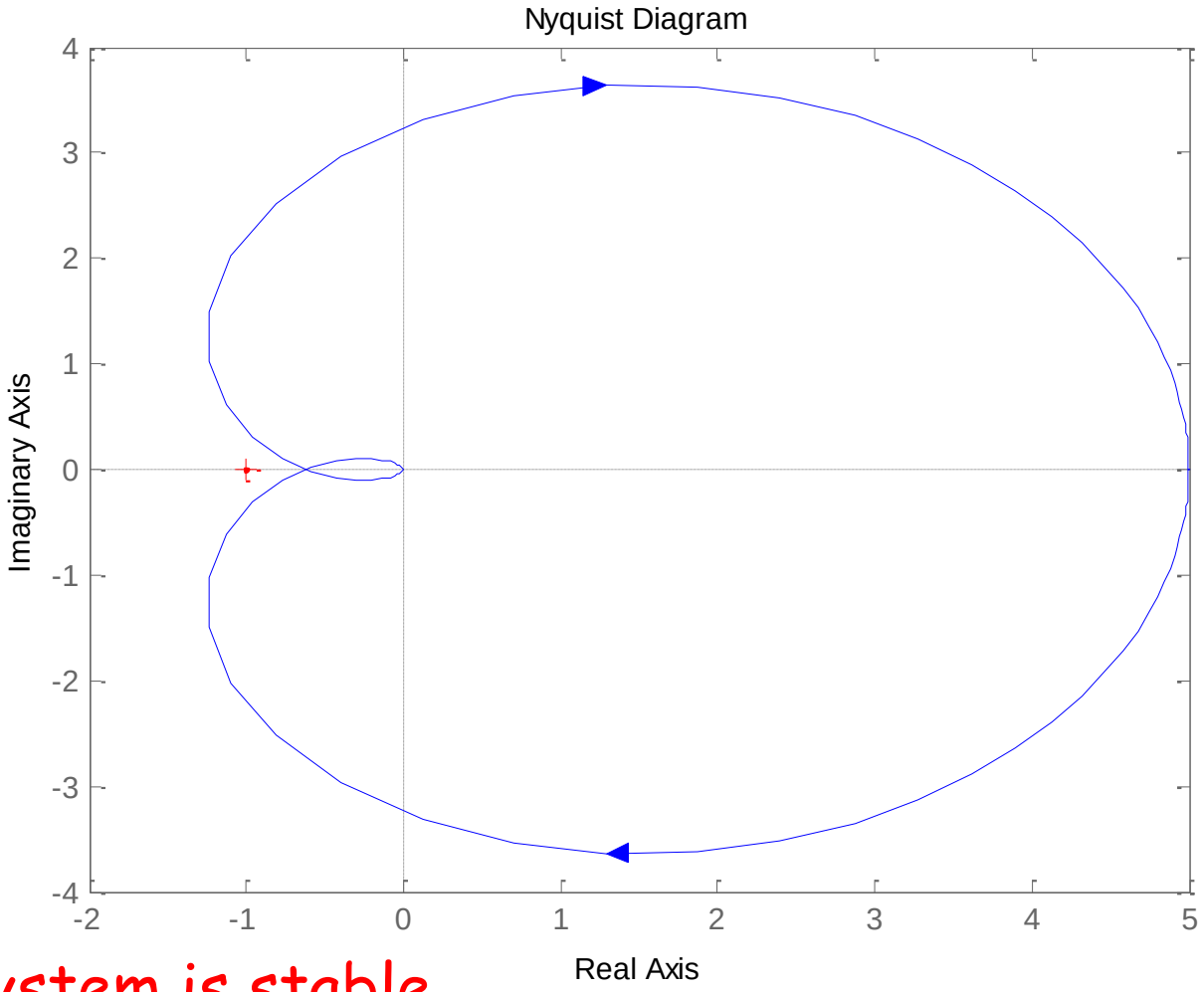
$$G(s)H(s) = \frac{5K}{(s+1)^3}$$

```
>>K=1;
>>num=5*K;
>>den=[1 3 3 1];
>>nyquist(num,den);
```

K=1

No pole of G(s)H(s) in RHP, so P=0;

Nyquist plot does not encircle (-1,j0), so N=0



Thus N=-P

The closed-loop system is stable

2. With Nyquist plot and Nyquist criterion:

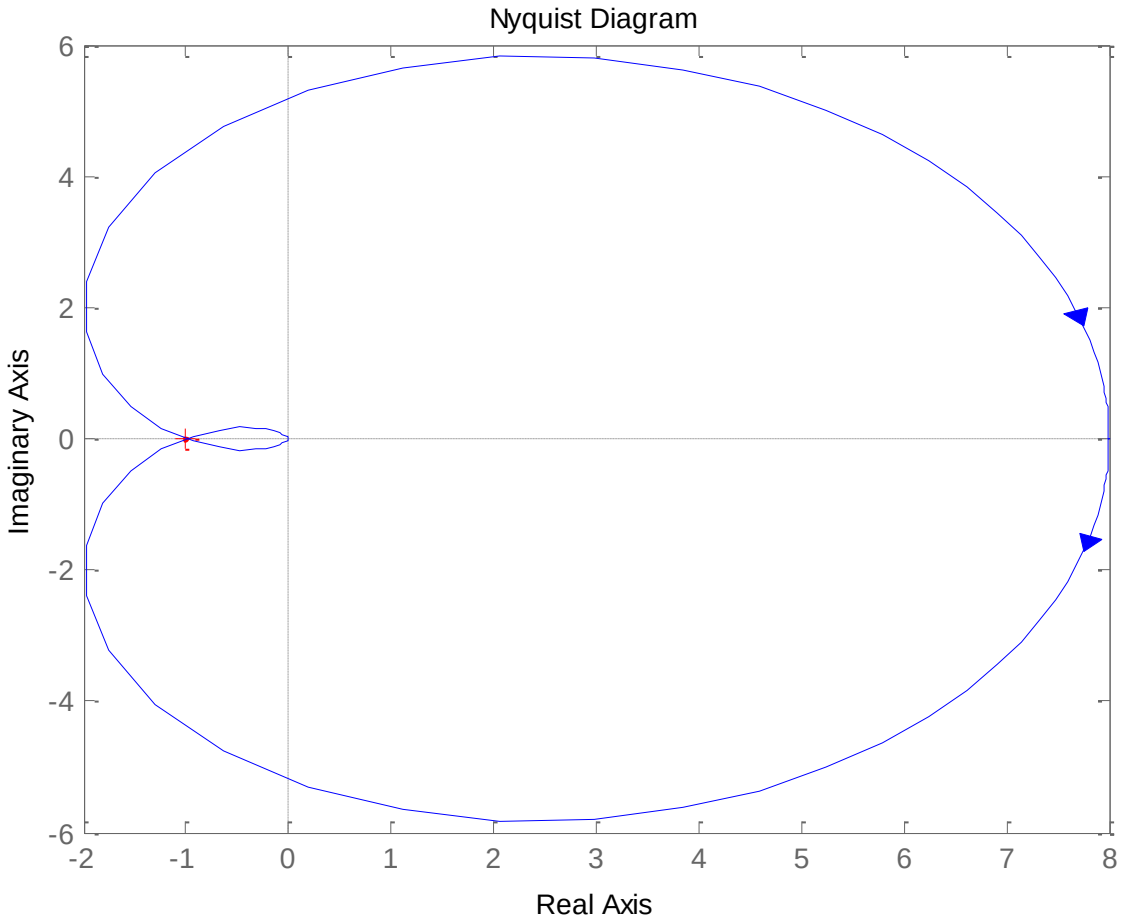
$$G(s)H(s) = \frac{5K}{(s+1)^3}$$

K=1.6

No pole of G(s)H(s) in RHP, so P=0;

The Nyquist plot just go through (-1,j0)

```
>>K=1.6;
>>num=5*K;
>>den=[1 3 3 1];
>>nyquist(num,den);
```



The closed-loop system is marginally stable

2. With Nyquist plot and Nyquist criterion:

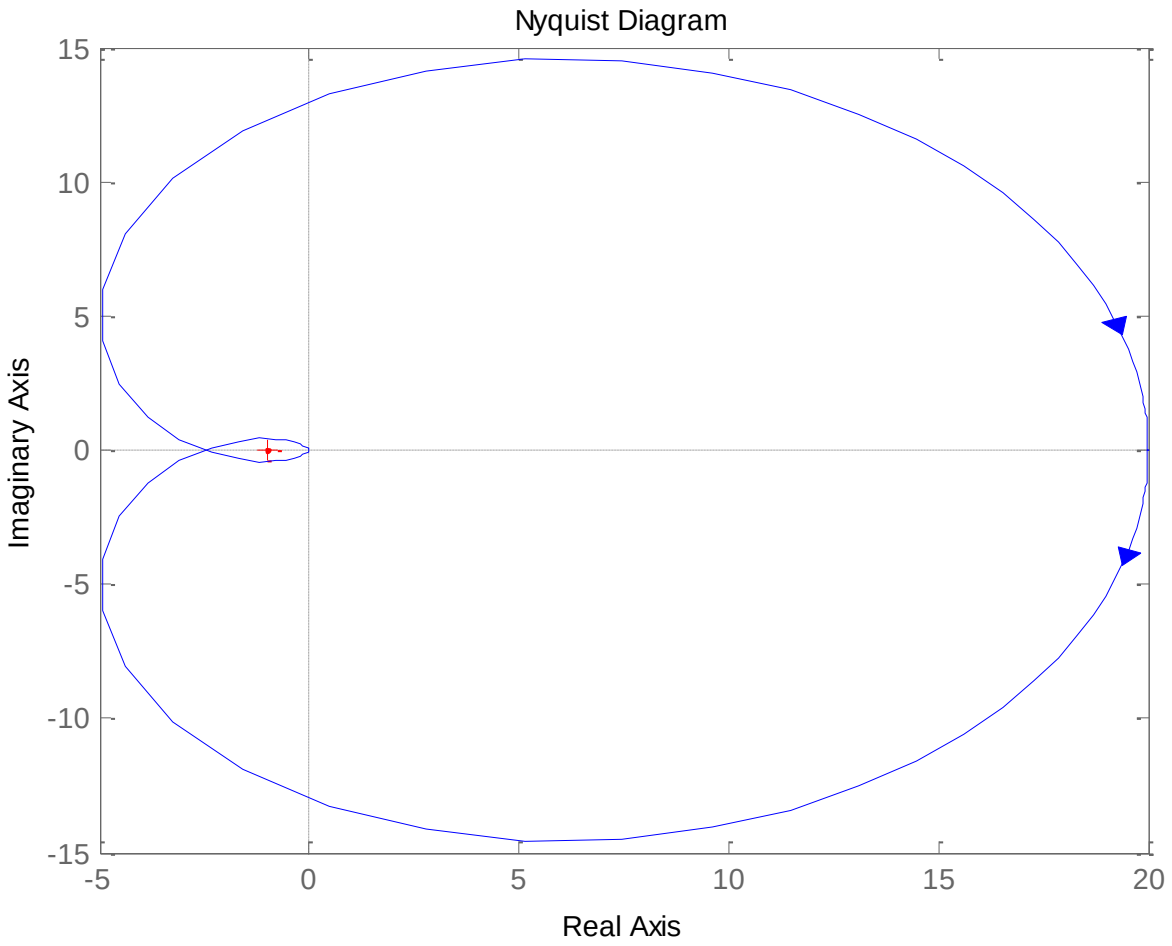
$$G(s)H(s) = \frac{5K}{(s+1)^3}$$

```
>>K=4;
>>num=5*K;
>>den=[1 3 3 1];
>>nyquist(num,den);
```

K=4

No pole of G(s)H(s) in RHP, so P=0;

Nyquist plot encircles (-1,j0) twice, so N=2



Thus Z=N+P=2

The closed-loop system has two poles in RHP and is unstable

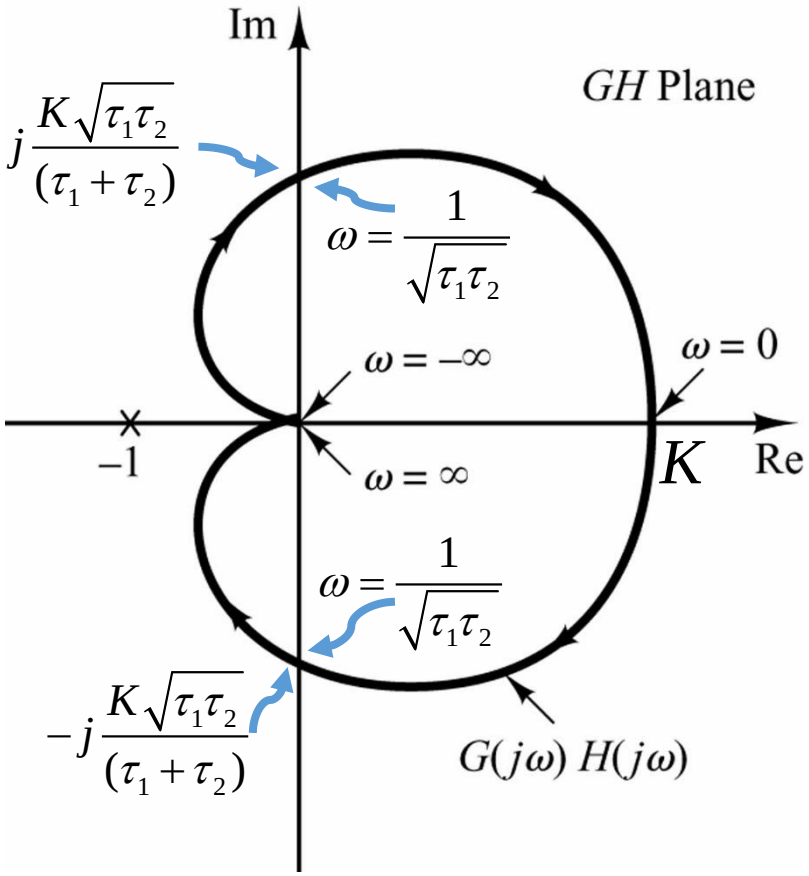
Example

Consider the system with OL TF (system with real poles)

$$G(s)H(s) = \frac{K}{(\tau_1 s+1)(\tau_2 s+1)}$$

$$G(j\omega)H(j\omega) = \frac{K}{(1+j\omega\tau_1)(1+j\omega\tau_2)}$$

$$= \frac{K}{(1-\omega^2\tau_1\tau_2)+j\omega(\tau_1+\tau_2)}$$



ω	0	$\frac{1}{\sqrt{\tau_1\tau_2}}$	∞
$\lvert GH\rvert$	K	$\frac{K\sqrt{\tau_1\tau_2}}{(\tau_1+\tau_2)}$	0
$\measuredangle GH$	0	-90°	-180°

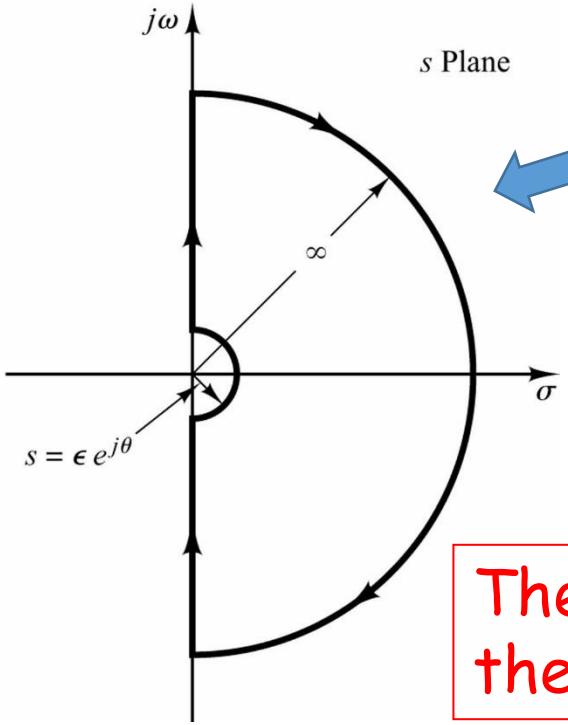
No OL poles in the RHP: P=0,
No encirclement of -1 by Γ_{GH} for any K>0: N=0,
N=-P=0 => The CL system is stable for all K>0.

Modified Nyquist Path Γ_s

The contour Γ_s in the s-plane should not pass through the singularities (i.e., poles and zeros) of G(s)H(s).

Therefore, if G(s)H(s) has poles and/or zeros on the jω-axis, then the contour Γ_s in the s-plane must be modified.

Usually, we use a semicircle of infinitesimal radius ε to overpass the singularity.

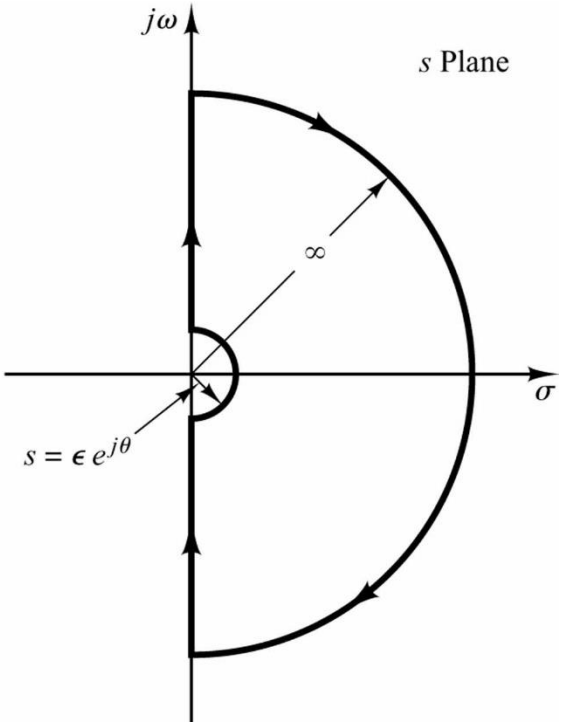


Avoiding poles or zeros at the origin using a small semicircle with radius ε around the origin.

Similarly, purely imaginary poles and zeros can be avoided using infinitesimal detour semicircles.

The stability criterion does not change for the modified path.

Modified Nyquist Path Γ_s



The points on the infinitesimal semicircle can be represented as

$$s = \varepsilon e^{j\theta}$$

where θ varies from -90° to +90°.

Similarly, the points on the outside (large) semicircle can be represented as

$$s = re^{j\theta}$$

where θ varies from +90° to -90°.

We have ε->0 and r->∞.

Consider the general OL TF

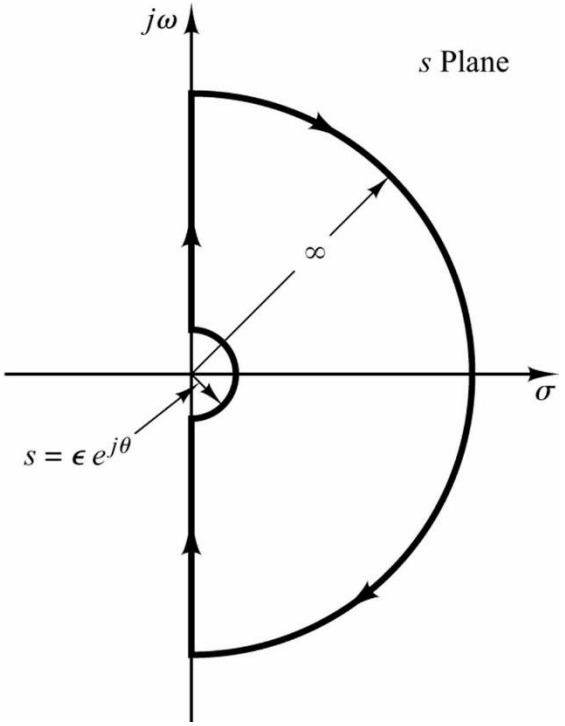
$$G(s)H(s) = \frac{K\prod_{i=1}^{Q}(1+T_i s)}{s^N \prod_{m=1}^{M}(1+\tau_m s)\prod_{k=1}^{R}\left(1+2\zeta_k\left(s/\omega_{n_k}\right)+\left(s/\omega_{n_k}\right)^2\right)}$$

m=Q, n=N+M+2R, K > 0, m < n

n – number of poles

m – number of zeros

Modified Nyquist Path Γ_s



At the detour semicircle around the origin (as ε->0) we have

$$\lim_{\varepsilon\to 0} G(s)H(s)\big|_{s=\varepsilon e^{j\theta}} = \lim_{\varepsilon\to 0}\frac{K}{\left(\varepsilon e^{j\theta}\right)^N}$$

$$= \lim_{\varepsilon\to 0}\frac{K}{\left(\varepsilon\right)^N} e^{-jN\times\theta}$$

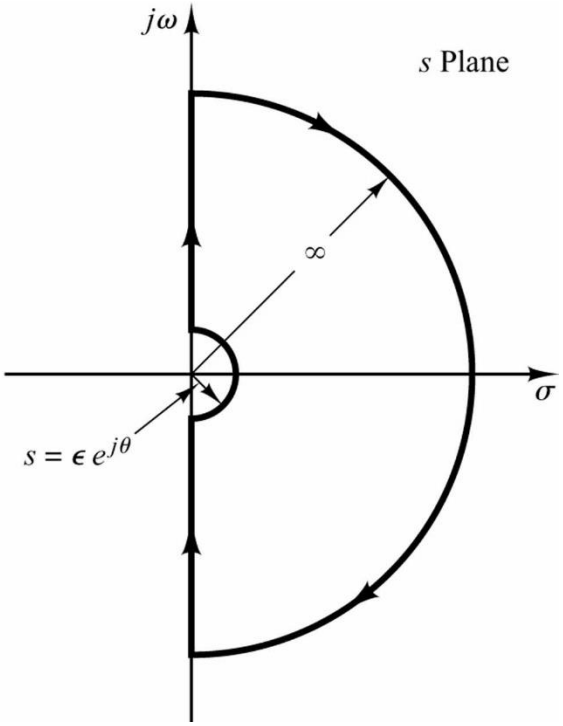
$$= \infty e^{-jN\times\theta}, \qquad -90^o < \theta < 90^o$$

Therefore, as ε->0

$$\left|G(s)H(s)\right|_{s=\varepsilon e^{j\theta}} \to \frac{K}{\left(\varepsilon\right)^N} \to \infty$$

and the angle varies from $+N\times 90^o$ to $-N\times 90^o$ => it is an arc with infinite radius between the specified angles.

Modified Nyquist Path Γ_s



At the portion of the contour which is a semicircle with infinite radius (r->∞) we have

$$\lim_{r\to\infty} G(s)H(s)\big|_{s=re^{j\theta}} = \lim_{r\to\infty} \frac{K}{\left(re^{j\theta}\right)^{n-m}}$$

$$= \lim_{r\to\infty} \frac{K}{\left(r\right)^{n-m}} e^{-j(n-m)\times\theta}$$

$$= 0e^{-j(n-m)\times\theta}, \qquad +90^o < \theta < -90^o$$

Therefore, as r->∞

$$\left|G(s)H(s)\right|_{s=re^{j\theta}} \to \frac{K}{\left(r\right)^{n-m}} \to 0$$

and the angle varies from $-(n-m)\times 90^o$ to $+(n-m)\times 90^o$

Example Consider the system with OL TF (system with two poles, one at the origin)

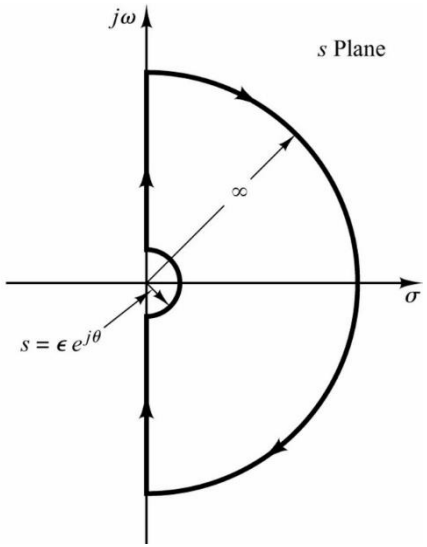
$$G(s)H(s) = \frac{K}{s(\tau s + 1)}$$

$$G(j\omega)H(j\omega) = \frac{K}{(j\omega)(1 + j\omega\tau)}$$

$$= \frac{K}{-\omega^2\tau + j\omega}$$

As ω varies from 0⁺ to +∞ (s=jω)

ω	0⁺	$1/\tau$	∞
$\lvert GH \rvert$	∞	$\frac{K\tau}{\sqrt{2}}$	0
$\measuredangle GH$	-90°	-135°	-180°



At the small semicircle around the origin we have

$$\lvert G(s)H(s) \rvert = \infty$$

and $\measuredangle G(s)H(s)$ varies from +90° to -90°.

At the semicircle with infinite radius we have

$$\lvert G(s)H(s) \rvert = 0$$

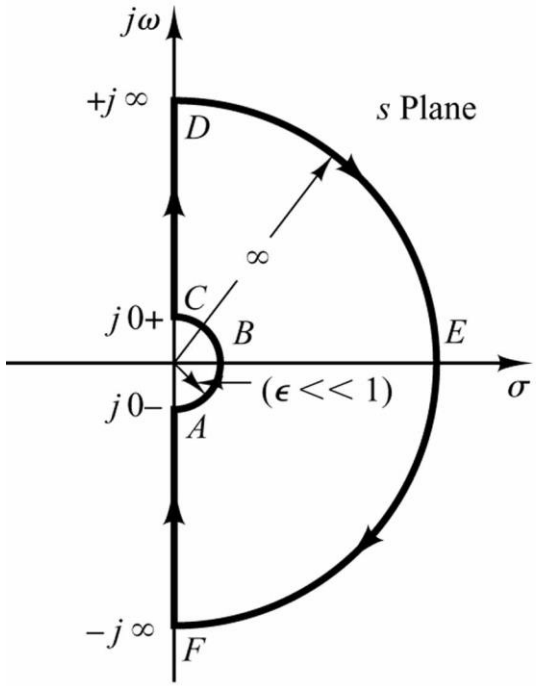
and $\measuredangle G(s)H(s)$ varies from -180° to +180°.

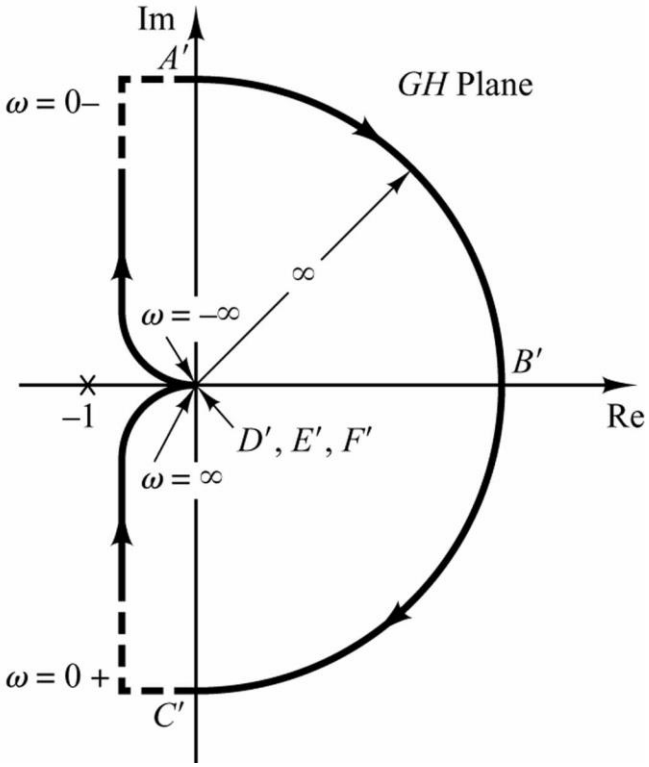
Example (system with two poles, one at the origin)

$$G(s)H(s) = \frac{K}{s(\tau s + 1)}$$

$$G(j\omega)H(j\omega) = \frac{K}{-\omega^2\tau + j\omega}$$

ω	0^+	$1/\tau$	∞
$\lvert GH \rvert$	∞	$\frac{K\tau}{\sqrt{2}}$	0
$\measuredangle GH$	-90°	-135°	-180°





No OL poles in the RHP: P=0,

No encirclement of -1 by Γ_{GH} for any K>0: N=0,

N=-P=0 => The CL system is stable for all K>0.

Example Consider the system with OL TF (system with three poles, one at the origin)

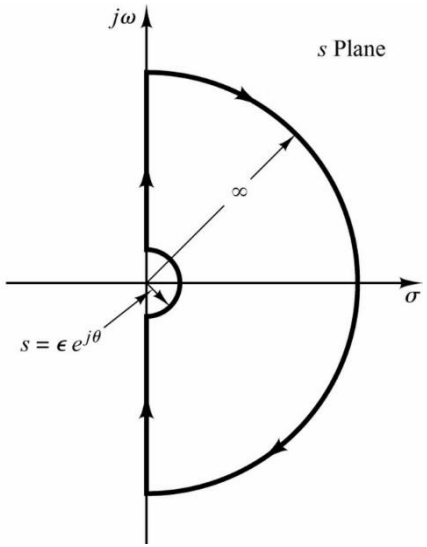
$$G(s)H(s)=\frac{K}{s(\tau_1 s+1)(\tau_2 s+1)}$$

$$G(j\omega)H(j\omega)=\frac{K}{(j\omega)(1+j\omega\tau_1)(1+j\omega\tau_2)}$$

$$=\frac{K}{-\omega^2(\tau_1+\tau_2)+j\omega(1-\omega^2\tau_1\tau_2)}$$

As ω varies from 0⁺ to +∞ (s=jω)

ω	0^+	$\frac{1}{\sqrt{\tau_1\tau_2}}$	∞
$\lvert GH\rvert$	∞	$\frac{K\tau_1\tau_2}{(\tau_1+\tau_2)}$	0
$\measuredangle GH$	-90°	-180°	-270°



At the small semicircle around the origin we have

$$\lvert G(s)H(s)\rvert=\infty$$

and $\measuredangle G(s)H(s)$ varies from +90° to -90°.

At the semicircle with infinite radius we have

$$\lvert G(s)H(s)\rvert=0$$

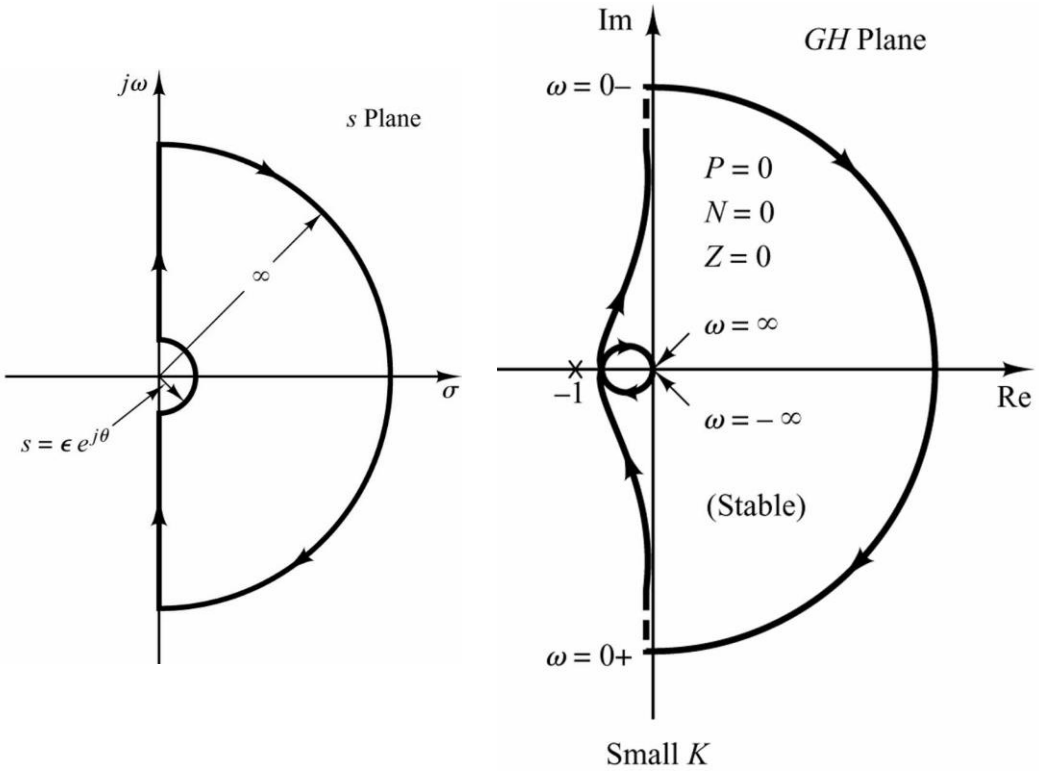
and $\measuredangle G(s)H(s)$ varies from -270° to +270°.

Example (system with three poles, one at the origin)

$$G(s)H(s)=\frac{K}{s(\tau_1 s+1)(\tau_2 s+1)}$$

$$G(j\omega)H(j\omega)=\frac{K}{-\omega^2(\tau_1+\tau_2)+j\omega(1-\omega^2\tau_1\tau_2)}$$

ω	0^+	$\frac{1}{\sqrt{\tau_1\tau_2}}$	∞
$\lvert GH\rvert$	∞	$\frac{K\tau_1\tau_2}{(\tau_1+\tau_2)}$	0
$\measuredangle GH$	-90°	-180°	-270°



At the small semicircle around the origin we have

$$\lvert G(s)H(s)\rvert=\infty$$

and $\measuredangle G(s)H(s)$ varies from +90° to –90°.

At the semicircle with infinite radius we have

$$\lvert G(s)H(s)\rvert=0$$

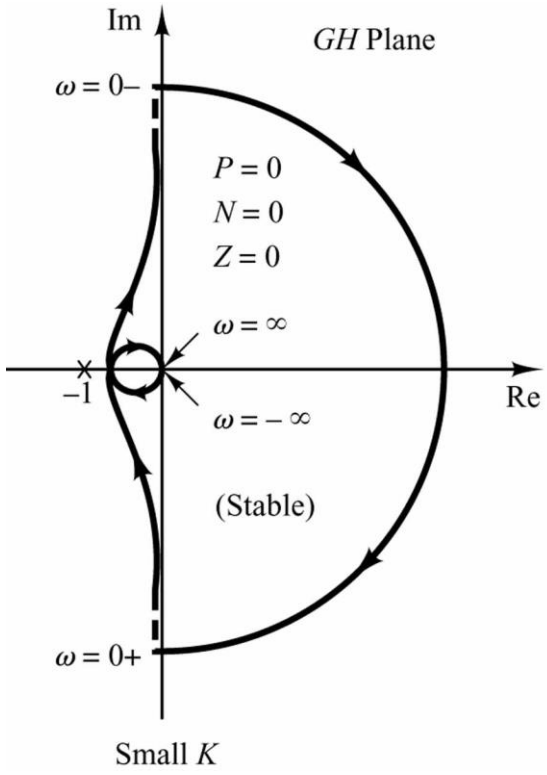
and $\measuredangle G(s)H(s)$ varies from -270° to +270°.

Example (system with three poles, one at the origin)

$$G(s)H(s) = \frac{K}{s(\tau_1 s+1)(\tau_2 s+1)}$$

$$G(j\omega)H(j\omega) = \frac{K}{-\omega^2(\tau_1+\tau_2)+j\omega(1-\omega^2\tau_1\tau_2)}$$

ω	0^+	$\frac{1}{\sqrt{\tau_1\tau_2}}$	∞
$\lvert GH\rvert$	∞	$\frac{K\tau_1\tau_2}{(\tau_1+\tau_2)}$	0
$\measuredangle GH$	-90°	-180°	-270°



No OL poles in the RHP: P=0,

If $-\frac{K\tau_1\tau_2}{(\tau_1+\tau_2)} > -1 \Rightarrow K < \frac{\tau_1+\tau_2}{\tau_1\tau_2}$

Then no encirclement of -1 by Γ_{GH} : N=0,

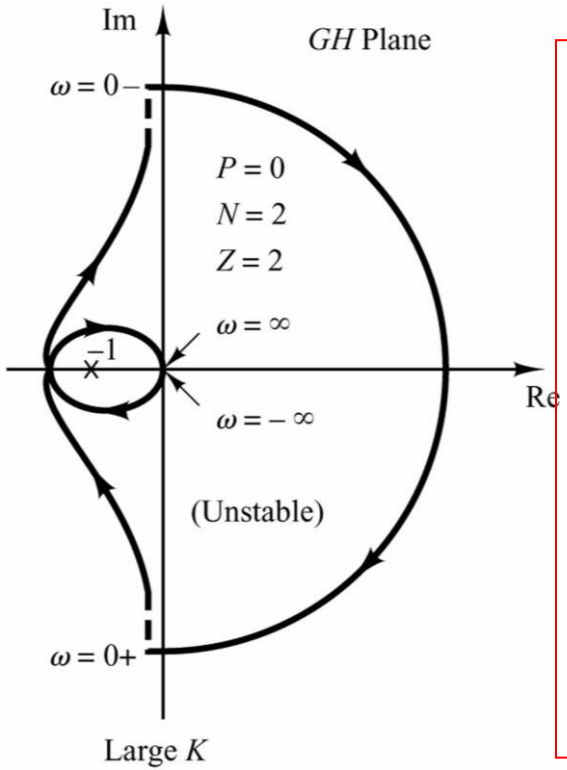
N=-P=0 => The CL system is stable.

Example (system with three poles, one at the origin)

$$G(s)H(s)=\frac{K}{s(\tau_1 s+1)(\tau_2 s+1)}$$

$$G(j\omega)H(j\omega)=\frac{K}{-\omega^2(\tau_1+\tau_2)+j\omega(1-\omega^2\tau_1\tau_2)}$$

ω	0^+	$\frac{1}{\sqrt{\tau_1\tau_2}}$	∞
$\lvert GH\rvert$	∞	$\frac{K\tau_1\tau_2}{(\tau_1+\tau_2)}$	0
$\measuredangle GH$	-90°	-180°	-270°



No OL poles in the RHP: P=0,

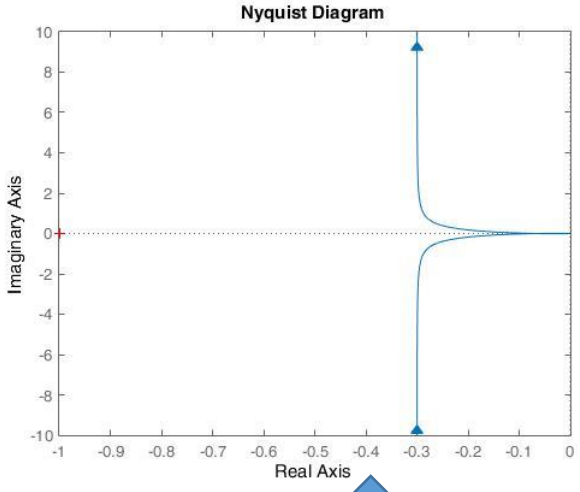
If $-\frac{K\tau_1\tau_2}{(\tau_1+\tau_2)}<-1 \Rightarrow K>\frac{\tau_1+\tau_2}{\tau_1\tau_2}$

Then Γ_{GH} encircles -1 2 times in the CW direction: N=2,

Z=N+P=2 => The CL system has Z=2 poles in the RHP and therefore is unstable.

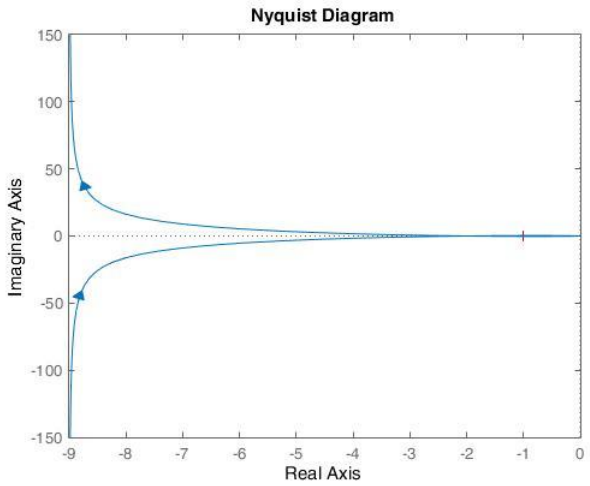
Example (system with three poles, one at the origin)

$$G(s)H(s)=\frac{K}{s(\tau_1 s+1)(\tau_2 s+1)},\ \tau_1=0.1,\ \tau_2=0.2,\ \frac{\tau_1+\tau_2}{\tau_1\tau_2}=15$$

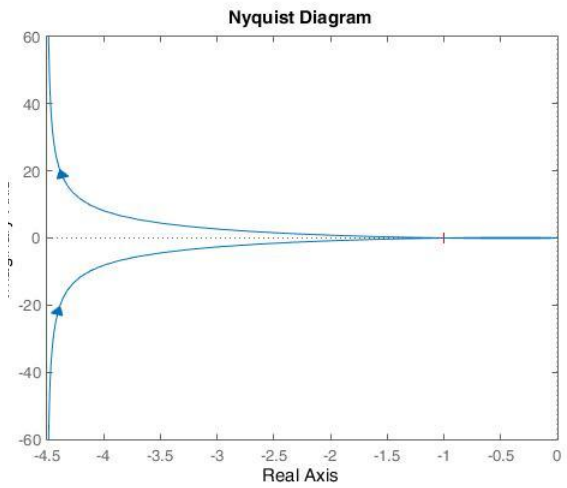


$K=1$

P=0, N=0, Z=0 stable

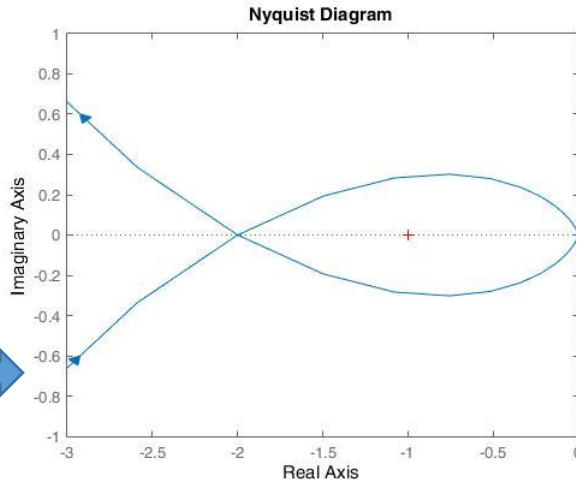


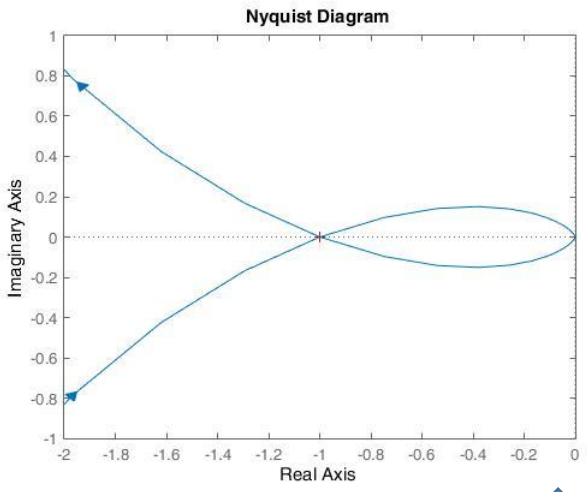
$K=30$



$K=15$

P=0, N=2, Z=2 unstable, two poles in the RHP





Γ_{GH} passes through the point -1 => marginally stable

Example Consider the system with OL TF (system with three poles, two at the origin)

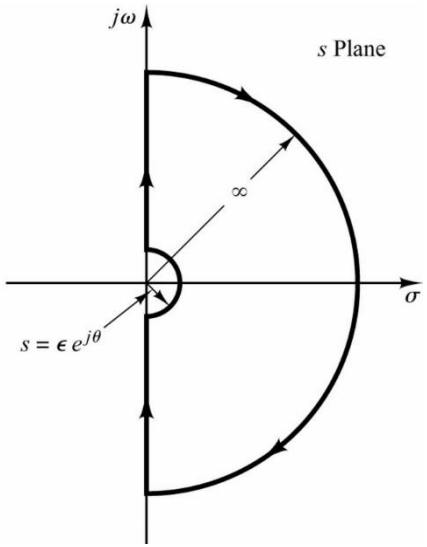
$$G(s)H(s) = \frac{K}{s^2(\tau s + 1)}$$

$$G(j\omega)H(j\omega) = \frac{K}{(j\omega)^2(1 + j\omega\tau)}$$

$$= \frac{-\left(K/\omega^2\right)}{1 + j\omega\tau}$$

As ω varies from 0⁺ to +∞ (s=jω)

ω	0⁺	$1/\tau$	∞
$\lvert GH \rvert$	∞	$\frac{K\tau^2}{\sqrt{2}}$	0
$\measuredangle GH$	-180°	-225°	-270°



At the small semicircle around the origin we have

$$\lvert G(s)H(s) \rvert = \infty$$

and $\measuredangle G(s)H(s)$ varies from +180° to -180°.

At the semicircle with infinite radius we have

$$\lvert G(s)H(s) \rvert = 0$$

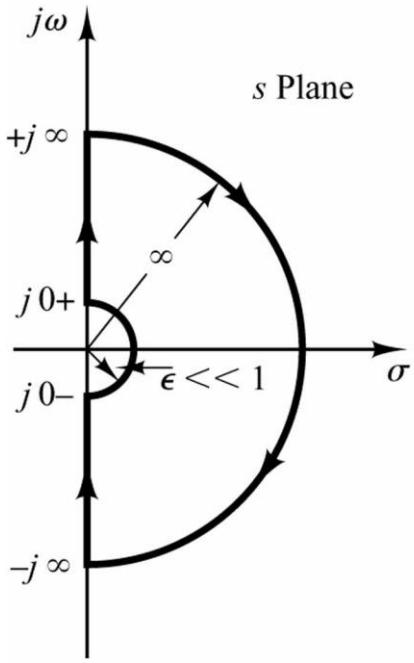
and $\measuredangle G(s)H(s)$ varies from -270° to +270°.

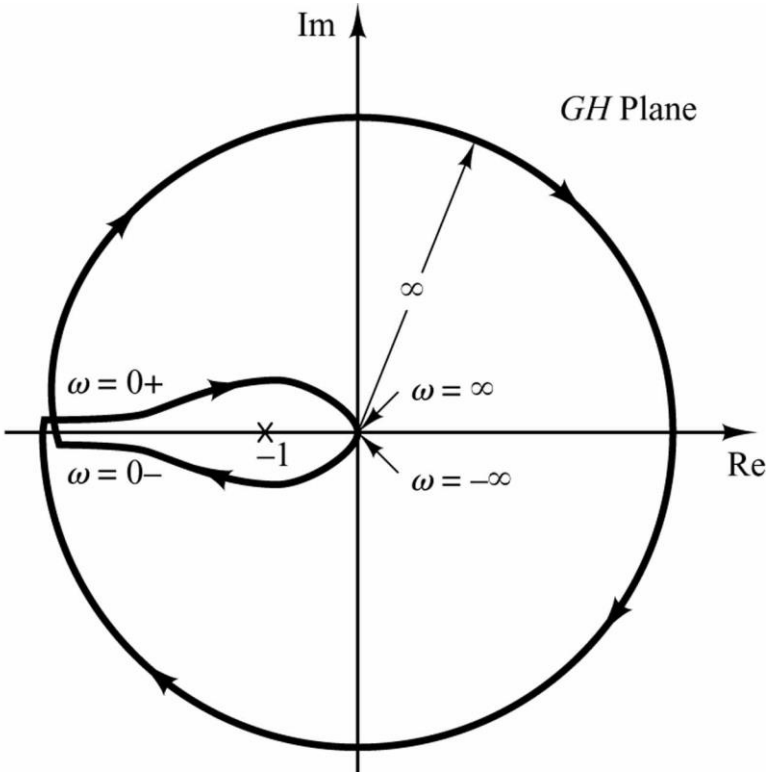
Example (system with three poles, two at the origin)

$$G(s)H(s) = \frac{K}{s^2(\tau s + 1)}$$

$$G(j\omega)H(j\omega) = \frac{-\left(K/\omega^2\right)}{1 + j\omega\tau}$$

ω	0^+	$1/\tau$	∞
$\lvert GH \rvert$	∞	$\frac{K\tau^2}{\sqrt{2}}$	0
$\sphericalangle GH$	-180°	-225°	-270°





No OL poles in the RHP: P=0,

Γ_{GH} encircles -1 two time in CW direction: N=2,

Z=N+P=2 => The CL system has two poles in the RHP and is always unstable.

Example Consider the system with OL TF (system with one pole in the RHP)

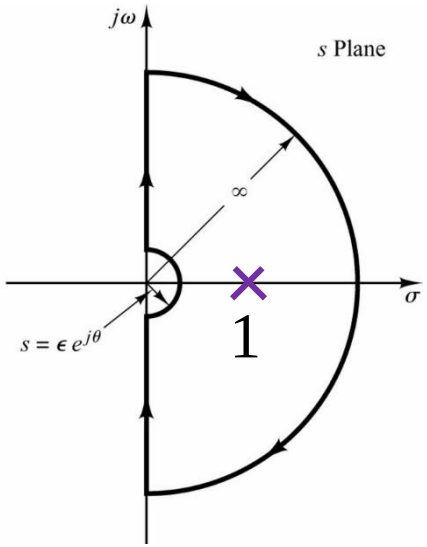
$$G(s)H(s)=\frac{K(s+3)}{s(s-1)}$$

$$G(j\omega)H(j\omega)=\frac{K(3+j\omega)}{(j\omega)(-1+j\omega)}$$

$$=-\frac{4K}{\omega^2+1}-j\frac{K(\omega^2-3)}{\omega(\omega^2+1)}$$

As ω varies from 0⁺ to +∞ (s=jω)

ω	0⁺	$\sqrt{3}$	∞
$\lvert GH\rvert$	∞	K	0
$\measuredangle GH$	+90° (-270°)	+180° (-180°)	+270° (-90°)



At the small semicircle around the origin we have

$$\lim_{\varepsilon\to 0}\left|G(s)H(s)\right|_{s=\varepsilon e^{j\theta}}=\lim_{\varepsilon\to 0}\left(\frac{-3K}{\varepsilon e^{j\theta}}\right)$$

$$=\infty e^{-j(180^o+\theta)}$$

At the semicircle with infinite radius we have

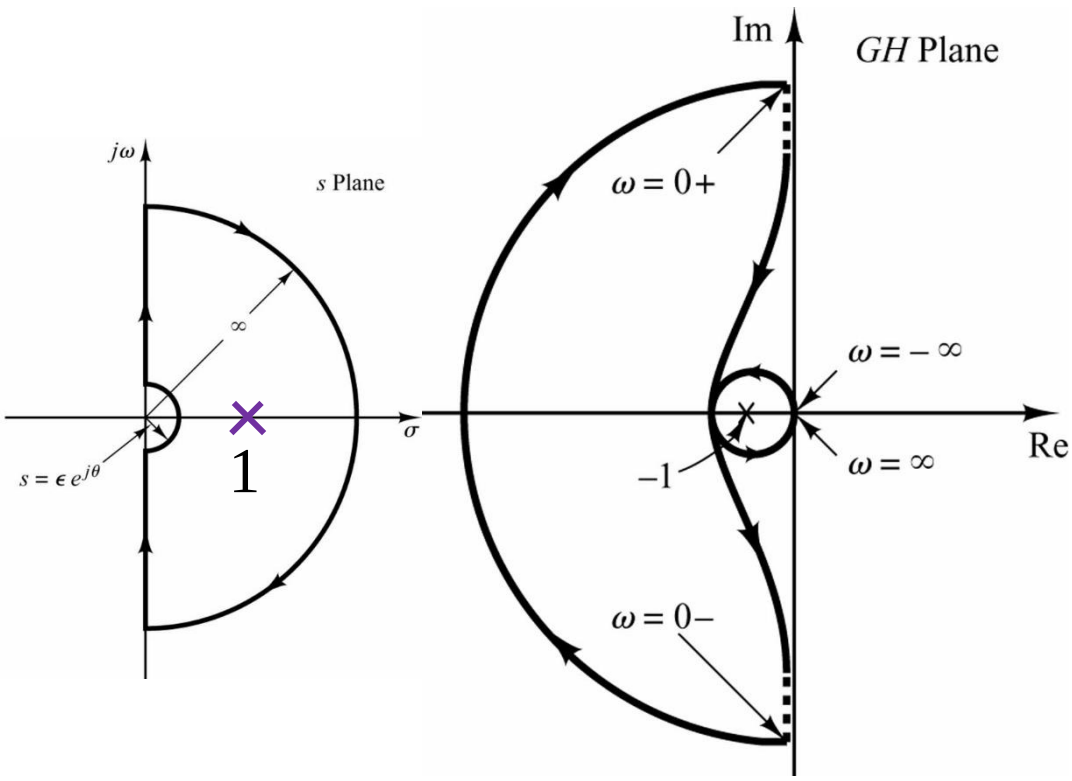
$$\lim_{r\to\infty}\left|G(s)H(s)\right|_{s=re^{j\theta}}=\lim_{r\to\infty}\left(\frac{K}{re^{j\theta}}\right)=0e^{-j\theta}$$

Example (system with one pole in the RHP)

$$G(s)H(s)=\frac{K(s+3)}{s(s-1)}$$

$$G(j\omega)H(j\omega)=-\frac{4K}{\omega^2+1}-j\frac{K(\omega^2-3)}{\omega(\omega^2+1)}$$

ω	0^+	$\sqrt{3}$	∞
$\lvert GH\rvert$	∞	K	0
$\measuredangle GH$	+90° (-270°)	+180° (-180°)	+270° (-90°)



At the small semicircle around the origin we have

$$\lvert G(s)H(s)\rvert=\infty$$

and $\measuredangle G(s)H(s)$ varies from -90° to -270°.

At the semicircle with infinite radius we have

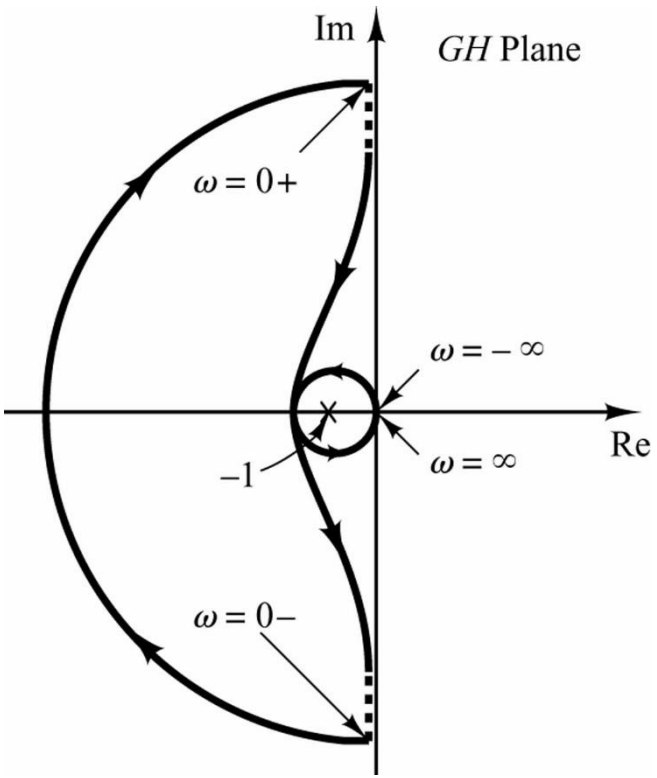
$$\lvert G(s)H(s)\rvert=0$$

and $\measuredangle G(s)H(s)$ varies from -90° to +90°.

Example (system with one pole in the RHP)

$$G(j\omega)H(j\omega) = -\frac{4K}{\omega^2+1} - j\frac{K(\omega^2-3)}{\omega(\omega^2+1)}$$

ω	0^+	$\sqrt{3}$	∞
$\lvert GH \rvert$	∞	K	0
$\measuredangle GH$	+90° (-270°)	+180° (-180°)	+270° (-90°)



The OL system has one pole in the RHP: P=1,
=> For stability we need one encirclement of -1 by Γ_{GH} in the CCW direction.

If K > 1 => N=-1 => the CL system is stable

If K < 1 => N=1 => Z=N+P=2 => The CL system has two poles in the RHP and is therefore unstable.

Nyquist Criterion for Systems with Minimum-Phase OL Transfer Functions

Recall that a minimum-phase transfer function has all its poles and zeros in the LHP.

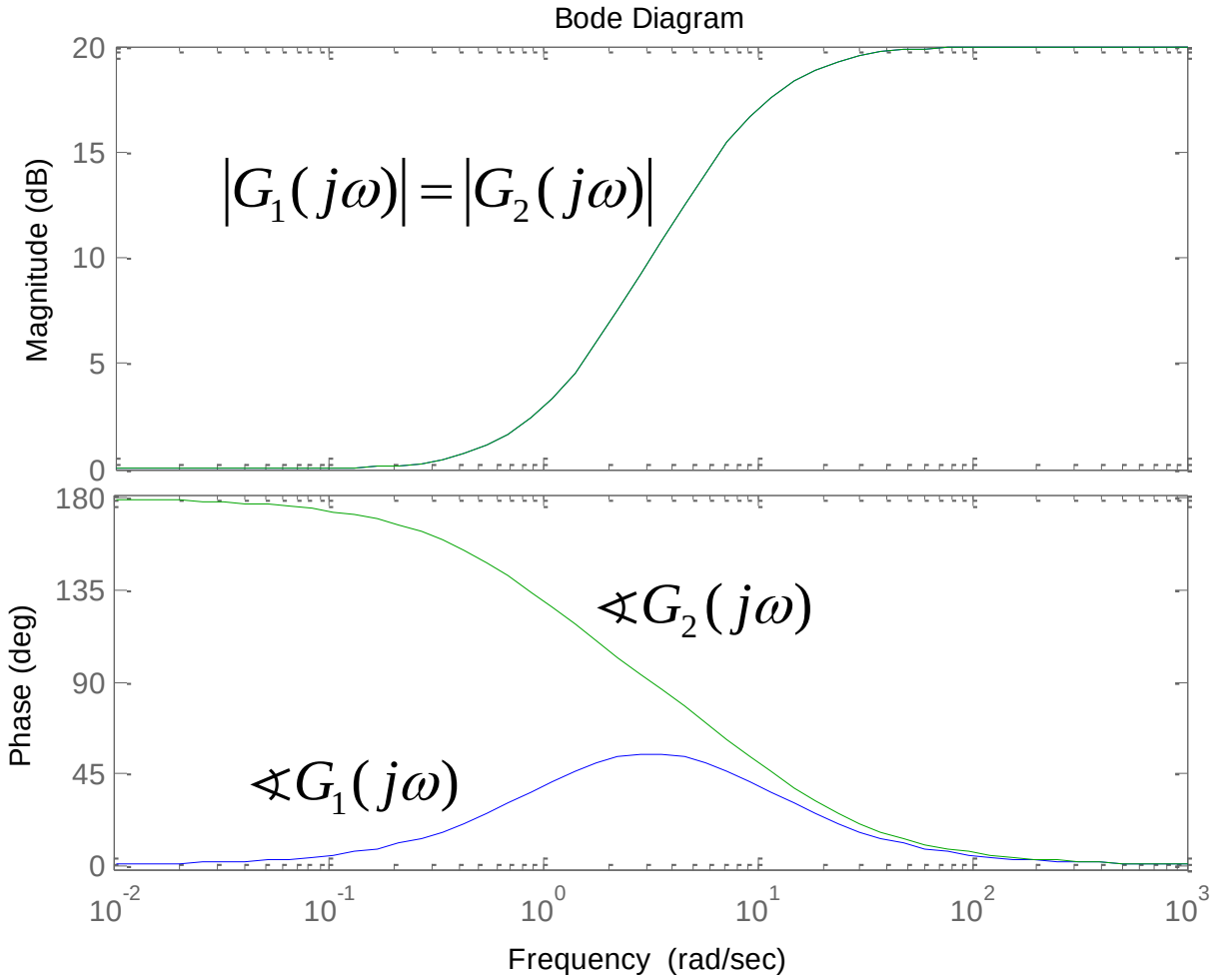
In other words, a minimum-phase transfer function does not have poles or zeros in the right-half s-plane or on the jω-axis, except a simple pole at s=0.

Consider the transfer functions

$$G_1(s) = \frac{10(s+1)}{(s+10)} \qquad G_2(s) = \frac{10(s-1)}{(s+10)}$$

Both transfer functions have the same magnitude for all frequencies $|G_1(j\omega)| = |G_2(j\omega)|$

But the phases of the two transfer functions are drastically different.



A minimum-phase system (all zeros and poles in the LHP) with a given magnitude curve will produce the smallest change in the associated phase, among the TF with the same magnitude curve.

Consider the loop transfer function:

$$L(s) = G(s)H(s)$$

If L(s) is minimum-phase, that is, L(s) does not have any poles or zeros in the right-half plane or on the jω-axis, except at s=0

Then P=0, where P is the number of poles of $\Delta(s)=1+G(s)H(s)$, which has the same poles as L(s).

Thus, the Nyquist criterion (N=-P) for a system with L(s) being minimum-phase is simplified to

$$\boxed{N = 0}$$

Nyquist criterion for systems with minimum-phase loop transfer function

$$N = 0$$

For a closed-loop system with minimum-phase loop transfer function L(s), the system is closed-loop stable, if the Nyquist plot (L(s) plot) that corresponds to the Nyquist path does not enclose the (-1,j0) point. If (-1,j0) is enclosed by the Nyquist plot, the system is unstable.

The stability property can be checked by plotting the segment of L(jω) from ω=0 to ∞.

Example Consider a unity feedback system with the loop transfer function

$$L(s) = G(s)H(s) = \frac{K}{s(s+2)(s+10)}$$

Analyze the stability of the closed-loop system.

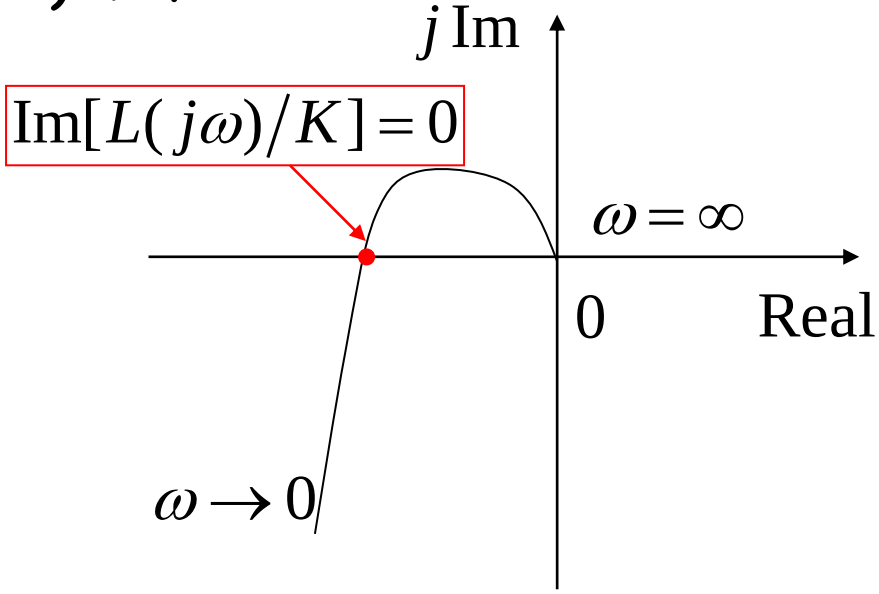
Solution.

Since L(s) is minimum-phase, we can analyze the closed-loop stability by investigating whether the Nyquist plot enclose the critical point (-1,j0) for L(jω)/K first.

$$\frac{L(j\omega)}{K} = \frac{1}{j\omega(j\omega+2)(j\omega+10)}$$

ω=∞: $\dfrac{L(j\infty)}{K} = 0\angle -270^\circ$

ω=0⁺: $\dfrac{L(j0)}{K} = \infty\angle -90^\circ$



$$\mathrm{Im}[L(j\omega)/K] = \mathrm{Im}[\frac{1}{j\omega(j\omega+2)(j\omega+10)}] = 0 \Rightarrow \omega = \pm\sqrt{20}\ rad/s$$

The frequency is positive, so $\omega = \sqrt{20}\ rad/s$

$$L(j\sqrt{20})/K = \frac{1}{j\sqrt{20}(j\sqrt{20}+2)(j\sqrt{20}+10)} = -0.004167$$

1. $K < 240 \Rightarrow L(j\sqrt{20}) > -1$

the Nyquist plot does not enclose (-1,jω); stable

2. $K = 240 \Rightarrow L(j\sqrt{20}) = -1$

the Nyquist plot goes through (-1,jω); marginally stable

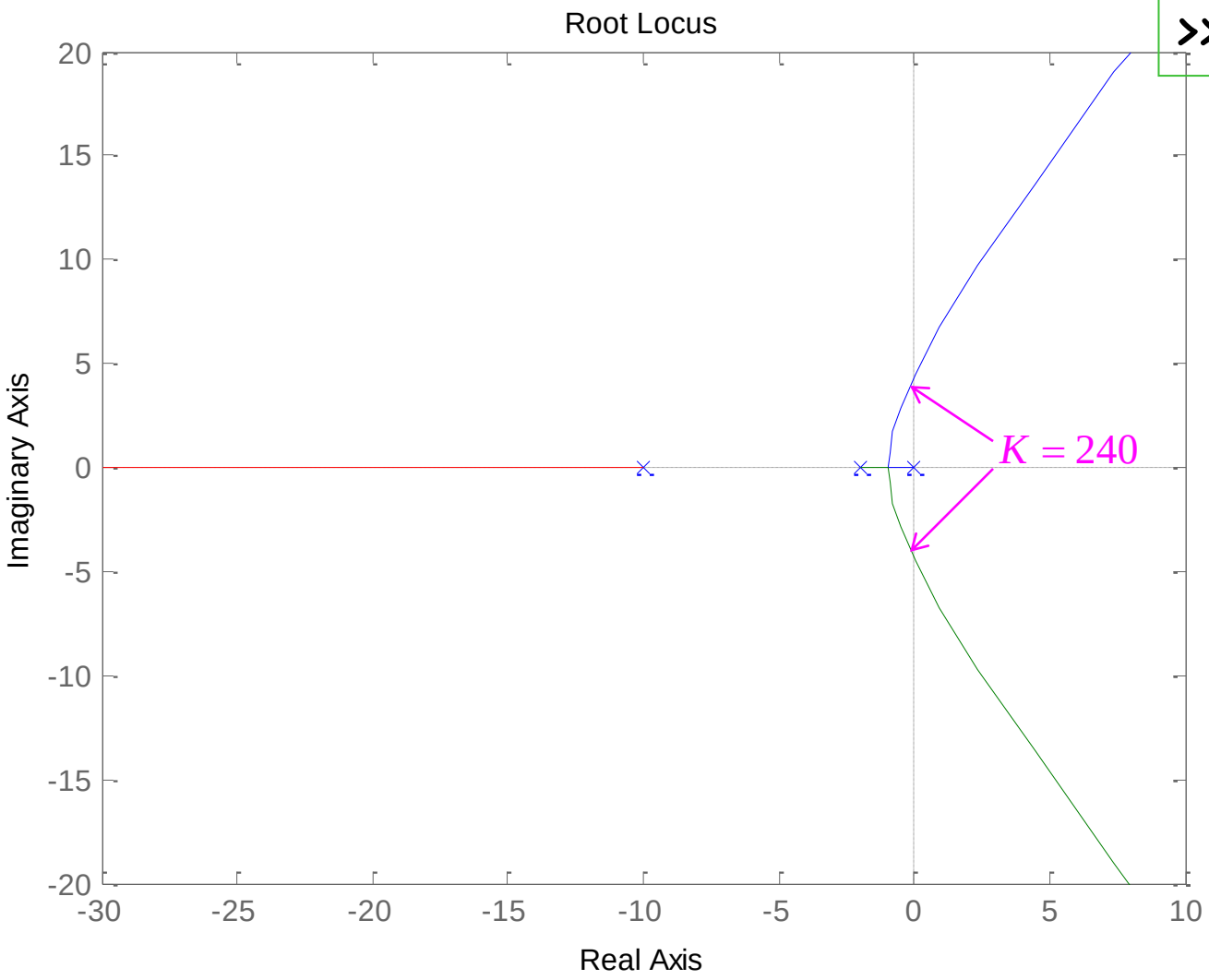
3. $K > 240 \Rightarrow L(j\sqrt{20}) < -1$

the Nyquist plot encloses (-1,jω). unstable

Utilizing the root locus technique

$$L(s) = G(s)H(s) = K\frac{1}{s(s+2)(s+10)}$$

```
>>z=[]
>>p=[0, -2, -10];
>>k=1
>>sys=zpk(z,p,k);
>>rlocus(sys);
```



Relative Stability and the Nyquist Criterion

Recall that relative stability indicates the degree of stability of a stable system. In other words, it describes how stable the system is and how close is it to instability.

In time-domain, the relative stability is measured in terms of the location of the CL poles relative to the jω-axis.

The farther left the CL poles from the jω-axis the more stable the system is.

Another indication for relative stability is the maximum overshoot (and the damping ratio).

More oscillatory systems with smaller damping are usually relatively less stable than the systems with less oscillations and larger damping.

The Nyquist Criterion can also be used to determine the relative stability of a given system.

Relative Stability and the Nyquist Criterion

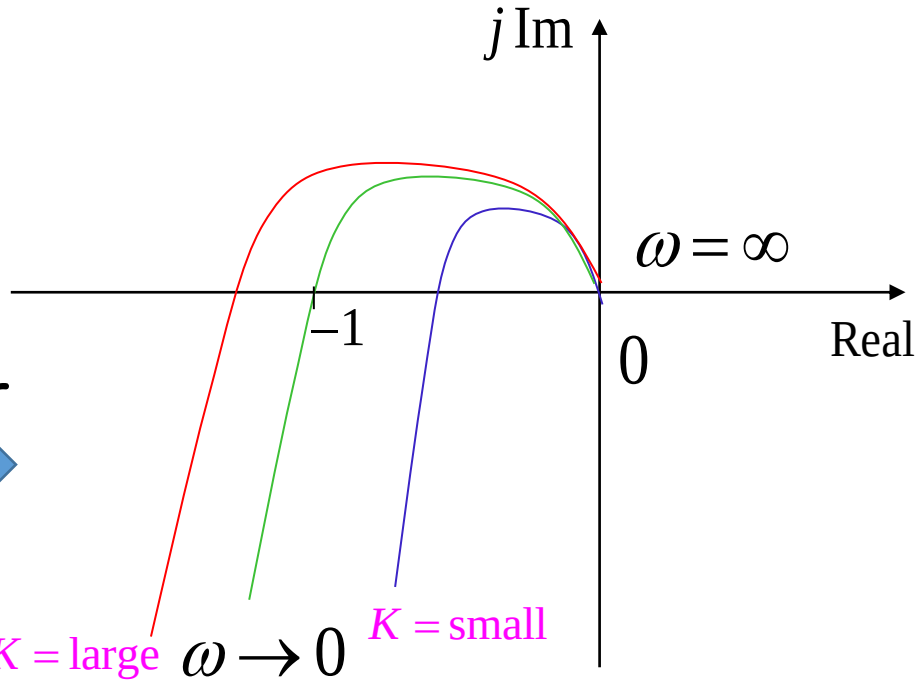
In frequency-domain, the relative stability is measured by resonance peak and how close the Nyquist plot of L(jω) is to enclosing the (-1,j0) point.

In discussing relative stability, we will mostly deal with systems with minimum phase OL TF, i.e., systems whose OL TFs do not have neither poles nor zeros in the RHP.

Consider the system

$$G(j\omega)H(j\omega) = \frac{K}{(j\omega)(1+j\omega\tau_1)(1+j\omega\tau_2)}$$

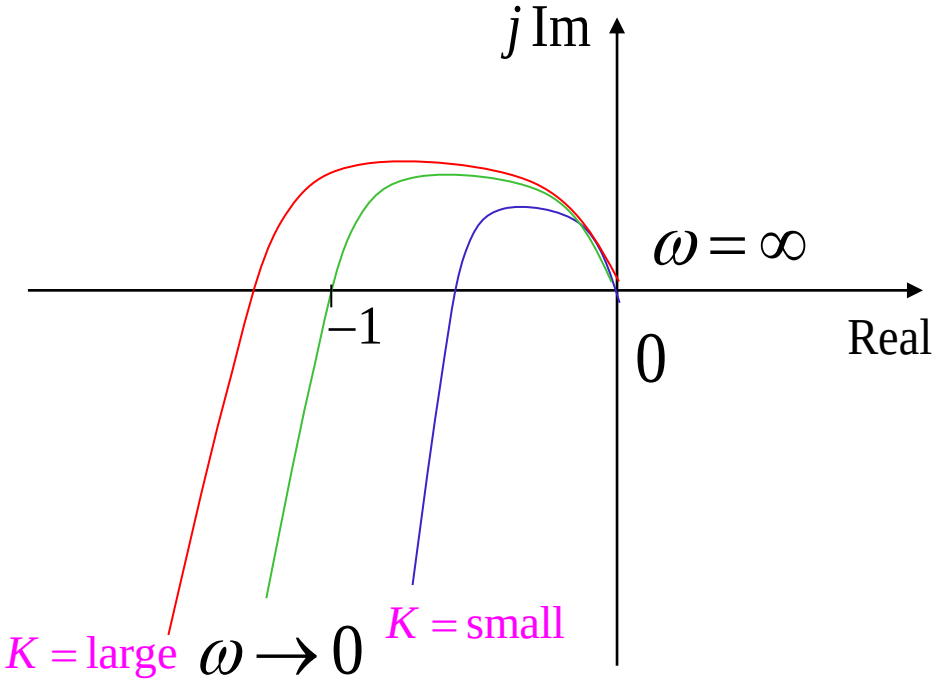
The Nyquist plots for different values of K are given by



Relative Stability and the Nyquist Criterion

The plot crosses the negative real axis at

$$u = -\frac{K\tau_1\tau_2}{(\tau_1+\tau_2)}$$



For $K < \dfrac{\tau_1+\tau_2}{\tau_1\tau_2}$

the system is stable

For $K = \dfrac{\tau_1+\tau_2}{\tau_1\tau_2}$

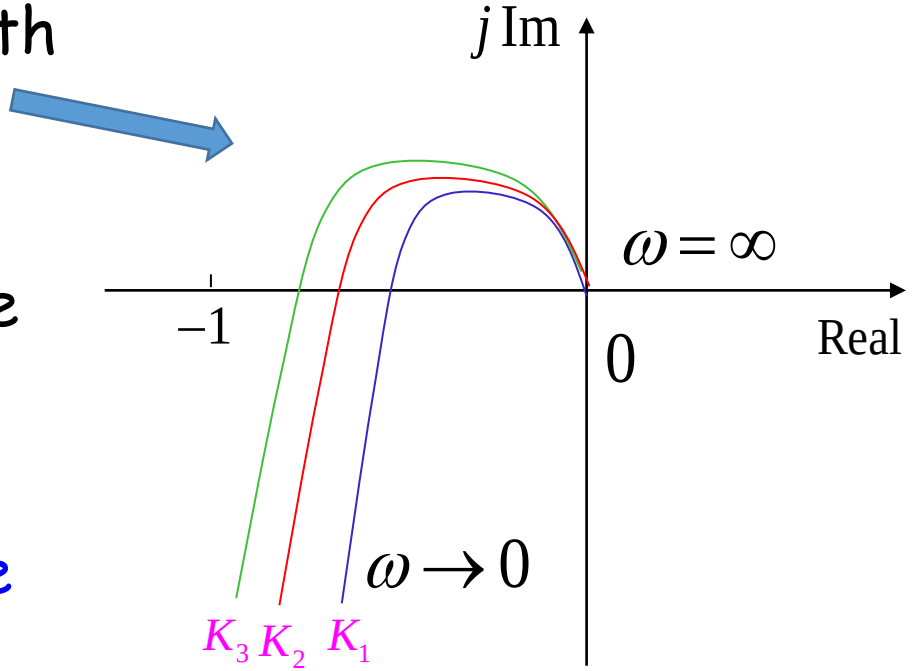
the system is marginally stable

For $K > \dfrac{\tau_1+\tau_2}{\tau_1\tau_2}$

the system is unstable

Relative Stability and the Nyquist Criterion

Consider the stable systems with Nyquist plots



In general, as G(jω)H(jω) gets closer to encircling -1, the more oscillatory and relatively less stable the system becomes.

The system with the blue curve has the best relative stability among the three systems.

The system with the green curve has the worst relative stability among the three systems.

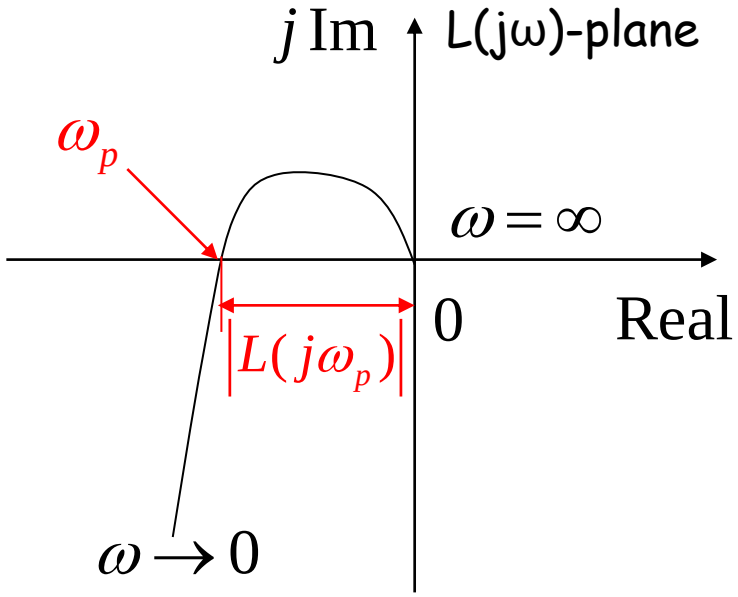
Two quantities, the gain margin and the phase margin, are usually used to measure the relative stability of a given system in the frequency domain.

Gain Margin (GM)
(for minimum-phase loop transfer functions)

Phase crossover

Phase crossover frequency ω_p

$$\measuredangle L(j\omega_p) = \measuredangle G(j\omega_p)H(j\omega_p) = -180^\circ$$



The gain margin K_g for a closed-loop system with loop transfer function L(jω)=G(jω)H(jω) is defined as

$$K_g = \frac{1}{|L(j\omega_p)|} = \frac{1}{|G(j\omega_p)H(j\omega_p)|}$$

The gain margin in dB is given by

$$\text{GM} = 20\log_{10} K_g = 20\log_{10} \frac{1}{|L(j\omega_p)|} = -20\log_{10} |L(j\omega_p)| \text{ dB}$$

$$\text{GM} = 20\log_{10} K_g = 20\log_{10} \frac{1}{\left|L(j\omega_p)\right|} = -20\log_{10} \left|L(j\omega_p)\right| \text{ dB}$$

If $\left|L(j\omega_p)\right| < 1, (K_g > 1)$ (stable), $\log_{10} \left|L(j\omega_p)\right| < 0 \Rightarrow$ GM > 0

As $\left|L(j\omega_p)\right| \downarrow$ (closer to the origin) $\Rightarrow$ GM $\uparrow$ (more stable)

As $\left|L(j\omega_p)\right| \uparrow$ (closer to -1) $\Rightarrow$ GM $\downarrow$ (less stable)

If $\left|L(j\omega_p)\right| = 1, (K_g = 1)$ (marginally stable), $\log_{10} \left|L(j\omega_p)\right| = 0 \Rightarrow$ GM=0

If $\left|L(j\omega_p)\right| > 1, (K_g < 1)$ (unstable), $\log_{10} \left|L(j\omega_p)\right| > 0 \Rightarrow$ GM<0

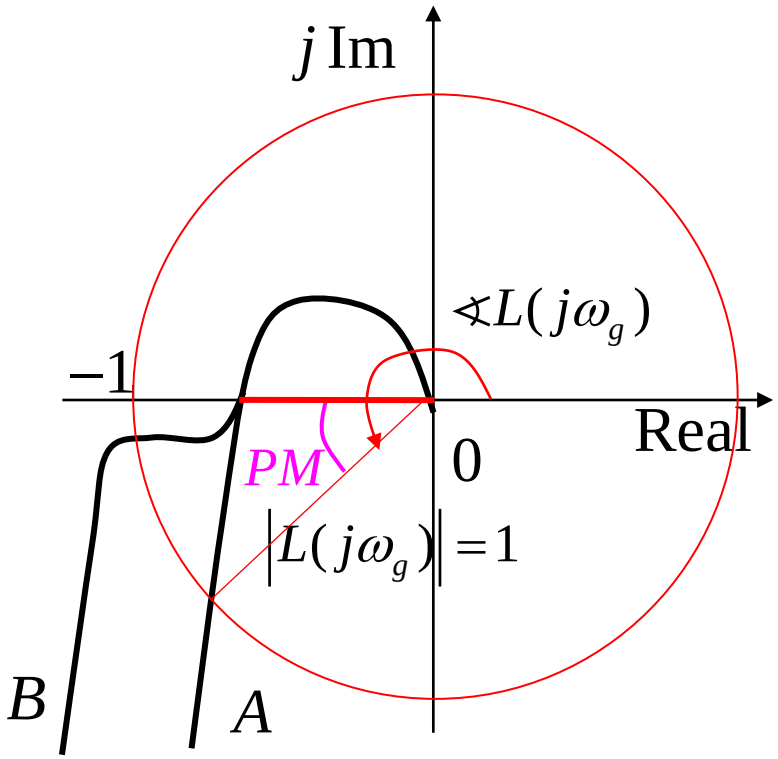
For stable systems the gain margin indicates how much the gain of the loop can be increased before the closed-loop system becomes unstable.

For unstable systems the gain margin indicates how much the gain of the loop must be decreased for the closed-loop system to become stable.

Phase Margin (PM)
(for minimum-phase loop transfer functions)

Gain margin alone is inadequate to indicate relative stability when system parameters other the loop gain are subject to variation.

With the same gain margin, the system represented by plot A is more stable than the system represented by plot B.



Gain crossover frequency ω_g

$$\left|L(j\omega_g)\right|=1$$

Phase margin:

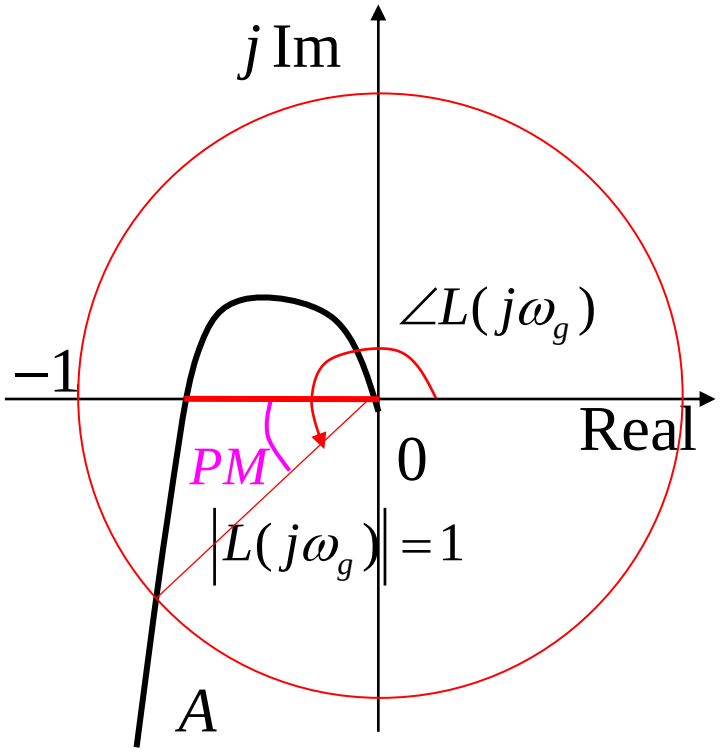
$$\text{PM}=\gamma=\angle L(j\omega_g)-180^\circ$$

Phase Margin (PM)
(for minimum-phase loop transfer functions)

Depending on how you calculated the phase of L(jω), the phase margin can be calculated as follows

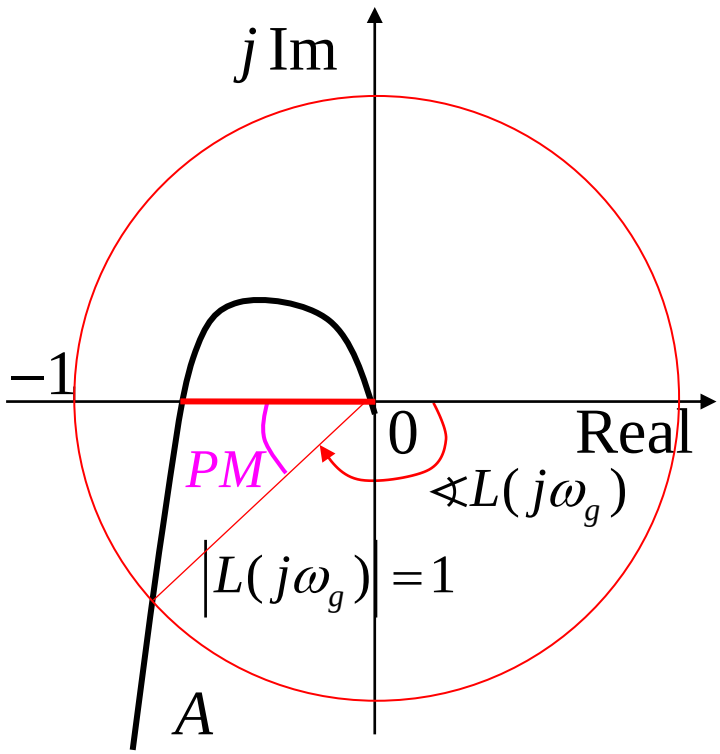
Phase margin:

$$\mathrm{PM} = \gamma = \angle L(j\omega_g) - 180^\circ$$



Phase margin:

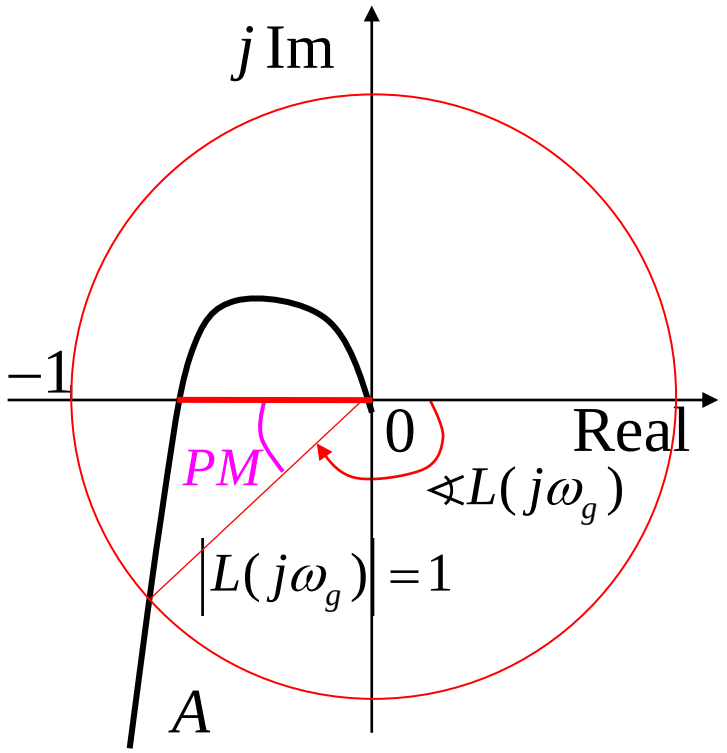
$$\mathrm{PM} = \gamma = \angle L(j\omega_g) + 180^\circ$$



Phase Margin (PM)
(for minimum-phase loop transfer functions)

The phase margin indicates the amount of phase shift of $L(j\omega)=G(j\omega)H(j\omega)$ at unity magnitude which can be tolerated before the system becomes unstable.

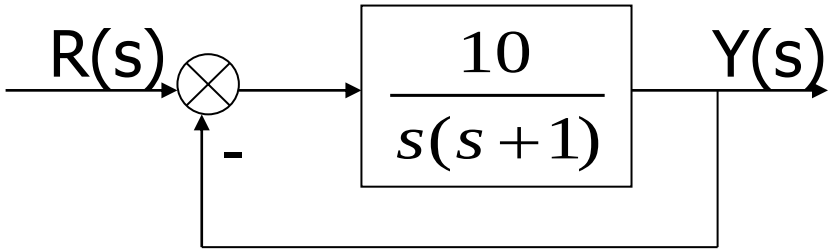
For stability of CL systems with minimum phase loop TF the phase margin must be positive.



Phase margin:

$$\text{PM}=\gamma=\measuredangle L(j\omega_g)+180^\circ$$

Example Consider the feedback control system



$$G(s)=\frac{10}{s(s+1)}$$

Determine the gain margin and the phase margin of the system.

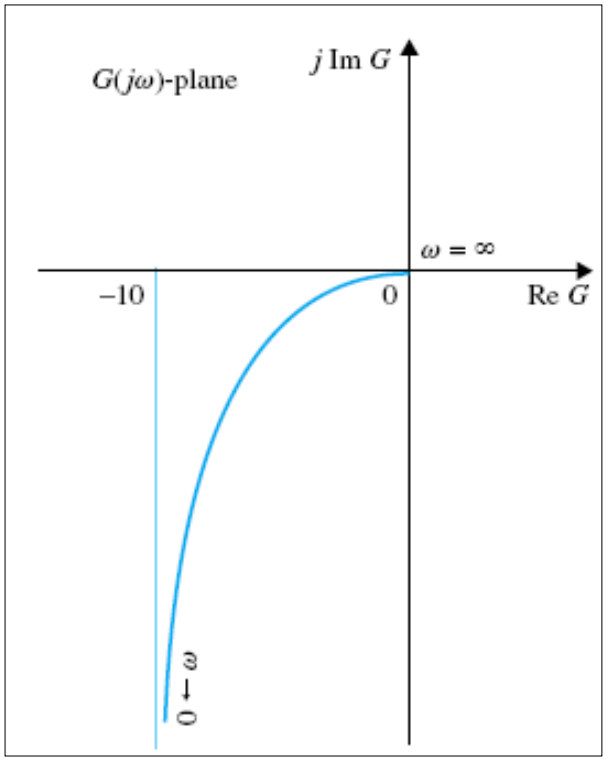
Solution. Substituting s=jω into G(s) yields: $G(j\omega)=\dfrac{10}{j\omega(j\omega+1)}$

$$G(j\omega)=\frac{10}{j\omega(j\omega+1)}=-\frac{10}{\omega^2+1}-j\frac{10}{\omega(\omega^2+1)}$$

Phase crossover: Intersection of the Nyquist plot of G(jω) with the real axis

$$\mathrm{Im}[G(j\omega)]=0 \Rightarrow \frac{10}{\omega(\omega^2+1)}=0 \Rightarrow \omega=\infty$$

The Nyquist plot of G(jω) *does not intersect the negative real axis* at any finite nonzero frequency. Therefore, the system has infinite gain margin.



Gain crossover: The frequency at which we have $|G(j\omega)|=1$. (Intersection of the Nyquist plot of $G(j\omega)$ with the unit circle.)

$$|G(j\omega)| = \left|\frac{10}{j\omega(j\omega+1)}\right| = \frac{10}{\omega\sqrt{\omega^2+1}} = 1 \quad \Rightarrow \quad \omega_g = 3.0842 \text{ rad/sec}$$

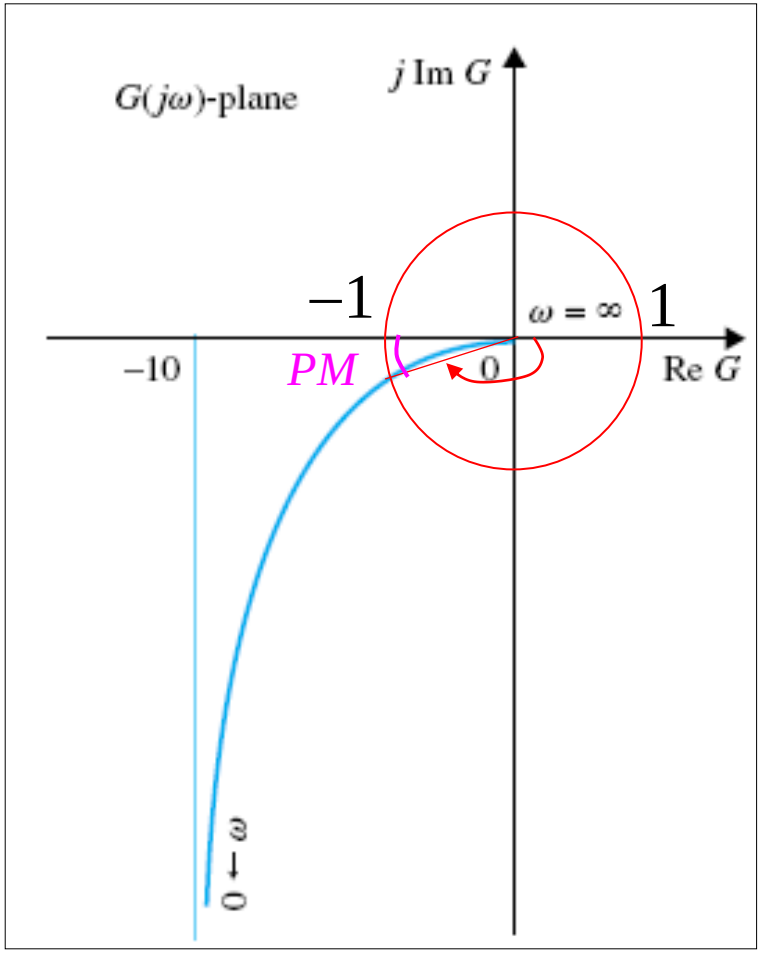
The phase angle of $G(j\omega)$ at $\omega=\omega_g$ is

$$\measuredangle G(j\omega_g) = \measuredangle G(j3.0842) = -90^o - \tan^{-1}\left(\frac{3.0842}{1}\right)$$

$$= -90^o - 72.0358^o = -162.0358^o$$

The phase margin is

$$\gamma = -162.0358^o + 180^\circ = 17.9642^o$$



Example Consider a system with a loop transfer function as

$$L(s) = \frac{2500}{s(s+5)(s+50)}$$

Determine its gain margin and phase margin.

Solution. Phase crossover frequency ω_p:

$$\mathrm{Im}[L(j\omega)] = 0 \Rightarrow \omega_p = 15.88 \text{ rad/sec}$$

Gain margin:

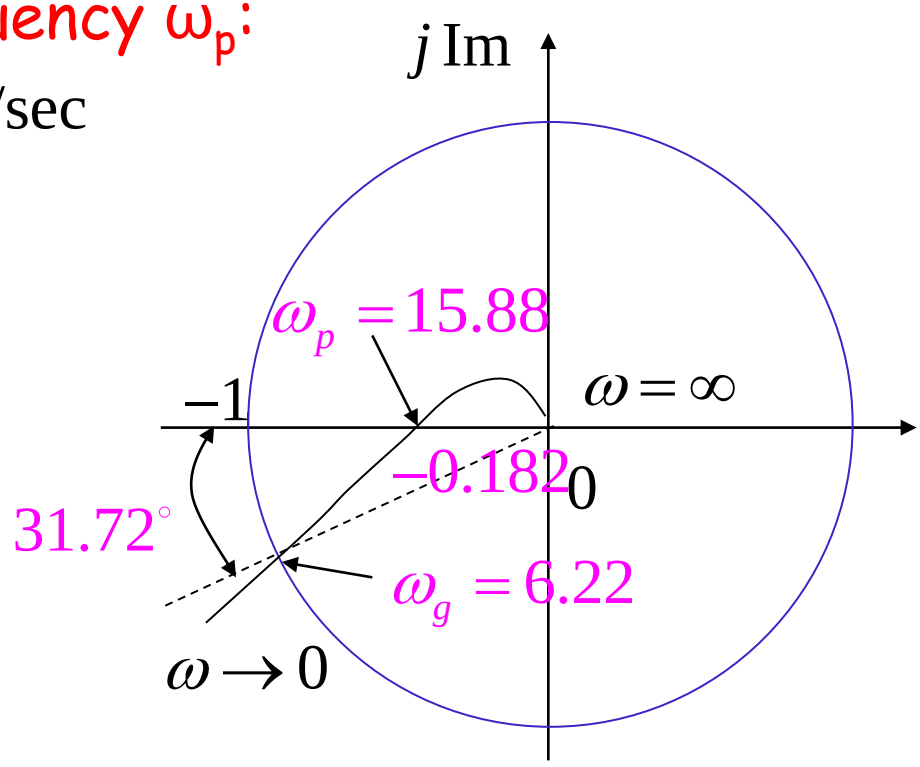
$$\left|L(j\omega_p)\right| = 0.182$$

$$K_g = \frac{1}{\left|L(j\omega_p)\right|} = 5.4945$$

$$\mathrm{GM} = -20\log_{10}\left|L(j\omega_p)\right| = 14.80 \text{ dB}$$

Gain crossover frequency ω_g:

$$\left|L(j\omega_g)\right| = 1 \Rightarrow \omega_g = 6.22 \text{ rad/sec}$$



Phase margin:

$$\gamma = \angle L(j\omega_g) - 180^\circ = 31.72^\circ$$

Gain and Phase Margins and the Bode Diagram
(for minimum-phase loop transfer functions)

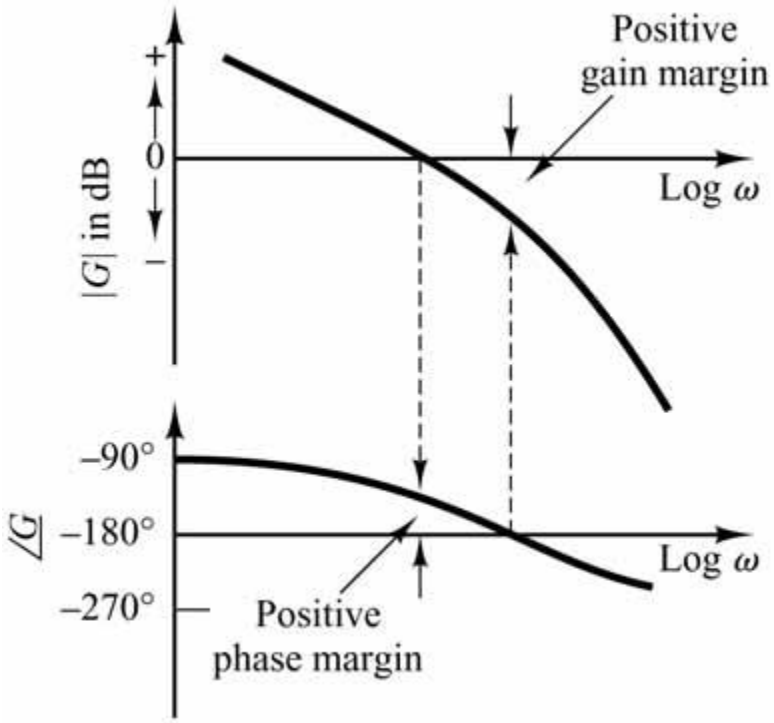
The gain margin and the phase margin can easily be determined from the bode diagrams.

Gain crossover: The frequency at which the magnitude plot cresses the 0-dB line.

Phase margin: The difference between the phase of the system and -180° at the gain crossover.

Phase crossover: The frequency at which the phase plot cresses the -180° line.

Gain margin (in dB): The difference between the 0-dB line and the magnitude of the system at the phase crossover.



Gain and Phase Margins and the Bode Diagram
(for minimum-phase loop transfer functions)

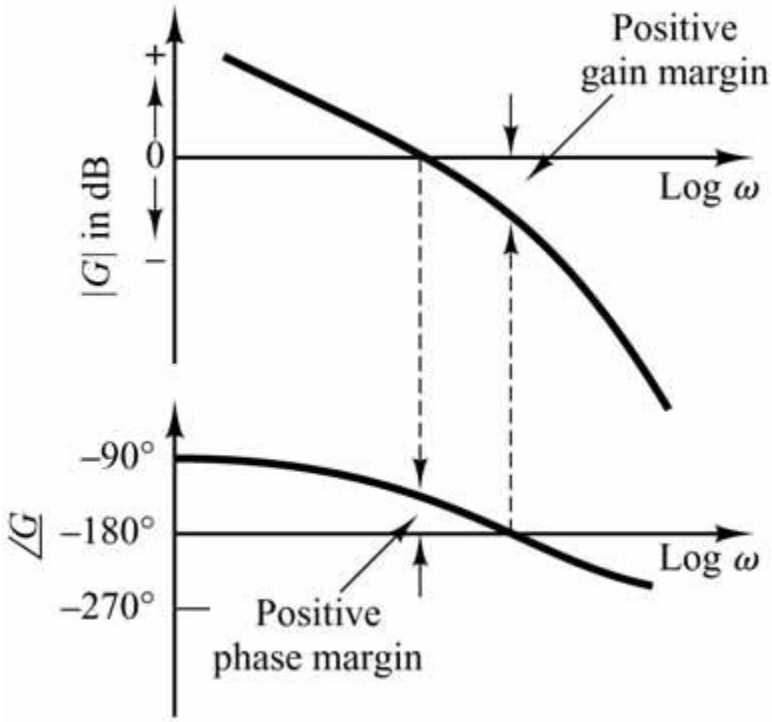
For stability of CL systems with minimum phase loop TF we need

$$K_g > 1 \Rightarrow \text{GM} > 0$$

(Positive GM in DB)

$$\text{PM} = \gamma = \angle L(j\omega_g) + 180^\circ > 0$$

(Positive phase margin)



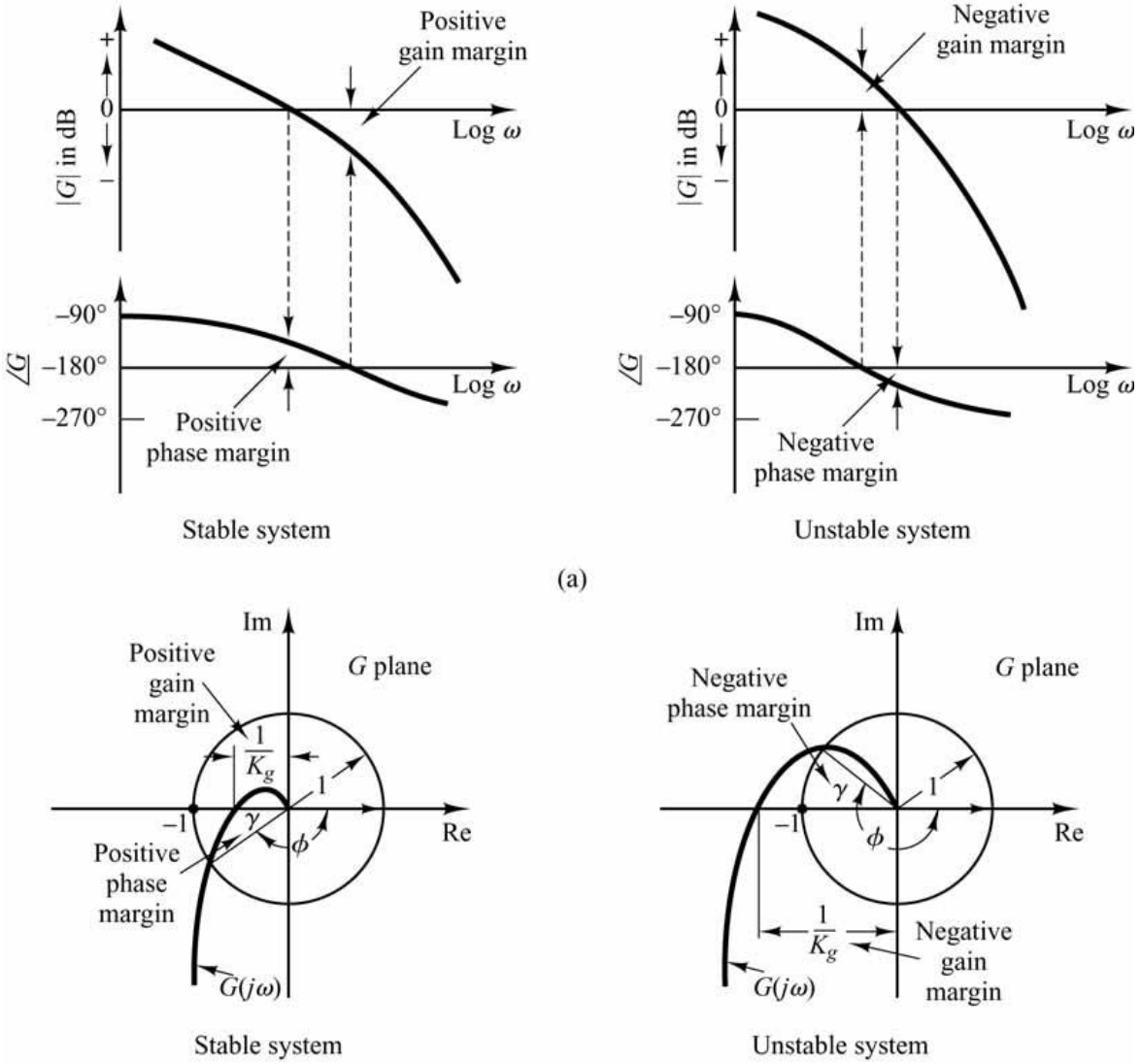
For a CL system with minimum-phase loop transfer function to be stable both the gain margin and the phase margin must be positive.

Gain and Phase Margins and the Bode Diagram
(for minimum-phase loop transfer functions)

Left hand side:

Stable system

positive gain and phase margins

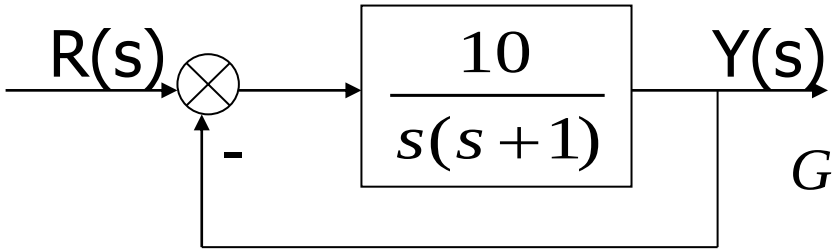


Right hand side:

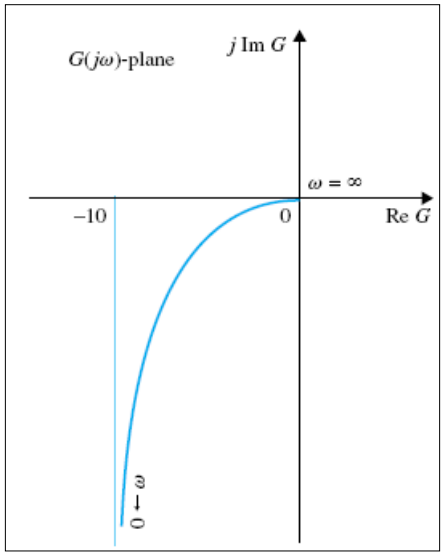
Unstable system

negative gain and phase margins

Example Consider again the system

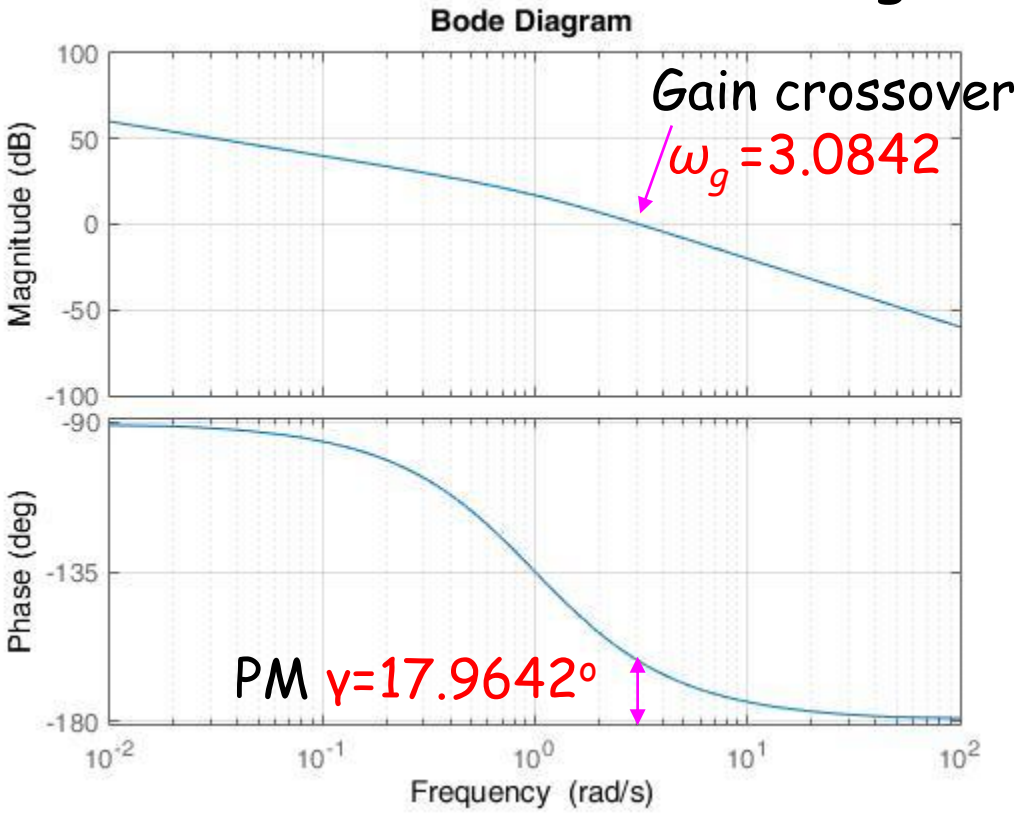


$$G(j\omega)=\frac{10}{j\omega(j\omega+1)}$$



Determine the gain margin and the phase margin of the system.

Solution. Sketch the Bode diagram



The phase angle of $G(j\omega_g)$ is

$$\angle G(j\omega_g)=-162.0358^o$$

The phase margin is

$$\gamma=-162.0358^o+180^\circ=17.9642^o$$

Gain crossover: ω_g =3.0842
Phase margin: γ=17.9642°

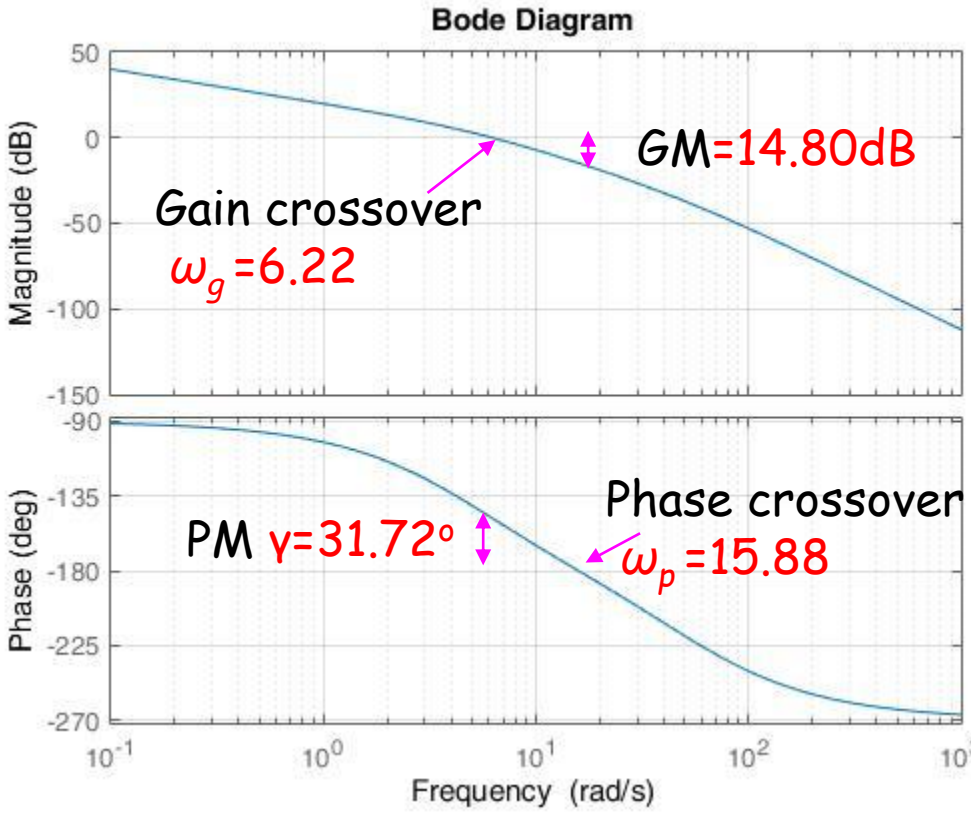
Phase crossover: No phase crossover
Gain margin: $K_g=\infty$

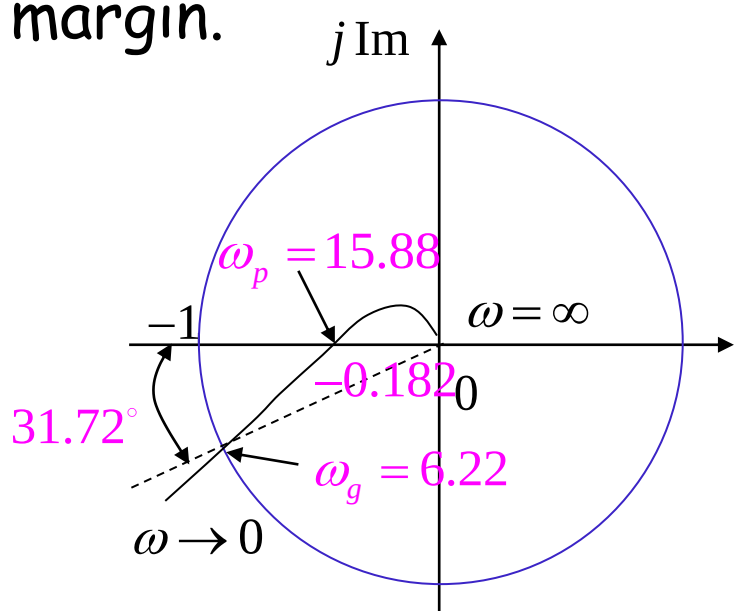
Example Consider a system with a loop transfer function

$$L(s) = \frac{2500}{s(s+5)(s+50)}$$

Determine its gain margin and phase margin.

Solution. Sketch the Bode diagram



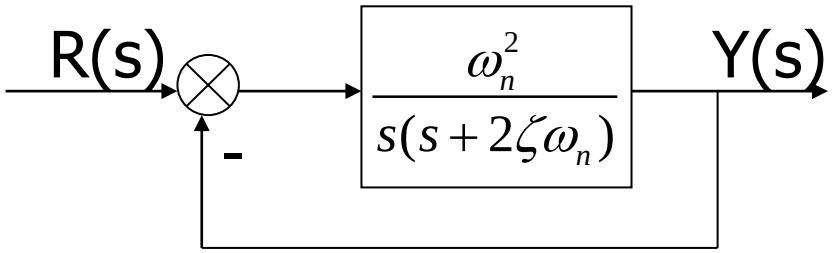


Gain crossover: ω_g=6.22
Phase margin: γ=31.72°

Phase crossover: ω_p=15.88
Gain margin (dB): GM=14.80 dB
Gain margin: K_g=5.4945

Phase Margin and Maximum Overshoot

Consider the second order feedback control system



$$G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$

It can be shown that there is a direct relationship between the phase margin γ and the damping ratio ζ .

In fact, for $0 \le \zeta \le 1/\sqrt{2}$ we have $\zeta \cong \gamma / 100$

This implies that there is a direct relationship between the phase margin γ and the maximum overshoot M_p in the time response.

Larger the phase margin => larger the damping ratio => smaller the overshoot.

Remarks on the Nyquist Plot

Advantages of Nyquist plot:

-Utilizing the Nyquist plot of the loop transfer function, the closed-loop stability can be easily determined with reference to the critical point (-1,j0).

- Nyquist plots can be used to analyze systems with either minimum phase or nonminimum phase loop transfer function.

Disadvantages of Nyquist plot:

-The Nyquist plot only is not convenient to carry out controller design.